Mediant Trees for Rational Approximation and Sturmian Word Generation

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Abstract

Given an oracle for a real number θ of sufficient precision, we present the following contributions. First, a closed-form formula for the unique optimal integer weight w^+ (Theorem 8.1): the weight whose weighted mediant of a Farey-adjacent bounded pair straddles θ. This yields an algorithm that constructs a *mediant tree* of increasingly precise rational approximations to θ, with every iterate (except, for rational θ, the final companion of the exact step) lying on the Stern–Brocot path of θ (Theorem 11.3). Second, structural theorems underpinning the algorithm's efficiency: exponential denominator growth (Lemma 11.14) and the no-consecutive-unit-weights property (Lemma 11.9) together imply that $O(\log N)$ iterations suffice to produce an approximant of size N (Theorem 11.15). Third, the mediant tree can be encoded as a *mediant word* — a string over $\{\mathsf{O}, \mathsf{E}\}$ — and this word is identical to the Sturmian word (the binary sequence $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ with slope $\alpha = 1/(1+\theta)$) at all interior positions, with the terminal symbol characterised by whether r is an overestimate or underestimate of θ (Theorem 12.6). Two structural invariants follow: every non-leaf mediant word begins with EE (Theorem 12.4), and the interior slice $\pi(r)[1{:}L{-}1]$ of any mediant word of length $L \geq 2$ is a palindrome (Theorem 12.5). Fourth, we show how to extend both algorithms to all real θ via simple algebraic reductions to the $\theta > 1$ case.

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1 Introduction

The approximation of irrational numbers by rationals is a classical subject in number theory. The Stern–Brocot tree and Farey sequences provide a systematic account of best rational approximants, and the continued-fraction expansion of a real number yields its convergents — the fractions closest to the target with bounded denominator. Sturmian words, introduced by Morse and Hedlund [1], give a combinatorial counterpart: the infinite binary sequence $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ with slope $\alpha \in (0,1)$ is the lower mechanical (Sturmian) word of slope α with intercept 0, having O-density α (other shifts are also balanced and aperiodic with the same density).

The present paper makes four contributions to this classical picture.

1. **Optimal weight formula.** We introduce the *bounded-pair* framework, in which two Farey-adjacent rational estimates bracket θ from above and below. Each weighted mediant step is a run of w steps in the Stern–Brocot tree in the same direction — the classical construction (see Graham, Knuth and Patashnik [2]). What is new is Theorem 8.1: a closed-form expression for the unique *optimal* integer weight w^+ — the one whose mediants straddle θ — derived from a sign-change bracket on the linear error function, enabling efficient iteration with no redundant steps.

2. **Mediant word representation.** The mediant tree can be encoded as a string over $\{\mathsf{O}, \mathsf{E}\}$, the *mediant word* (Section 10). This representation connects rational approximation to combinatorics on words.

3. **Sturmian identity.** The mediant word produced by the mediant tree is identical to the Sturmian word with slope $1/(1+\theta)$ (Theorem 12.6), agreeing at every position except possibly the last symbol of each finite prefix. The identification of Christoffel words with Sturmian prefixes is classical (Lothaire [3]); the novelty is the direct proof via the mediant construction and the precise last-letter criterion.

4. **Compressed transducer.** Each iterate produced by the algorithm (except, for rational θ, the final companion r^+ of the exact step) is a convergent or semi-convergent of θ (a node on the Stern–Brocot path of θ) — the best bracketing fraction achievable in one step from the current bounded pair. The mediant tree is therefore a *compressed transducer* for the Sturmian word: the approximation path ($O(\log N)$ real-arithmetic iterations, each requiring $O(1)$ oracle evaluations; proved via exponential denominator growth, Theorem 11.15) encodes the entire construction history, after which the N symbols are enumerated by a purely integer recursive traversal with no further real arithmetic.

The algorithm requires an oracle for θ — a means of evaluating $\lceil o\theta \rceil$ to sufficient precision at each step — but is otherwise entirely general. Given such an oracle, the mediant tree can represent *any* real number as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$, together with a tag identifying the regime of Table 1, making it a universal encoder from reals to binary words.

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Organisation. Part I (Sections 2–5) collects background on the Stern–Brocot tree, Farey sequences, Sturmian words, and Christoffel words. Part II develops the algorithm and its theory: Section 6 introduces the primitive definitions (bounded pair, approximant, λ, ε); Section 7 introduces the derived quantities (Δ, Θ, step types) used throughout Sections 8–12; Section 8 derives the optimal weight formula; Section 9 develops the mediant construction and straddling inequality; Section 10 introduces mediant words; Section 10.4 describes the mediant tree; Section 11 analyses the rate of convergence (including denominator growth and the $O(\log N)$ depth bound, Section 11); Section 12 establishes the connection to Sturmian words; Section 13 extends the construction to all real θ; Section 14 concludes with open questions.

Notation. Throughout, $\theta > 1$ is the target real number (the case $\theta \notin (1, \infty)$ is handled by reduction in Section 13). A rational approximant is written $r = e/o$ with numerator e (*even-count*) and denominator o (*odd-count*); $e, o > 0$ always, and $\gcd(e, o) = 1$ unless otherwise stated.

Alphabet. Mediant words are strings over $\{\mathsf{E}, \mathsf{O}\}$, where E stands for *even* and O for *odd*.

Depth index. The algorithm progresses through depths $d = 0, 1, 2, \ldots$. At each depth d (for rational θ, up to but not including the terminal depth) the algorithm holds a *bounded pair* (r_{2d}, r_{2d+1}), where r_{2d+1} is always the better estimate (smaller relative error $|\varepsilon|$) and r_{2d} the worse. We call r_{2d} the *older sibling* and r_{2d+1} the *younger sibling* of the pair at depth d; the older sibling is always the worse estimate and the younger sibling always the better. The flat linear subscripts $2d$ and $2d + 1$ reflect the sequential history of all estimates produced: r_0, r_1 are the initial integer bounds, and each depth $d \geq 1$ contributes two new estimates r_{2d} and r_{2d+1} (one of which may coincide with r_{2d-2} when $w^- = 0$).

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Intermediate quantities (depth d to $d + 1$). Between depth d and depth $d + 1$ the algorithm computes several quantities that are intermediate in the sense that they are derived from the pair (r_{2d}, r_{2d+1}) and used to produce the next pair. These carry *superscript* decorations $+$, $-$ (or $*$ for a combined value) to distinguish them from the depth-indexed estimates:

- w^+: the *optimal weight*, i.e. the weight computed in Step 4 of the algorithm (Section 8). At rational approximation steps it is the unique integer such that $\mathrm{Mediant}(r_{2d}, r_{2d+1}, w^+ - 1)$ and $\mathrm{Mediant}(r_{2d}, r_{2d+1}, w^+)$ straddle θ. At the exact step no weight straddles θ strictly; Step 4 then returns the integer one above the weight whose mediant equals θ (Lemma 8.4).
- $w^- = w^+ - 1$: the companion (sub-optimal) weight.
- r^+, r^-: the two candidate mediants at weights w^+ and w^-; the better one becomes r_{2d+3} and the worse r_{2d+2}.
- ε^+, ε^-: the relative errors of r^+ and r^-.
- λ^+, λ^-: the ratios $(e/o)/\theta$ for r^+ and r^-.
- e^+, e^- and o^+, o^-: the numerators and denominators of r^+ and r^-.

Superscript convention. The + superscript marks the candidate mediant produced at the *optimal* weight w^+, and the − superscript marks the candidate at the *sub-optimal* weight $w^- = w^+ - 1$. At *rational approximation* steps (those at which the weight that straddles θ is unique; see *Step types* in Section 7), w^+ is the unique weight at which the mediant switches to the opposite side of θ, so r^+ lies on the opposite side of θ from r_{2d}, while r^- lies on the same side as r_{2d}. At the *exact* step (which occurs only for rational θ and is the last step), $r^- = \theta$ exactly (on neither side), and r^+ still lies strictly on the opposite side of θ from r_{2d}, i.e. on the side of r_{2d+1} (Lemma 8.4). The superscript/subscript distinction is systematic: a $+/-$ *superscript* always marks an intermediate quantity between two depths; a numeric *subscript* always marks a position in the flat estimate sequence. These two systems are complementary and never overlap. When the depth matters we write $w_d^\pm$ for the propagated weights of Definition 10.2; in §§7–8 the depth is fixed and the subscript is omitted.

Gender. A rational approximant $r = e/o$ is called *female* if it overestimates θ (i.e. $e/o > \theta$, signed error > 0) and *male* if it underestimates θ (i.e. $e/o < \theta$, signed error < 0). At every approximation depth (every depth for irrational θ; every depth before the terminal one for rational θ) the pair contains exactly one male and one female member — the *bracketing invariant*. At the terminal depth for rational θ the better member is θ itself, which is neither male nor female; the other member is male if the exact step had $B < 0$ and female if $B > 0$. The exact step, in which $r^- = \theta$ becomes the younger sibling, is gender-changing only in the formal sense of the definition below.

A generation at depth $d \geq 0$ is called *gender-changing* if r^- becomes the younger sibling of the depth-$(d+1)$ pair (i.e. r^- is the better estimate), and *gender-preserving* otherwise (i.e. r^+ is the younger sibling). Lemma 11.9 (Section 11) states that $w_0^+ \geq 2$ always, and that no two consecutive weights w_d^+ equal 1. Unit-weight ($w_d^+ = 1$) generations are gender-preserving, but not conversely.

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Per-estimate scalar quantities. For a single estimate $r = e/o$ the key scalars are:

- $\lambda(r) = (e/o)/\theta$: the ratio of r to θ; equals 1 iff $r = \theta$.
- $\varepsilon(r) = \lambda(r) - 1$: the *relative error*; negative for underestimates, positive for overestimates.
- $\Delta(r, r') = e'o - eo'$: the *Farey determinant* of two estimates; $|\Delta| = 1$ iff r and r' are Farey adjacent.
- $\Theta(r) = o(e - \theta o)$: the *scaled error* of estimate $r = e/o$ (Definition 7.1); for the current pair, $\Theta := \Theta(r_{2d+1})$. Equals $\theta\, o^2\, \varepsilon$ and $\Theta = 0$ iff $r = \theta$ exactly.

Part I
Background

2 The Stern–Brocot Tree and Mediants

The *Stern–Brocot tree* is a complete binary search tree containing every positive rational exactly once in lowest terms (see, e.g., Graham, Knuth, and Patashnik [2]). It is constructed by repeated *mediant* insertion: starting from the pair $(0/1, 1/0)$, the mediant of two adjacent fractions a/b and c/d is $(a+c)/(b+d)$, and inserting it between them yields two new adjacent pairs. Every pair of fractions on adjacent branches of the Stern–Brocot tree satisfies $bc - ad = 1$; this is the Farey adjacency condition (Definition 3.1, Section 3). The Stern–Brocot tree encodes continued-fraction expansions as paths from the root: a partial quotient of k corresponds to k steps in the same direction before the first reversal.

The path from the root of the Stern–Brocot tree to θ passes through the *convergents* p_k/q_k of θ, defined as the fractions produced by truncating the continued-fraction expansion $\theta = [a_0; a_1, a_2, \ldots]$. The nodes visited between consecutive convergents p_{k-1}/q_{k-1} and p_k/q_k are the *semi-convergents*:

Definition 2.1 (Convergents and semi-convergents)**.** Let $\theta = [a_0; a_1, a_2, \ldots]$ with $p_{-1}/q_{-1} = 1/0$ and convergents p_k/q_k for $k \geq 0$. The *semi-convergents* between consecutive convergents p_k/q_k and p_{k+1}/q_{k+1} are the fractions

$$\frac{p_{k-1} + j\,p_k}{q_{k-1} + j\,q_k}, \quad 1 \leq j < a_{k+1}.$$

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(These appear on the SB path strictly between p_k/q_k and p_{k+1}/q_{k+1}.) The integer spine $1/1, 2/1, \ldots, (\lfloor\theta\rfloor - 1)/1$ consists of the initial SB-path nodes strictly before the first convergent $p_0/q_0 = \lfloor\theta\rfloor/1$. The convergents and semi-convergents together, plus the integer spine, form exactly the nodes on the Stern–Brocot path of θ. We use this broad definition throughout; in particular every SB-path node that is not a convergent is called a semi-convergent, with no additional size condition on j.

3 Farey Sequences

The *Farey sequence* F_n is the increasing sequence of fractions in $[0, 1]$ with denominator at most n, arranged in order.

Definition 3.1 (Farey adjacency)**.** Two fractions a/b and c/d (in lowest terms, $a/b < c/d$) are *Farey adjacent* if

$$bc - ad = 1.$$

Equivalently, $|bc - ad| = 1$ for an unordered pair. Consecutive terms in any Farey sequence F_n are Farey adjacent, and Farey-adjacent fractions lie on adjacent branches of the Stern–Brocot tree. Farey adjacency is equivalent to the condition that no rational with denominator at most $\max(b, d)$ lies strictly between a/b and c/d.

The mediant $(a+c)/(b+d)$ of two Farey-adjacent fractions is itself Farey adjacent to each of them; their mediant $(a+c)/(b+d)$ is also the unique fraction with smallest denominator strictly between them. This adjacency-preservation is the key structural property exploited by our algorithm.

4 Sturmian Words

Sturmian words were introduced by Morse and Hedlund [1] as the simplest aperiodic symbolic sequences. Let $\alpha \in (0,1)$ be irrational. The *lower Sturmian word* with slope α is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \geq 0.$$

Encoding $s_n = 1$ as O and $s_n = 0$ as E, this is a binary sequence with O-density exactly α. Since $\alpha < 1$, we have $s_0 = \lfloor \alpha \rfloor = 0$, so the Sturmian word always begins with E.

Definition 4.1 (Sturmian word)**.** The *Sturmian word* $\mathbf{s}(\alpha)$ with slope $\alpha \in (0,1)$ is the infinite word over $\{\mathsf{O}, \mathsf{E}\}$ defined by $s_n = \mathsf{O}$ if $\lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor$, and $s_n = \mathsf{E}$ otherwise.

Sturmian words are *balanced*: no two factors of the same length differ in their O-count by more than one. For irrational α, the lower mechanical (Sturmian) word of slope α with intercept 0 is the canonical balanced aperiodic sequence; shifts of it are also balanced and aperiodic [1].

The two-word structure is fundamental: a Sturmian word with slope $\alpha \in (0, 1/2)$ consists of exactly two distinct *primitive words* (unbreakable building blocks), of lengths $\lfloor 1/\alpha \rfloor$ and $\lceil 1/\alpha \rceil$, concatenated in the Sturmian pattern. Concretely, with $\alpha = 1/(1+\theta)$ for $\theta > 1$, the two primitive words are $\mathsf{E}^{\lfloor \theta \rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil \theta \rceil}\mathsf{O}$ — exactly the leaf words of the mediant tree (Section 10.1).

Since $\alpha < 1/2$ for $\theta > 1$, the sequence avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. Conversely, when $\alpha > 1/2$ (i.e. $\theta < 1$), consecutive O's are unavoidable.

Sturmian morphisms. The two *Sturmian morphisms* $\sigma_E \colon \mathsf{E} \mapsto \mathsf{E},\ \mathsf{O} \mapsto \mathsf{EO}$ and $\sigma_O \colon \mathsf{E} \mapsto \mathsf{OE},\ \mathsf{O} \mapsto \mathsf{O}$ map Sturmian words to Sturmian words (of a different slope in general).

5 Christoffel Words

Definition 5.1 (Christoffel words)**.** For coprime positive integers e and o, the *lower Christoffel word* $\mathrm{Ch}^-(e,o)$ is the unique primitive word of length $L = e + o$ over $\{\mathsf{E}, \mathsf{O}\}$ containing exactly e copies of E and o copies of O that ends in O, with O-positions given by

$$\mathrm{Ch}^-(e,o)[i] = \mathsf{O} \iff \lfloor (i+1)o/L \rfloor > \lfloor i \cdot o/L \rfloor.$$

The *upper Christoffel word* $\mathrm{Ch}^+(e,o)$ is the reverse of $\mathrm{Ch}^-(e,o)$: it ends in E instead of O. Both words share the same palindromic interior U: $\mathrm{Ch}^-(e,o) = \mathsf{E} \cdot U \cdot \mathsf{O}$ and $\mathrm{Ch}^+(e,o) = \mathsf{O} \cdot U \cdot \mathsf{E}$ (Berstel [5], Chapter 1).

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$\mathrm{Ch}^-(e,o)$ is the lower Sturmian word of slope o/L truncated to length L and normalised to end in O; $\mathrm{Ch}^+(e,o)$ is the upper Sturmian word of the same slope, ending in E (Berstel [5], Chapter 1).

Each Christoffel word is characterised purely by its fraction o/L: given $\gcd(e,o)=1$, the pair (e,o) is uniquely determined by $o/(e+o)$.

Theorem 5.2 (Christoffel concatenation (Berstel [5], Theorem 1.5.1))**.** *Let* $\mathrm{Ch}^-(e_a,o_a)$ *and* $\mathrm{Ch}^-(e_b,o_b)$ *be two lower Christoffel words with* $o_a/L_a < o_b/L_b$ *(where* $L_a = e_a+o_a$*,* $L_b = e_b+o_b$*) satisfying the Farey adjacency condition* $o_bL_a - o_aL_b = 1$*. Then for any integer* $w \geq 1$*:*

$$\begin{aligned}\mathrm{Ch}^-(e_a,o_a)^w \cdot \mathrm{Ch}^-(e_b,o_b) &= \mathrm{Ch}^-(w\,e_a+e_b,\ w\,o_a+o_b),\\ \mathrm{Ch}^-(e_a,o_a) \cdot \mathrm{Ch}^-(e_b,o_b)^w &= \mathrm{Ch}^-(e_a+w\,e_b,\ o_a+w\,o_b).\end{aligned}$$

In the first identity the smaller-slope word is repeated; in the second the larger-slope word is repeated.

Part II
The Algorithm

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6 Definitions and Notation

Throughout Sections 6–12, let $\theta > 1$ be a fixed real number. Section 13 handles all other real numbers by reduction to this case. We write $\mathbb{N}$ for the positive integers and $\lfloor\cdot\rfloor$, $\lceil\cdot\rceil$ for floor and ceiling.

Definition 6.1 (Approximant)**.** An *approximant* is an ordered pair $r=(e,o)\in\mathbb{N}^2$ denoting the rational number e/o. The *canonical* form satisfies $\gcd(e,o)=1$; this is assumed unless otherwise stated. A non-canonical representation is permitted when only the value e/o matters. An approximant $r=(e,o)$ is a *rational estimate* of θ when used in that context.

Definition 6.2 (Mediant)**.** Given two approximants $r=(e,o)$ and $r'=(e',o')$ and a non-negative integer $w\geq 0$, the *mediant* of r and r' with weight w is the approximant

$$\mathrm{Mediant}(r,\,r',\,w) = (e+w\,e',\ o+w\,o'),$$

which is in canonical form whenever r and r' are Farey-adjacent (Lemma 9.1(4)); in general it need not be. The *Farey determinant* of the pair (r,r') is

$$\Delta(r,r') = e'o - eo'.$$

The pair (r,r') is *Farey-adjacent* when $|\Delta(r,r')|=1$ (Definition 3.1). The degenerate case $w=0$ gives $\mathrm{Mediant}(r,r',0)=r$. Throughout this paper, when $\mathrm{Mediant}(r,r',w)$ is used as a rational approximation of θ, the convention is that r' is the better estimate in the sense that $|\varepsilon'|<|\varepsilon|$, i.e. $|r'/\theta-1|<|r/\theta-1|$.

Definition 6.3 (Good estimate)**.** An estimate e/o is a *good estimate* of θ if

$$e \in \{\lfloor o \cdot \theta \rfloor,\ \lceil o \cdot \theta \rceil\},$$

i.e. e/o is one of the two nearest rationals to θ with denominator o. A good estimate with $e = \lceil o\theta \rceil$ satisfies $e/o \geq \theta$ and is an *upper bound*; one with $e = \lfloor o\theta \rfloor$ satisfies $e/o \leq \theta$ and is a *lower bound*.

Definition 6.4 (Bounded pair)**.** Two good estimates e_l/o_l and e_u/o_u form a *bounded pair* if e_l/o_l is a lower bound, e_u/o_u is an upper bound,

$$\frac{e_l}{o_l} < \theta \leq \frac{e_u}{o_u},$$

and they satisfy the *Farey adjacency condition*

$$e_u \cdot o_l - e_l \cdot o_u = 1.$$

Definition 6.5 (λ and ε)**.** For an estimate e/o of θ, define the *relative ratio*

$$\lambda = \frac{e/o}{\theta}$$

and the *relative error*

$$\varepsilon = \lambda - 1 = \frac{e/o}{\theta} - 1.$$

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An upper-bound good estimate satisfies $\lambda \geq 1$ (so $\varepsilon \geq 0$); a lower-bound good estimate satisfies $\lambda \leq 1$ (so $\varepsilon \leq 0$). In both cases $|\varepsilon|$ is small when o is large.

Definition 6.6 (Age ordering)**.** For approximants r_i and r_j produced by the mediant tree, define

$$\mathrm{age}(r_i, r_j) = (r_{\min(i,j)},\ r_{\max(i,j)}).$$

Writing $(u, v) = \mathrm{age}(r_i, r_j)$ gives $u = r_{\min(i,j)}$ and $v = r_{\max(i,j)}$, so u was produced no later than v in the mediant tree (u is the *older* estimate; siblings produced at the same step are ordered by subscript).

Definition 6.7 (Height ordering)**.** For approximants r and r' with $\varepsilon \neq \varepsilon'$, define

$$\mathrm{height}(r, r') = (r_{\mathrm{lo}},\ r_{\mathrm{hi}}),$$

where $\varepsilon_{\mathrm{lo}} < \varepsilon_{\mathrm{hi}}$ (signed-error order). Writing $(u, v) = \mathrm{height}(r, r')$ gives $\varepsilon_u < \varepsilon_v$. When r and r' bracket θ, u is the *underestimate* ($u/\theta < 1$) and v is the *overestimate* ($v/\theta \geq 1$); in general only the signed-error ordering is guaranteed.

Definition 6.8 (Fit ordering)**.** For a target θ and approximants r, r' with $|\varepsilon| \neq |\varepsilon'|$ (where $\varepsilon = e/o - \theta$ evaluated at each), define

$$\mathrm{fit}(\theta,\, r,\, r') = (r_{\mathrm{worse}},\; r_{\mathrm{better}}),$$

where $|\varepsilon_{\mathrm{worse}}| > |\varepsilon_{\mathrm{better}}|$ (absolute-error order). Writing $(u, v) = \mathrm{fit}(\theta, r, r')$ gives $|\varepsilon_u| > |\varepsilon_v|$: u is the *worse* estimate and v is the *better* estimate.

Definition 6.9 (Alpha transformation)**.** For real or rational numbers $x, y \geq 0$, define

$$\mathrm{alpha}(x, y) = \Big(\frac{1}{1+y},\; \frac{1}{1+x}\Big).$$

The key invariant is that $x \mapsto 1/(1+x)$ is strictly decreasing, so

$$\frac{1}{1+y} \leq \frac{1}{1+x} \iff x \leq y.$$

Writing $(u, v) = \mathrm{alpha}(x, y)$, the pair (u, v) preserves the size relations of (x, y): $u \leq v \iff x \leq y$. If x and y are Farey-adjacent approximants then so are u and v (Lemma 9.2), with $\Delta(u, v) = \Delta(x, y)$. The map alpha corresponds to the change of variable $\theta \mapsto \alpha = 1/(1+\theta)$ that relates the mediant algorithm for θ to the Sturmian slope α; it is named alpha (not α) to avoid clash with that symbol.

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Index convention

Estimates are indexed so that r_{2d} and r_{2d+1} form the bounded pair at depth d, with r_{2d+1} always the better estimate (smaller $|\varepsilon|$). Their children are r_{2d+2} and r_{2d+3}, the bounded pair at depth $d+1$. We write λ_{2d}, ε_{2d}, etc. for the corresponding values. In a bounded pair we always orient so that $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, i.e. r_{2d+1} is the better estimate of the two.

Farey adjacency and the Stern–Brocot tree

The Farey adjacency condition in Definition 6.4 ensures that the pair lies on adjacent branches of the Stern–Brocot tree (Section 2) and that no rational with smaller denominator lies strictly between e_l/o_l and e_u/o_u. In terms of Δ, the condition is precisely $|\Delta| = 1$; the sign depends on which of r_{2d}, r_{2d+1} is the upper bound.

Domain restriction

The definitions in this section assume $\theta > 1$. Section 13 shows how to reduce all other real numbers to this case.

Rational vs. irrational θ

For rational θ, $-A/B$ can be an integer w^-; the depth is then *exact* (Section 7, *Step types*). No integer satisfies the strict straddling condition: weights w^- and $w^+ = w^- + 1$ give product zero, and all others give a positive product. The mediant at weight w^- equals θ, Step 4 computes $w^+ = w^- + 1$, Step 5 places θ second, and the next depth is terminal (Lemma 8.4). Hence for rational θ the algorithm terminates after finitely many steps. All claims of Sections 9–12 hold at every approximation depth and at every depth up to termination, with the following exceptions.

- Theorems 12.6, 11.4, 11.8, and Corollary 12.7 require θ irrational.
- *At the exact step.* The hypotheses of Theorem 8.1, Corollary 8.2, and the strict bracketing hypothesis of Theorem 9.3 and Corollary 9.4 are unmet. The conclusions of Theorem 9.3 and the second inequality of Corollary 9.4 still hold by direct argument; the first conclusion of Corollary 9.4 fails ($\varepsilon^-\varepsilon^+ = 0$); the conclusion of Corollary 8.2 holds if and only if $B < 0$. Theorems 11.6 and 11.8 are also excluded: at an exact depth, $\theta = r_{2d} \oplus r_{2d+1}$ (Lemma 8.4(3)), so r_{2d+1} is one of the two Stern–Brocot parents of θ, and the convergent/semi-convergent label of r_{2d+1} is CF-convention-dependent; Theorem 11.8 requires irrational θ.
- *At the terminal depth.* $\operatorname{sgn}(\Theta) = \operatorname{sgn}(\Delta)$ fails ($\Theta = 0$, $|\Delta| = 1$); the pair is a bounded pair if and only if $B < 0$ (with B taken at the exact step); Theorem 12.2 applies to both r_{2d+2} and $r_{2d+3} = \theta$ (part (3) of that theorem covers $r = \theta$); and $r_{2d+2} = r^+$ is a Stern–Brocot descendant of θ, so Theorems 11.3, 11.4, and 11.8 are restricted to approximation depths (the latter two require irrational θ and so never encounter the terminal depth).

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The pair produced at the exact step is $(r_{2d+2}, r_{2d+3}) = (r^+, \theta)$, where r^+ lies strictly on the side of r_{2d+1}, opposite to r_{2d}. If $B > 0$, both members satisfy $r \geq \theta$ and the pair is not a bounded pair. If $B < 0$, it is a bounded pair with θ as its upper member.

7 Derived Quantities

The following quantities are defined in terms of the primitives of Section 6 and are used throughout Sections 9–12.

Definition 7.1 (Δ and Θ)**.** Given approximants $r_{2d} = e_{2d}/o_{2d}$ and $r_{2d+1} = e_{2d+1}/o_{2d+1}$ (in practice the pair held by the algorithm at depth d, which at every approximation depth is a bounded pair with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$), define

$$\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$$

and

$$\Theta = o_{2d+1}(e_{2d+1} - \theta \cdot o_{2d+1}).$$

The invariants Δ and Θ

More generally, define $\Theta(r) = o(e - \theta o)$ for any estimate $r = e/o$; then $\Theta := \Theta(r_{2d+1})$ for the current pair. Geometrically, $\Theta(r)/o = e - \theta o$ is the signed vertical distance from the lattice point (o, e) to the line $e = \theta o$, and $|\Theta(r)|$ is twice the area of the triangle with vertices $(0, 0)$, $(o, \theta o)$, (o, e). Equivalently, $\Theta = \theta\, o_{2d+1}^2 \varepsilon_{2d+1}$ is the signed absolute error of r_{2d+1} scaled by its denominator squared. One checks that $|\Theta|/o_{2d+1} = |e_{2d+1} - \theta o_{2d+1}| < 1$ (the good-estimate condition), and that $\operatorname{sgn}(\Theta) = \operatorname{sgn}(\Delta)$ at every non-terminal step (i.e. when $\Theta \neq 0$).

Step types

Let (r_{2d}, r_{2d+1}) be the pair held by the algorithm at depth d, and write $A = e_{2d} - \theta\, o_{2d}$ and $B = e_{2d+1} - \theta\, o_{2d+1}$ for the scaled errors of its members (the quantities of Section 8), so that $\Theta = o_{2d+1} B$. We call depth d

- *terminal* if $\Theta = 0$, i.e. $r_{2d+1} = \theta$: the algorithm halts at Step 3 and produces no children;
- *approximation* if $\Theta \neq 0$. An approximation depth is either
  - *rational approximation* if $-A/B \notin \mathbb{Z}$: Theorem 8.1 and Corollary 8.2 apply, and w^+ is the unique weight whose consecutive mediants straddle θ; or
  - *exact* if $-A/B = w^- \in \mathbb{Z}$: no integer weight straddles θ strictly, the mediant at weight w^- equals θ, and Lemma 8.4 (Section 8) gives the full description. The exact step can occur only for rational θ, and when it does the next depth is terminal.

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The word "non-terminal" is reserved for approximation depths (rational approximation or exact) as a group.

Bounded pairs and the mediant tree

A bounded pair is a purely analytic object: any two good estimates that bracket θ and satisfy the Farey adjacency condition qualify, regardless of how they were found. The mediant construction (Section 9) provides a canonical way to *generate* bounded pairs, but the concept is wider. The invariants Δ and Θ belong to the pair itself; the Christoffel word structure (Section 5) is a property of how the pair was reached via the mediant tree. In particular, $\Delta = \pm 1$ holds for any bounded pair; $\operatorname{sgn}(\Theta) = \operatorname{sgn}(\Delta)$ holds when $\Theta \neq 0$ (i.e. at non-terminal steps).

8 The Optimal Weight Formula

A key question is: for which integer $w \geq 1$ do the two consecutive weighted mediants

$$r^- = \text{Mediant}(r_{2d}, r_{2d+1}, w - 1), \qquad r^+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$$

straddle θ, i.e. satisfy the bracketing condition $(\lambda^- - 1)(\lambda^+ - 1) < 0$?

Writing $A = e_{2d} - \theta\, o_{2d}$ and $B = e_{2d+1} - \theta\, o_{2d+1}$ for the scaled errors, the bracketing condition becomes $(A+(w-1)B)(A+wB) < 0$. Observe that $A+(w-1)B$ and $A+wB$ are consecutive evaluations of the linear function $f(w) = A + wB$ at $w-1$ and w: their product is negative iff f changes sign between $w-1$ and w, i.e. iff the zero $-A/B$ of f lies in the interval $(w-1, w)$. Assuming $-A/B \notin \mathbb{Z}$ (the hypothesis of Theorem 8.1), this is equivalent to $w = \lfloor -A/B \rfloor + 1$, and exactly one such w exists. (Note: $B = e_{2d+1} - \theta\, o_{2d+1} = \Theta/o_{2d+1}$, so $B \neq 0$ iff $\Theta \neq 0$; the theorem is only invoked at non-terminal steps where $\Theta \neq 0$, ensuring $-A/B$ is well-defined.) The unique integer satisfying this bracketing condition is given by the following closed-form expression.

Theorem 8.1 (Optimal weight)**.** *Let (r_{2d}, r_{2d+1}) be a bounded pair bracketing θ with $-A/B \notin \mathbb{Z}$ (in particular when θ is irrational), where*

$$A = e_{2d} - \theta\, o_{2d}, \qquad B = e_{2d+1} - \theta\, o_{2d+1},$$

and set $\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$, $\Theta = o_{2d+1}B$. Then the unique integer $w \geq 1$ for which $r^- = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w-1)$ and $r^+ = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w)$ satisfy the bracketing condition $(\lambda^- - 1)(\lambda^+ - 1) < 0$ is

$$w^+ = \left\lfloor \frac{-A}{B} \right\rfloor + 1 = \left\lfloor \frac{\Delta}{\Theta} - \frac{o_{2d}}{o_{2d+1}} \right\rfloor + 1.$$

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Proof. We prove the closed-form formula $w^+ = \lfloor -A/B \rfloor + 1$ directly from the bracketing condition, using only a linear sign argument.

Since r_{2d} and r_{2d+1} are on opposite sides of θ (so $A, B \neq 0$ — both are non-terminal estimates) we have $\mathrm{sgn}(A) = -\mathrm{sgn}(B)$, so $A/B < 0$ and $c := -A/B > 0$. The two weighted mediants are

$$r^- = \frac{e_{2d} + (w-1)\, e_{2d+1}}{o_{2d} + (w-1)\, o_{2d+1}}, \qquad r^+ = \frac{e_{2d} + w\, e_{2d+1}}{o_{2d} + w\, o_{2d+1}},$$

with signed errors (numerators of ε^-, ε^+ up to positive denominators)

$$A + (w-1)B \qquad \text{and} \qquad A + wB.$$

Bracketing condition. The pair (r^-, r^+) brackets θ iff ε^- and ε^+ have opposite signs, i.e. $(A+(w-1)B)(A+wB) < 0$. Writing $c = -A/B$, this product equals $B^2\,(w-1-c)(w-c)$, which is negative iff $w-1-c$ and $w-c$ have opposite signs, i.e.

$$w - 1 < c < w,$$

and since $c = -A/B \notin \mathbb{Z}$ by hypothesis, this is equivalent to $w - 1 = \lfloor c \rfloor$, i.e. $w = \lfloor c \rfloor + 1$. The hypothesis ensures exactly one integer w satisfies the condition, namely

$$w^+ = \lfloor -A/B \rfloor + 1.$$

We also write $w^- = w^+ - 1 = \lfloor -A/B \rfloor$ for the companion weight.

Expression in terms of Δ and Θ. Recall $\Theta = o_{2d+1}B$ and $A\,o_{2d+1} - B\,o_{2d} = -\Delta$, giving $A = (Bo_{2d} - \Delta)/o_{2d+1}$, so

$$\frac{-A}{B} = \frac{\Delta}{o_{2d+1}B} - \frac{o_{2d}}{o_{2d+1}} = \frac{\Delta}{\Theta} - \frac{o_{2d}}{o_{2d+1}},$$

and therefore

$$w^+ = \left\lfloor \frac{\Delta}{\Theta} - \frac{o_{2d}}{o_{2d+1}} \right\rfloor + 1,$$

completing the proof.

Termination. When $\Theta = 0$, r_{2d+1} is an exact rational representation of θ; no further improvement is possible and the algorithm terminates. □

When $\Theta = 0$, r_{2d+1} is an exact rational representation of θ and no further improvement is possible; the algorithm terminates.

Corollary 8.2 (Children at w^+ form a bounded pair)**.** *Under the conditions of Theorem 8.1, the two children* $r^- = \text{Mediant}(r_{2d}, r_{2d+1}, w^-)$ *and* $r^+ = \text{Mediant}(r_{2d}, r_{2d+1}, w^+)$ *form a bounded pair.*

Proof. Farey adjacency $|e^+o^- - e^-o^+| = 1$ follows from Lemma 9.1(3), since $r^- = M_{w^-}$ and $r^+ = M_{w^+}$ are consecutive mediants ($w^+ = w^- + 1$). It remains to show both children are good estimates. Using $f(w) = A + wB$ from the Section 8 preamble, for each $w^\dagger \in \{w^-, w^+\}$ set $r^\dagger = \text{Mediant}(r_{2d}, r_{2d+1}, w^\dagger)$ with components $(e^\dagger, o^\dagger)$; then $f(w^\dagger) = e^\dagger - \theta\, o^\dagger$ (the scaled error of $r^\dagger$). The bracketing condition $f(w^-)\, f(w^+) < 0$ means the two values have opposite signs, so

$$|f(w^-)|\, o^+ + |f(w^+)|\, o^- = |f(w^-)\, o^+ - f(w^+)\, o^-| = |e^-\, o^+ - e^+\, o^-| = |\Delta| = 1.$$

Both terms on the left are in fact positive: the bracketing condition $f(w^-)\, f(w^+) < 0$ implies $f(w^-) \neq 0$ and $f(w^+) \neq 0$, so $|f(w^-)|$ and $|f(w^+)|$ are both nonzero; and $o^-, o^+ \geq 1$. Since two positive terms sum to 1, each is strictly less than 1, i.e. $|e^- - \theta o^-| < 1$ and $|e^+ - \theta o^+| < 1$, confirming both children are good estimates. Moreover, $\gcd(e^+, o^+) = \gcd(e^-, o^-) = 1$ (Lemma 9.1(4)), so the children are approximants in canonical form. □

Lemma 8.3 (Tree approximants are good estimates)**.** *Every approximant produced by the mediant tree is a good estimate of* θ*.*

Proof. By induction on depth. The inductive hypothesis is: the pair at depth d is a bounded pair of good estimates (valid for all depths at which the algorithm steps, i.e. while $\Theta \neq 0$). The seed pair (r_0, r_1) satisfies this by assumption.

At each subsequent step, let (r_{2d}, r_{2d+1}) be the depth-d bounded pair. Two cases arise.

Case 1: $-A/B \notin \mathbb{Z}$. The bracketing condition $f(w^-)\, f(w^+) < 0$ implies both f-values are nonzero. The proof of Corollary 8.2 gives $|f(w^-)|\, o^+ + |f(w^+)|\, o^- = 1$, with both terms positive and $o^-, o^+ \geq 1$, so each term is strictly less than 1. Hence $|e^- - \theta\, o^-| < 1$ and

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$|e^+ - \theta\, o^+| < 1$: both children are good estimates. By Corollary 8.2, they form a bounded pair.

Case 2: $-A/B = w^- \in \mathbb{Z}$. Then $f(w^-) = 0$, so $r^- = \theta$ exactly, which is trivially a good estimate. For the other child, $f(w^+) = f(w^-) + B = B$, and $|B| < 1$ because r_{2d+1} is a good estimate (inductive hypothesis). Thus $|e^+ - \theta\, o^+| = |B| < 1$, so r^+ is also a good estimate. (In this case the next depth is terminal; see Lemma 8.4.) □

Lemma 8.4 (The exact step)**.** *Let depth d be exact (*Step types*, Section 7): (r_{2d}, r_{2d+1}) is a bounded pair with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, $\Theta \neq 0$, and $-A/B = w^- \in \mathbb{Z}$. Let $r^- = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^-)$ and $r^+ = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^+)$ with $w^+ = w^- + 1$. Then:*

1. *$w^- \geq 1$, and Step 4 computes $\lfloor \Delta/\Theta - o_{2d}/o_{2d+1} \rfloor + 1 = w^- + 1 = w^+$; in particular $w^+ \geq 2$.*
2. *No integer w satisfies $f(w-1)\, f(w) < 0$.*
3. *$r^- = \theta$ exactly and $f(w^+) = e^+ - \theta\, o^+ = B$; moreover $o^- B = \Delta$, so $|B| = 1/o^-$ and $\operatorname{sgn} B = \operatorname{sgn} \Delta$.*
4. *r^+ lies strictly on the same side of θ as r_{2d+1}, i.e. strictly on the opposite side from r_{2d}.*
5. *$\varepsilon^- = 0$, so $\varepsilon^- \varepsilon^+ = 0$; $\varepsilon^+ \neq 0$ has the sign of ε_{2d+1}, and $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$.*
6. *Both children are good estimates, $e^+ o^- - e^- o^+ = \Delta$, and Step 5 assigns $r_{2d+3} = r^- = \theta$ and $r_{2d+2} = r^+$.*
7. *$(r_{2d+2}, r_{2d+3}) = (r^+, \theta)$ is a bounded pair (lower member r^+, upper member θ) if and only if $B < 0$. If $B > 0$ both members are $\geq \theta$ and the pair is not a bounded pair in either orientation.*
8. *Depth $d+1$ is terminal: $\Theta(r_{2d+3}) = 0$, while $\Delta(r_{2d+2}, r_{2d+3}) = e_{2d+3} o_{2d+2} - e_{2d+2} o_{2d+3} = -\Delta = \mp 1$.*

Proof. Since (r_{2d}, r_{2d+1}) is a bounded pair with $\Theta \neq 0$, and $r_{2d} \neq \theta$ (because $|\varepsilon_{2d}| > |\varepsilon_{2d+1}| \geq 0$), A and B are nonzero with opposite signs; hence $w^- = -A/B > 0$, i.e. $w^- \geq 1$. (1) $\Delta = B o_{2d} - A o_{2d+1}$, so $\Delta/\Theta - o_{2d}/o_{2d+1} = -A/B = w^-$. (2) With $A = -w^- B$, $f(w-1)f(w) = B^2(w-1-w^-)(w-w^-) \geq 0$, a product of two consecutive integers times B^2. (3) $f(w^-) = 0$ gives $e^- = \theta\, o^-$, and $f(w^+) = f(w^-) + B = B$. Lemma 9.1(3) gives $e^+ o^- - e^- o^+ = \Delta$; substituting $e^- = \theta o^-$ yields $o^-(e^+ - \theta o^+) = \Delta$. (4) $f(w^+) = B$ has the sign of B, which is opposite to that of A. (5) $\varepsilon^- = f(w^-)/(\theta o^-) = 0$, $\varepsilon^+ = B/(\theta o^+)$ and $\varepsilon_{2d+1} = B/(\theta\, o_{2d+1})$ with $o^+ > o_{2d+1}$. (6) $r^- = \theta$ is trivially a good estimate, and $|e^+ - \theta o^+| = |B| < 1$ because r_{2d+1} is a good estimate; the Farey identity is Lemma 9.1(3), and the assignment follows from $|\varepsilon^-| = 0 < |\varepsilon^+|$. (7) If $B < 0$ then $r^+ < \theta = r^-$, $e^+ = \lfloor o^+ \theta \rfloor$, $e^- = \lceil o^- \theta \rceil$, and by (3) $\Delta = o^- B < 0$, so $\Delta = -1$ and $e^- o^+ - e^+ o^- = -\Delta = 1$. If $B > 0$ then $r^+ > \theta = r^-$. (8) $\Theta(r_{2d+3}) = o^-(e^- - \theta o^-) = 0$, and the determinant is that of (6) with the order reversed. □

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For example, $\theta = 5/3$ with $(r_0, r_1) = (1/1, 2/1)$ has $B = 1/3 > 0$, $w^- = 2$ and terminal pair (7/4, 5/3), which is not a bounded pair; $\theta = 4/3$ with $(r_0, r_1) = (2/1, 1/1)$ has $B = -1/3 < 0$, $w^- = 2$ and terminal pair (5/4, 4/3), which is a bounded pair.

The continuous optimum and the discrete formula coincide

To see why w^+ is the natural choice, consider treating w as a real variable and finding the stationary point of the error product $f(w-1)\,f(w)$, where $f(w) = A + wB$. Differentiating with respect to w and setting the derivative to zero:

$$\frac{d}{dw}\big[f(w-1)\,f(w)\big] = B\big[(A+wB) + (A+(w-1)B)\big] = 0.$$

Solving gives the unique real critical point

$$w^* = \tfrac{1}{2} - \frac{A}{B} = c + \tfrac{1}{2},$$

where $c = -A/B > 0$. The product $f(w-1)\,f(w)$ has roots at $w = c$ (where $f(w) = 0$) and $w = c+1$ (where $f(w-1) = 0$), and $w^* = c + \frac{1}{2}$ is their midpoint — the unique real maximum of $|f(w-1)\,f(w)|$ between the roots. The unique integer w for which $f(w-1)$ and $f(w)$ have opposite signs satisfies $c < w < c+1$, i.e. $w = \lfloor c \rfloor + 1 = w^+$. Since $w^* = c + \frac{1}{2}$ also lies in $(c,\, c+1)$, and w^+ is the unique integer in that interval, w^+ is the nearest integer to w^*: both the continuous stationary point and the sign-change bracket single out the same w^+.

Remark 8.5. This analysis applies at rational approximation steps, where $c = -A/B \notin \mathbb{Z}$ so the interval $(c,\, c+1)$ contains exactly one integer. At the exact step $c \in \mathbb{Z}$ and the open interval $(c,\, c+1)$ contains no integer; instead $w^- = c$ and $w^+ = c+1$ are the two bracket endpoints, and both children are good estimates (Lemma 8.4(1,2)).

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Algorithm. The iteration proceeds as follows.

1. Set $d = 0$ and $(r_0, r_1) = \mathrm{fit}(\theta,\, \lfloor\theta\rfloor/1,\, \lceil\theta\rceil/1)$.
2. Compute Δ and Θ for the current pair (r_{2d}, r_{2d+1}).
3. If $\Theta = 0$, terminate.
4. Compute $w^+ = \lfloor \Delta/\Theta - o_{2d}/o_{2d+1} \rfloor + 1$ and set $w^- = w^+ - 1$.
5. Set
$$r^- = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^-), \qquad r^+ = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^+).$$
Increment d and set $(r_{2d}, r_{2d+1}) = \mathrm{fit}(\theta,\, r^-,\, r^+)$. Repeat from step 2.

When $w^+ \geq 2$, the denominator of each new mediant strictly exceeds both parent denominators. When $w^+ = 1$ only one new fraction r^+ is produced (the older sibling r^- coincides with r_{2d}, so the generation is gender-preserving), but r^+ has strictly larger denominator. In either case the denominators of the better estimates grow without bound and the iteration converges to θ (Corollary 11.2).

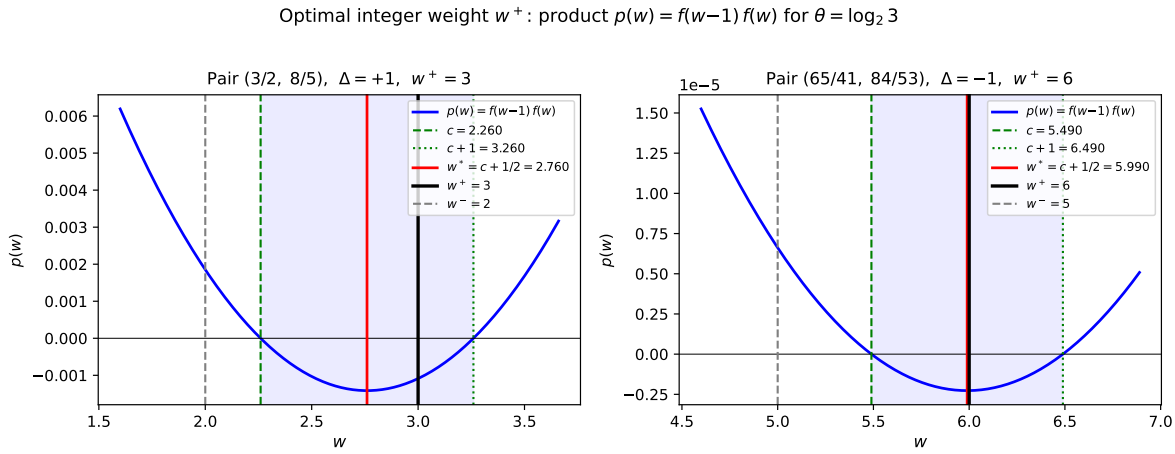


Figure 1: The product $p(w) = f(w-1)\,f(w)$ for two pairs from the mediant tree for $\theta = \log_2 3$, illustrating the optimal integer weight w^+. *Left:* pair $(3/2,\ 8/5)$ with $\Delta = +1$, $c \approx 2.260$, and $w^+ = 3$. *Right:* pair $(65/41,\ 84/53)$ with $\Delta = -1$, $c \approx 5.490$, and $w^+ = 6$. In each case the roots of p are at c (where $f(w) = 0$, green dashed) and $c+1$ (where $f(w-1) = 0$, green dotted); the continuous stationary point $w^* = c + \frac{1}{2}$ (red) is their midpoint; the unique integer in the shaded interval $(c,\ c+1)$ is w^+ (black); and its companion $w^- = w^+ - 1$ (grey dashed) lies just outside the interval on the left. Together w^- and w^+ are the two weights whose mediants bracket θ. The two panels illustrate both signs of Δ.

9 The Mediant Construction

The classical *mediant* of two fractions e_{2d}/o_{2d} and e_{2d+1}/o_{2d+1} is $(e_{2d} + e_{2d+1})/(o_{2d} + o_{2d+1})$. We generalise this to a weighted family parametrised by a non-negative integer w.

Lemma 9.1 (Farey-adjacency of mediants)**.** *Let (r, r') be a Farey-adjacent pair and $w \geq 0$. Write $M_w = \mathrm{Mediant}(r, r', w)$. Then:*

1. *$\Delta(M_w, r') = \Delta(r, r')$, so M_w is Farey-adjacent to r' for all $w \geq 0$.*
2. *$\Delta(r, M_w) = w \cdot \Delta(r, r')$, so M_w is Farey-adjacent to r if and only if $w = 1$.*
3. *$\Delta(M_{w-1}, M_w) = \Delta(r, r')$ for all $w \geq 1$, so consecutive mediants M_{w-1} and M_w are Farey-adjacent.*
4. *M_w is in lowest terms, i.e. $\gcd(e + we', o + wo') = 1$.*

Proof. Write $M_w = (e + we', o + wo')$. For (1): $\Delta(M_w, r') = e'(o + wo') - (e + we')o' = e'o - eo' = \Delta(r, r')$, so $|\Delta(M_w, r')| = 1$. For (2): $\Delta(r, M_w) = (e + we')o - e(o + wo') = w(e'o - eo') = w \cdot \Delta(r, r')$. For (3): $\Delta(M_{w-1}, M_w) = (e + we')(o + (w-1)o') - (e + (w-1)e')(o + wo') = e'o - eo' = \Delta(r, r')$, so $|\Delta(M_{w-1}, M_w)| = 1$. For (4): any common divisor g of $e + we'$ and $o + wo'$ also divides $e'(o + wo') - o'(e + we') = e'o - eo' = \Delta(r, r')$, which has absolute value 1, so $g = 1$. □

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Lemma 9.2 (Farey adjacency under $x \mapsto 1/(1+x)$)**.** *If $r = (e, o)$ and $r' = (e', o')$ are Farey-adjacent approximants, then $\hat{\alpha} = (o, e+o)$ and $\hat{\alpha}' = (o', e'+o')$ (i.e. $1/(1+r) = o/(e+o)$ and $1/(1+r') = o'/(e'+o')$) are also Farey-adjacent.*

Proof. $\Delta(\hat{\alpha}, \hat{\alpha}') = o'(e+o) - o(e'+o') = o'e - oe' = -\Delta(r, r')$, so $|\Delta(\hat{\alpha}, \hat{\alpha}')| = |\Delta(r, r')| = 1$. □

Theorem 9.3 (Mediant convergence)**.** *Let (r_{2d}, r_{2d+1}) be a bounded pair for θ with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, and let $w \geq 1$ be a positive integer such that*

$$\big(\text{Mediant}(r_{2d}, r_{2d+1}, w-1),\ \text{Mediant}(r_{2d}, r_{2d+1}, w)\big)$$

strictly brackets θ (i.e. $\varepsilon^- \varepsilon^+ < 0$, where $\varepsilon^\pm = e^\pm - \theta\, o^\pm$ are the signed errors of the two candidates). Write $r^- = \text{Mediant}(r_{2d}, r_{2d+1}, w-1)$ and $r^+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$ for these two candidates. Then

$$|\varepsilon^-| \leq |\varepsilon_{2d}| \qquad \textit{and} \qquad |\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|.$$

(When $w = 1$, $r^- = r_{2d}$ exactly so $|\varepsilon^-| = |\varepsilon_{2d}|$; when $w \geq 2$, $|\varepsilon^-| < |\varepsilon_{2d}|$ strictly.) Moreover $|e^+ o^- - e^- o^+| = 1$ (Lemma 9.1(3), since $r^- = M_{w-1}$ and $r^+ = M_w$), so (r^-, r^+) is Farey-adjacent. The two mediants are assigned by $(r_{2d+2}, r_{2d+3}) = \text{fit}(\theta, r^-, r^+)$ (Definition 6.8): r_{2d+3} is the better estimate (smaller $|\varepsilon|$) and r_{2d+2} is the worse. Since r^+ always satisfies $|\varepsilon^+| < |\varepsilon_{2d+1}|$, we have $|\varepsilon_{2d+3}| < |\varepsilon_{2d+1}|$ regardless of which candidate is assigned to r_{2d+3}.

Proof. The denominators satisfy $o^- = o_{2d} + (w-1)o_{2d+1} < o_{2d} + w\, o_{2d+1} = o^+$, so r^- and r^+ are distinct. The ratio λ^- is a weighted combination of λ_{2d} and λ_{2d+1}:

$$\lambda^- = \frac{o_{2d}\,\lambda_{2d} + (w-1)o_{2d+1}\,\lambda_{2d+1}}{o_{2d} + (w-1)o_{2d+1}}.$$

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When $w = 1$ this gives $\lambda^- = \lambda_{2d}$ exactly, so $|\varepsilon^-| = |\varepsilon_{2d}|$. When $w \geq 2$, r^- lies strictly between r_{2d} and r_{2d+1}. Write $f(w) = A + wB$ where $A = e_{2d} - \theta o_{2d}$ and $B = e_{2d+1} - \theta o_{2d+1}$ are the scaled errors of r_{2d} and r_{2d+1}; then $f(w) = e^+ - \theta\, o^+$. From the bracketing hypothesis $f(w-1)f(w) < 0$ it follows that $B \neq 0$ (otherwise f would be constant and could not change sign), and that the zero $c = -A/B$ of f lies in $(w-1, w)$. Thus $f(w-1)$ has the sign of A (the side of r_{2d}) and $f(w)$ has the sign of B (the side of r_{2d+1}). Since the bracketing condition places r^- on the same side of θ as r_{2d}, the ratio r^-/θ lies between r_{2d}/θ and 1, so $|\lambda^- - 1| < |\lambda_{2d} - 1|$, i.e. $|\varepsilon^-| < |\varepsilon_{2d}|$ (this strict inequality holds for $w \geq 2$; for $w = 1$, $r^- = r_{2d}$ and equality holds, as noted in the statement). For r^+ (weight $w \geq 1$), the bracketing condition places r^+ on the same side of θ as r_{2d+1}, so r^+/θ lies between r_{2d+1}/θ and 1, giving $|\lambda^+ - 1| < |\lambda_{2d+1} - 1|$, i.e. $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$. The Farey adjacency condition for (r^-, r^+) follows by direct expansion:

$$\begin{aligned} e^+ o^- - e^- o^+ &= (e_{2d} + w\, e_{2d+1})(o_{2d} + (w-1)o_{2d+1}) - (e_{2d} + (w-1)e_{2d+1})(o_{2d} + w\, o_{2d+1}) \\ &= \big[(w-1) - w\big] e_{2d}\, o_{2d+1} + \big[w - (w-1)\big] e_{2d+1}\, o_{2d} \\ &= -e_{2d}\, o_{2d+1} + e_{2d+1}\, o_{2d} = \Delta, \end{aligned}$$

so $|e^+ o^- - e^- o^+| = |\Delta| = 1$, and the Farey adjacency of (r^-, r^+) is inherited directly from that of (r_{2d}, r_{2d+1}). □

Corollary 9.4 (Straddling inequality)**.** *Under the conditions of Theorem 9.3,*

$$\varepsilon^- \varepsilon^+ < 0 \qquad \textit{and} \qquad |\varepsilon^-||\varepsilon^+| < |\varepsilon_{2d}|\,|\varepsilon_{2d+1}|.$$

Proof. The first inequality holds because r^- and r^+ straddle θ, so their errors have opposite signs. For the second: $|\varepsilon^-| \le |\varepsilon_{2d}|$ and $|\varepsilon^+| < |\varepsilon_{2d+1}|$ (Theorem 9.3), so $|\varepsilon^-||\varepsilon^+| \le |\varepsilon_{2d}||\varepsilon^+| < |\varepsilon_{2d}|\,|\varepsilon_{2d+1}|$. □

Remark 9.5 (The exact step). At the exact step (rational θ, $-A/B = w^- \in \mathbb{Z}$) the strict bracketing hypothesis of Theorem 9.3 fails, because $f(w^-) = 0$. All conclusions of the theorem nevertheless hold: $|\varepsilon^-| = 0 < |\varepsilon_{2d}|$, $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$, (r^-, r^+) is Farey-adjacent, and $|\varepsilon_{2d+3}| = 0 < |\varepsilon_{2d+1}|$ (Lemma 8.4). In Corollary 9.4 the second inequality holds (its left side is 0), but the first fails: $\varepsilon^-\varepsilon^+ = 0$.

10 Mediant Words and the Mediant Tree

The mediant construction admits a compact combinatorial encoding. Every estimate produced by the algorithm is either a *leaf* or a *branch*, and the recursive structure of all estimates defines a binary tree whose nodes carry both rational values and combinatorial words over $\{\mathsf{O}, \mathsf{E}\}$.

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10.1 Leaf words

Definition 10.1 (Leaf mediant word)**.** Let $\{\mathsf{O}, \mathsf{E}\}^*$ denote the set of finite strings over the two-letter alphabet $\{\mathsf{O}, \mathsf{E}\}$. A *leaf* estimate has denominator $o = 1$, i.e. it is a base estimate $e/1$ for some positive integer e. Its *mediant word* is

$$\pi(e/1) = \underbrace{\mathsf{E}\cdots\mathsf{E}}_{e}\mathsf{O},$$

a string of e copies of E followed by one O.

The two leaf estimates that initialise the algorithm are $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, ordered as (r_0, r_1) so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). Their mediant words are $\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil\theta\rceil}\mathsf{O}$ respectively. We assume $\theta \notin \mathbb{Z}$ (for $\theta \in \mathbb{Z}$ the approximant $\theta/1$ is exact and the algorithm is not needed). Then $\lceil\theta\rceil = \lfloor\theta\rfloor + 1$, so $\pi(r_0)$ and $\pi(r_1)$ differ by exactly one E. If $\{\theta\} = 1/2$ the two leaves have equal $|\varepsilon|$; we adopt the tie-break $r_1 = \lfloor\theta\rfloor/1$ (so $r_0 = \lceil\theta\rceil/1$), so depth 0 is an exact step with $w^- = 1$. Each word $\mathsf{E}^k\mathsf{O}$ contains exactly one O, so it is not a power of a shorter word; both $\pi(r_0)$ and $\pi(r_1)$ are therefore primitive. Moreover, $\mathsf{E}^k\mathsf{O}$ is the lower Christoffel word $\mathrm{Ch}^-(k, 1)$ (Definition 5.1): its O appears at position k, matching the Christoffel formula. Thus both leaf words are lower Christoffel words of their respective fractions.

10.2 Branch words and the weighted operators

Each branch step produces two new mediant words from the parent pair $(u, v) = (\pi(r_{2d-2}), \pi(r_{2d-1}))$ by concatenation:

Definition 10.2 (Branch mediant word)**.** Let $d \geq 1$ be such that depth $d-1$ is an approximation depth, so that the algorithm produced the pair (r_{2d}, r_{2d+1}) from (r_{2d-2}, r_{2d-1}). This includes the terminal pair for rational θ; (r_{2d}, r_{2d+1}) is a bounded pair unless depth d is terminal and the exact step had $B > 0$. The pair consists of estimates r_{2d} and r_{2d+1}, with r_{2d+1} always the better estimate (smaller $|\varepsilon|$). Each estimate r_{2d} or r_{2d+1} has two *ordered parents* (r_{2d-2}, r_{2d-1}) and a non-negative integer *weight* w (equal to zero only when $w^+_{d-1} = 1$, in which case $r_{2d} = r_{2d-2}$). The two estimates r_{2d} and r_{2d+1} carry weights w^-_{d-1} and w^+_{d-1}. We distinguish two uses of the word *weight*:

- The *inherited weights* w_{2d} and w_{2d+1} of the two estimates r_{2d} and r_{2d+1} are the weights with which they were formed from the pair (r_{2d-2}, r_{2d-1}) at depth $d-1$ — specifically $\{w_{2d}, w_{2d+1}\} = \{w^-_{d-1}, w^+_{d-1}\}$ with $w_{2d} = w^+_{d-1} - 1$ and $w_{2d+1} = w^+_{d-1}$ (or vice versa, depending on which child is the better estimate). Each inherited weight is used to form exactly one of the two estimates: $r_{2d} = \mathrm{Mediant}(r_{2d-2}, r_{2d-1}, w_{2d})$ and $r_{2d+1} = \mathrm{Mediant}(r_{2d-2}, r_{2d-1}, w_{2d+1})$.

- The *propagated weights* w^-_d and w^+_d of a bounded pair (r_{2d}, r_{2d+1}) are the optimal weights computed *from* that pair to produce the next pair (r_{2d+2}, r_{2d+3}) — computed by Step 4 (Theorem 8.1 at rational approximation depths; Lemma 8.4(1) at the exact depth).

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These differ in general: the inherited weights w_{2d} and w_{2d+1} record how r_{2d} and r_{2d+1} were born; the propagated weights w^-_d and w^+_d determine where the algorithm goes next. Write $u = \pi(r_{2d-2})$ and $v = \pi(r_{2d-1})$ for the *parent* mediant words (at depth $d-1$; these become the parents of the depth-d pair).

The two branch words at each depth share the same parents and differ only in the multiplicity of v: one estimate uses weight w^+_{d-1} and the other uses weight w^-_{d-1}. Which is the better estimate depends on the errors: writing ε^- and ε^+ for the errors of the two candidates (Definition 6.5),

$$w_{2d+1} - w_{2d} = \begin{cases} +1 & \text{if } |\varepsilon^+| < |\varepsilon^-|, \\ -1 & \text{otherwise.} \end{cases}$$

In either case $|\pi(r_{2d+1})| - |\pi(r_{2d})| = \pm|\pi(r_{2d-1})|$.

The branch mediant words are (using $w = w^+_{d-1}$, the propagated optimal weight of the parent pair (r_{2d-2}, r_{2d-1}), and Δ the determinant of that parent pair):

- $\Delta > 0$ (r_{2d-1} is the upper bound, v plays the role of E):

$$\pi(r^-) = v^{w^+_{d-1}-1} \cdot u,$$
$$\pi(r^+) = v^{w^+_{d-1}} \cdot u.$$

- $\Delta < 0$ (r_{2d-1} is the lower bound, v plays the role of O):

$$\pi(r^-) = u \cdot v^{w^+_{d-1}-1},$$
$$\pi(r^+) = u \cdot v^{w^+_{d-1}}.$$

Which of r^- and r^+ is the better estimate r_{2d+1} (and which is r_{2d}) is determined by comparing $|\varepsilon^-|$ and $|\varepsilon^+|$; the assignment is not fixed in advance.

Concrete example (for $\theta = \log_2 3$: $w = 2$, depth-0 leaves $\pi(r_0) = \mathtt{EO}$, $\pi(r_1) = \mathtt{EEO}$, $\Delta > 0$). Write $\Phi^w(v, u) = (v^w \cdot u,\ v^{w-1} \cdot u)$ for the pair of branch words at weight w (the $\Delta > 0$ case):

$$\Phi^2\big(\mathtt{EEO},\, \mathtt{EO}\big) = \big(\mathtt{EEO}^2\,\mathtt{EO},\ \mathtt{EEO}\,\mathtt{EO}\big) = \big(\mathtt{EEOEEOEO},\ \mathtt{EEOEO}\big),$$

so $\pi(r_3) = \mathtt{EEOEEOEO}$ and $\pi(r_2) = \mathtt{EEOEO}$ (the second component of $\Phi^2(\mathtt{EEO}, \mathtt{EO})$, i.e. the weight-$w^- = 1$ child).

10.3 Mediant words are Christoffel words

A word is *primitive* if it is not a power x^k of a strictly shorter word x (with $k \geq 2$).

Lemma 10.3 (Every mediant word is a lower Christoffel word)**.** *For every estimate $r = (e, o)$ produced by the mediant tree, $\pi(r) = \mathrm{Ch}^-(e, o)$, the lower Christoffel word at slope $o/(e+o)$. In particular, $\pi(r)$ is never an upper Christoffel word, since by Theorem 12.1 every mediant word ends in O, whereas upper Christoffel words end in E.*

Proof. By structural induction on the mediant tree.

Base case. By §10.1, $\pi(e/1) = \mathsf{E}^e\mathsf{O} = \mathrm{Ch}^-(e, 1)$.

Inductive step. Let (r_{2d}, r_{2d+1}) be a bounded pair with $\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$. By induction $\pi(r_{2d}) = \mathrm{Ch}^-(e_{2d}, o_{2d})$ and $\pi(r_{2d+1}) = \mathrm{Ch}^-(e_{2d+1}, o_{2d+1})$, both lower Christoffel words (ending in O), with slopes o_{2d}/L_{2d} and o_{2d+1}/L_{2d+1} (where $L_k = e_k + o_k$) on opposite sides of $\alpha = 1/(1+\theta)$.

When $\Delta > 0$: r_{2d+1} is the upper estimate so $e_{2d+1}/o_{2d+1} > \theta$, giving $o_{2d+1}/L_{2d+1} < \alpha < o_{2d}/L_{2d}$. The new word is $v^w u = \pi(r_{2d+1})^w \cdot \pi(r_{2d})$: the smaller-slope word $\pi(r_{2d+1})$ repeated w times, followed by $\pi(r_{2d})$ once. Since (r_{2d}, r_{2d+1}) is Farey-adjacent, Lemma 9.2 gives $o_{2d}L_{2d+1} - o_{2d+1}L_{2d} = 1$, the Farey hypothesis of Theorem 5.2 (with $e_a = e_{2d+1}$, $e_b = e_{2d}$), so this equals $\mathrm{Ch}^-(w\,e_{2d+1} + e_{2d},\ w\,o_{2d+1} + o_{2d})$.

When $\Delta < 0$: r_{2d+1} is the lower estimate so $e_{2d+1}/o_{2d+1} < \theta$, giving $o_{2d}/L_{2d} < \alpha < o_{2d+1}/L_{2d+1}$. Thus $\pi(r_{2d})$ has the smaller slope and $\pi(r_{2d+1})$ has the larger slope. The new word is $uv^w = \pi(r_{2d}) \cdot \pi(r_{2d+1})^w$: the single smaller-slope word followed by the larger-slope word repeated w times. Since (r_{2d}, r_{2d+1}) is Farey-adjacent, Lemma 9.2 gives $o_{2d+1}L_{2d} - o_{2d}L_{2d+1} = 1$, the Farey hypothesis of Theorem 5.2 (with $e_a = e_{2d}$, $e_b = e_{2d+1}$), so this equals $\mathrm{Ch}^-(e_{2d} + w\,e_{2d+1},\ o_{2d} + w\,o_{2d+1})$.

In both cases $\gcd(e_{2d} + we_{2d+1},\ o_{2d} + wo_{2d+1}) = 1$ (Lemma 9.1(4), since (r_{2d}, r_{2d+1}) is Farey-adjacent), confirming the result is a valid lower Christoffel word. □

The Christoffel identification is the key fact connecting the mediant algorithm to the combinatorics of Sturmian words. The recurrence $\pi(r_{2d+3}) = v^{w_d^\pm} \cdot u$ (or $u \cdot v^{w_d^\pm}$), where $u = \pi(r_{2d})$ and $v = \pi(r_{2d+1})$ are the parent words at depth d and $w_d^\pm$ is the propagated weight of the pair (r_{2d}, r_{2d+1}), matches the standard sequence generation rule for Sturmian words (Widmer [4], Proposition 2.4.6).

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Remark 10.4. The Christoffel tree — the binary tree whose nodes are lower Christoffel words under the same concatenation rules — is a classical object (Berstel et al. [5], Chapter 2). Lemma 10.3 says that the mediant algorithm traverses exactly this tree: every estimate produced is a node of the Christoffel tree, and the branch rule is the Christoffel concatenation. The novelty is not the tree itself but the *driving criterion*: the optimal weight w^+ (Theorem 8.1) selects the unique child that straddles θ, making the traversal goal-directed rather than exhaustive.

10.4 The mediant tree

The mediant tree is the practical device for generating mediant words of arbitrary length and precision. Given a target θ and a desired precision, one runs the algorithm until the bounded pair (r_{2d}, r_{2d+1}) achieves the required length $e_{2d+1} + o_{2d+1}$. The mediant word $\pi(r_{2d+1})$ is then read off the tree by the recursive rule: an interior node $\mathrm{Mediant}(s, t, w)$ with $\Delta > 0$ yields $\pi(t)^w \cdot \pi(s)$, and with $\Delta < 0$ yields $\pi(s) \cdot \pi(t)^w$. Deeper trees produce longer prefixes of the Sturmian word with slope $1/(\theta + 1)$ (Section 12).

Definition 10.5 (Mediant tree)**.** The *mediant tree* of an estimate r is defined recursively:

- If $r \in \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$, then r is a *leaf node*.
- Otherwise r is an *interior node* $\mathrm{Mediant}(s, t, w)$ with left subtree s, right subtree t, and integer weight $w \geq 0$, representing the rational

$$r = \frac{e_s + w\, e_t}{o_s + w\, o_t}.$$

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The two leaves are the unique consecutive integers bracketing θ, forming the initial bounded pair (r_0, r_1), ordered so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). Each bounded pair (r_{2d}, r_{2d+1}) in the algorithm's sequence consists of two mediant trees sharing the same leaves; as the iteration advances the trees grow deeper, recording the full construction history of each estimate.

A detailed worked example for $\theta = \log_2 3$, tracing the first four steps of the algorithm and the corresponding mediant word decompositions, is given in Appendix A.

10.5 The mediant word encodes the fraction

By Lemma 10.3, the mediant word $\pi(r)$ of an estimate $r = e/o$ is uniquely determined by its fraction: $\pi(r) = \mathrm{Ch}^-(e, o)$. The fraction e/o records the full construction history of r via the Euclidean structure of the mediant algorithm. Conversely, the fraction e/o can be read off from $\pi(r)$ by counting E's and O's. As the iteration advances, the ratio of E's to O's in $\pi(r_{2d+1})$ converges to θ, with strict improvement guaranteed across iterations (Theorem 11.1) but not within a single depth (the worse estimate r_{2d} need not improve).

11 Convergence Analysis

Theorem 11.1 (Strict decrease of $|\varepsilon|$)**.** *At each iteration of the algorithm, $|\varepsilon_{2d+1}|$ strictly decreases.*

Proof. By Theorem 9.3, $|\varepsilon^+| < |\varepsilon_{2d+1}|$. The next iterate assigns the better of r^- and r^+ as r_{2d+3}, so $|\varepsilon_{2d+3}| \le |\varepsilon^+| < |\varepsilon_{2d+1}|$. (At the exact step for rational θ, $r^- = \theta$ exactly so $|\varepsilon^-| = 0 < |\varepsilon_{2d+1}|$ (Lemma 8.4(3)); the better child is r^- and the conclusion holds trivially.) □

Corollary 11.2 (Convergence to θ)**.** *The sequence of better estimates produced by the algorithm satisfies $\varepsilon_{2d+1} \to 0$.*

Proof. From the proof of Corollary 8.2 (at a rational approximation step; at the exact step the same identity holds with $f(w^-) = 0$, since $|f(w^-)o^+ - f(w^+)o^-| = |\Delta| = 1$ for any two consecutive weights and the product $f(w^-)f(w^+) \le 0$), with $f(w^\dagger) = A + w^\dagger B$ the linear error function and o^+, o^- the two child denominators:

$$|f(w^-)|\, o^+ + |f(w^+)|\, o^- = |\Delta| = 1.$$

Since $f(w^\dagger) = e^\dagger - \theta\, o^\dagger$ and $\varepsilon^\dagger = f(w^\dagger)/(\theta\, o^\dagger)$ for $r^\dagger = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^\dagger)$, substituting $|f(w^\dagger)| = |\varepsilon^\dagger|\, \theta\, o^\dagger$ gives

$$\big(|\varepsilon^-| + |\varepsilon^+|\big)\, \theta\, o^-\, o^+ = 1,$$

so in particular the better child satisfies

$$|\varepsilon_{\mathrm{better}}| \le \frac{1}{2\,\theta\, o^-\, o^+}.$$

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(When $w^+ = 1$, so $w^- = 0$: $r^- = r_{2d}$ coincides with the parent and is not a new fraction; the better child is r^+ in this case since $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$. If the better child is r^-, then $w^- \ge 1$ and $o^- = o_{2d} + (w^+ - 1)\, o_{2d+1} > o_{2d+1}$. In either case the bound $|\varepsilon_{\mathrm{better}}| \le 1/(2\theta\, o^-\, o^+)$ holds.) Since $o^+ = o_{2d} + w^+\, o_{2d+1} \ge o_{2d} + o_{2d+1} > o_{2d+1}$, the denominator of the better child strictly exceeds o_{2d+1}; by induction the denominators diverge. Since $|\varepsilon_{\mathrm{better}}| \le 1/(2\,\theta\, o^-\, o^+)$ and $o^+ \to \infty$, we have $\varepsilon_{2d+1} \to 0$. For rational θ the sequence is finite and its last term has $\varepsilon = 0$. □

Theorem 11.3 (MT fractions lie on the Stern–Brocot path)**.** *Every fraction appearing as r_{2d} or r_{2d+1} at an approximation depth, together with θ itself when θ is rational, is a convergent or semi-convergent of θ, i.e. lies on the Stern–Brocot path of θ. (For rational θ the other member $r_{2d+2} = r^+$ of the terminal pair is a child of θ in the Stern–Brocot tree and is not on the path.)*

Proof. The initial bounds $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$ are consecutive integers, hence Farey adjacent, and both lie on the Stern–Brocot path of θ.

At each depth d, the algorithm forms the two mediants $r^{\pm} = \text{Mediant}(r_{2d}, r_{2d+1}, w^{\pm})$ with $w^- = w^+ - 1$. For $w \geq 2$ these are *weighted* mediants, not ordinary mediants of (r_{2d}, r_{2d+1}), so the one-step argument does not apply directly. Instead, use the identity

$$P + wQ = (P + (w-1)Q) \oplus Q, \qquad \det(P + (w-1)Q,\, Q) = \det(P, Q),$$

which shows that $\text{Mediant}(r_{2d}, r_{2d+1}, w)$ is the ordinary mediant of $(r_{2d} + (w-1)\, r_{2d+1},\ r_{2d+1})$, a Farey-adjacent pair with the same determinant as (r_{2d}, r_{2d+1}) (Lemma 9.1(1)). By induction on $w = 1, \ldots, w^+$: the pair $(r_{2d} + (w-1)\, r_{2d+1},\ r_{2d+1})$ is Farey adjacent and straddles θ for each $w \leq w^+$ at a rational approximation step, and for each $w \leq w^-$ at the exact step (where the node at $w = w^-$ is $\theta = r^-$, the end of the path). By the standard Stern–Brocot lemma, any Farey-adjacent pair (P, Q) of non-negative fractions defines a unique SB-tree interval, and θ lies in the interval of (P, Q) if and only if $P \oplus Q$ lies on the SB path to θ; so its ordinary mediant $r_{2d} + w\, r_{2d+1}$ is a node on the Stern–Brocot path of θ. The case $w^- = 0$ (i.e. $w^+ = 1$) is immediate: $r^- = r_{2d}$ is already on the path by inductive hypothesis, and $r^+ = r_{2d} + r_{2d+1}$ is the unique ordinary mediant of the Farey pair (r_{2d}, r_{2d+1}). Both r^- and r^+ are produced this way at a rational approximation step, so both lie on the path. (At the exact step $r^- = \theta = r_{2d+3}$ is the endpoint of the SB path and lies on the path by definition; $r^+ = r_{2d+2}$ lies on the same side of θ as r_{2d+1} (Lemma 8.4(4)) and is a child of θ in the SB tree, hence not on the path — which is why the theorem statement excludes it explicitly.) □

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Theorem 11.4 (MT never skips a convergent)**.** *Assume θ is irrational. Every CF convergent p_k/q_k of θ appears as r_{2d} or r_{2d+1} in some MT pair (a bounded pair at an approximation depth, or the terminal pair).*

Proof. A convergent p_k/q_k is characterised as the *last* node in a run of same-side steps on the SB path of θ, i.e. the node just before the path crosses to the other side. In the induction of Theorem 11.3, the path nodes visited within stage d are $r_{2d} + w\, r_{2d+1}$ for $w = 1, \ldots, w^+$. For $w < w^+$ the node lies on the same side of θ as r_{2d}. When $w^+ \geq 2$, $r^- = r_{2d} + (w^+ - 1)\, r_{2d+1}$ has the same gender as r_{2d} and is the last node on r_{2d}'s side, so r^- is a convergent. (When $w^+ = 1$ the generation is gender-preserving and r_{2d} itself is carried forward as the older sibling; it is a convergent from the previous stage.)

We show that every convergent appears in the MT sequence. After the integer spine and the two leaves, the Stern–Brocot path of θ is the concatenation of the stage blocks $\{r_{2d} + w\, r_{2d+1} : 1 \leq w \leq w_d^+\}$, $d = 0, 1, 2, \ldots$. (After the last node r^+ of block $d-1$, the SB interval is (r_{2d}, r_{2d+1}), so the next path node is $r_{2d} \oplus r_{2d+1}$, the first node of block d.) Within a block, the nodes with $w < w_d^+$ lie on the same side of θ as r_{2d}, and r^+ lies on the other side. The nodes that end a run—and hence are convergents—are: $r^- = r_{2d} + (w_d^+ - 1)\, r_{2d+1}$ (when $w_d^+ \geq 2$); r^+; and, when $w_d^+ = 1$, r_{2d} itself (carried forward from the previous block). All of these are members of an MT pair. Together with the leaf $\lfloor\theta\rfloor/1 = p_0/q_0$, every convergent appears. □

The next two results give precise characterisations of which MT fractions are semi-convergents rather than convergents.

Remark 11.5 (Integer leaves). $\lfloor\theta\rfloor/1 = p_0/q_0$ is always a convergent. If $\lceil\theta\rceil/1$ is the better leaf (i.e. $|\varepsilon_{\lceil\theta\rceil}| < |\varepsilon_{\lfloor\theta\rfloor}|$, which holds iff $\{\theta\} > 1/2$), then $\{\theta\} > 1/2$ forces $a_1 = 1$, so $\lceil\theta\rceil/1 = p_1/q_1$ is the first convergent. If $\lfloor\theta\rfloor/1$ is the better leaf (i.e. $\{\theta\} < 1/2$), then $\lceil\theta\rceil/1$ is a semi-convergent iff $a_1 > 1$, and a convergent iff $a_1 = 1$. In either ordering, the better leaf r_1 is always a convergent. Leaves carry no weight, so the weight condition of Theorem 11.6 does not apply at $d = 0$; instead the $d = 0$ case is handled directly in the proof.

Theorem 11.6 (Semi-convergent characterisation of r_{2d+1})**.** *For all $d \geq 0$ such that depth d is not exact, r_{2d+1} is a semi-convergent of θ if and only if $w_d^+ = 1$, i.e. the propagated weight of the pair (r_{2d}, r_{2d+1}) equals 1.*

Proof. Write $A_d = e_{2d} - \theta o_{2d}$ and $B_d = e_{2d+1} - \theta o_{2d+1}$ for the signed errors of the current pair, so $w_d^+ = \lfloor |A_d|/|B_d| \rfloor + 1$.

Case $d \geq 1$**.** Let r_{d-1}^-, r_{d-1}^+ be the children formed at depth $d-1$, with $o_{d-1}^- < o_{d-1}^+$. The new pair is $(r_{2d}, r_{2d+1}) = (r_{d-1}^-, r_{d-1}^+)$ or (r_{d-1}^+, r_{d-1}^-) according to which has smaller $|\varepsilon|$; by convention r_{2d+1} is the better estimate, so either $r_{2d+1} = r_{d-1}^+$ (a gender-preserving generation) or $r_{2d+1} = r_{d-1}^-$ (a gender-changing generation).

Case $r_{2d+1} = r_{d-1}^+$ (gender-preserving generation: r_{d-1}^+ is the younger sibling). After the path visits r_{d-1}^+, the SB interval narrows to $(r_{d-1}^-, r_{d-1}^+) = (r_{2d}, r_{2d+1})$, so the next path node is $r_{2d} \oplus r_{2d+1}$. By definition of $w_d^+ = \lfloor |A_d|/|B_d| \rfloor + 1$, $w_d^+ = 1$ iff $|A_d| < |B_d|$, i.e. $r_{2d} \oplus r_{2d+1}$ lies on the same side as r_{2d+1}, i.e. the SB path continues on r_{2d+1}'s side; when this holds, r_{d-1}^+ is not the last node of its run. Hence r_{d-1}^+ is the last node of its run (a convergent) iff $w_d^+ \geq 2$, and a semi-convergent iff $w_d^+ = 1$.

Case $r_{2d+1} = r_{d-1}^-$ (gender-changing generation: r_{d-1}^- is the younger sibling). Since $r_{2d+1} = r_{d-1}^-$ is the better estimate, we have $o_{2d+1} = o_{d-1}^- < o_{d-1}^+ = o_{2d}$ (as $w_{d-1}^- < w_{d-1}^+$ implies $o_{d-1}^- < o_{d-1}^+$). By Theorem 11.3, the SB path visits r_{d-1}^- at weight $w = w_{d-1}^+ - 1$ within stage $d-1$, immediately before crossing to r_{d-1}^+ at weight w_{d-1}^+. Hence r_{d-1}^- is the last same-side node before the crossing, i.e. a convergent. For the weight condition: since $r_{2d+1} = r_{d-1}^-$ is the better estimate, $|\varepsilon_{2d+1}| < |\varepsilon_{2d}|$, i.e. $|B_d|/o_{2d+1} < |A_d|/o_{2d}$ in terms of relative errors. In terms of scaled errors, $|A_d| = \theta\, o_{2d}\, |\varepsilon_{2d}|$ and $|B_d| = \theta\, o_{2d+1}\, |\varepsilon_{2d+1}|$, so

$$\frac{|A_d|}{|B_d|} = \frac{o_{2d}}{o_{2d+1}} \cdot \frac{|\varepsilon_{2d}|}{|\varepsilon_{2d+1}|} > 1,$$

since both factors exceed 1 ($o_{2d} > o_{2d+1}$ and $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$). Hence $|A_d| > |B_d|$ and $w_d^+ = \lfloor |A_d|/|B_d| \rfloor + 1 \geq 2$. The statement holds in this case too.

Case $d = 0$**.** The seed pair is $(r_0, r_1) = (\lfloor\theta\rfloor/1, \lceil\theta\rceil/1)$ or $(\lceil\theta\rceil/1, \lfloor\theta\rfloor/1)$, ordered so that r_1 is the better estimate (smaller $|\varepsilon|$). Write $\gamma = \{\theta\} \in (0,1)$ for the fractional part of θ. For the integer leaves $o_0 = o_1 = 1$, the signed errors satisfy $|A_0| = \theta\, |\varepsilon_{r_0}|$ and $|B_0| = \theta\, |\varepsilon_{r_1}|$. The values $|\varepsilon_{r_0}|$ and $|\varepsilon_{r_1}|$ are $\{1-\gamma, \gamma-0\} = \{1-\gamma, \gamma\}$ in some order (for $\lfloor\theta\rfloor/1$ the relative error is γ/θ, for $\lceil\theta\rceil/1$ it is $(1-\gamma)/\theta$; scaled by θ these are γ and $1-\gamma$). With $|B_0| < |A_0|$ by definition of the better-estimate ordering, we have $|A_0|/|B_0| = \max(\gamma, 1-\gamma)/\min(\gamma, 1-\gamma) > 1$, so $w_0^+ = \lfloor |A_0|/|B_0| \rfloor + 1 \geq 2$. Since $w_0^+ \geq 2 \neq 1$, the biconditional at $d = 0$ requires r_1 not to be a semi-convergent. By Remark 11.5, r_1 is always a convergent at $d = 0$. Hence the statement holds.

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In all cases the characterisation is complete.

Exclusion of exact depths. At an exact depth, $\theta = r_{2d} \oplus r_{2d+1}$ (Lemma 8.4(3)), so r_{2d+1} is one of the two Stern–Brocot parents of θ. The theorem characterises r_{2d+1} as a convergent or semi-convergent by asking whether it is the last same-side node before the SB path reaches θ. But the SB path to a rational θ terminates at θ, and which of its two parents is the "last same-side node" (i.e. a convergent) depends on which of the two standard CF expansions of θ is used. Since the paper does not fix a CF convention for rational θ, the convergent/semi-convergent label of r_{2d+1} at an exact depth is undefined, and the exact depth is excluded from the statement. □

Remark 11.7. Note that w_d^+ (the propagated weight of (r_{2d}, r_{2d+1})) differs in general from w_{2d+1} (the inherited weight of r_{2d+1}). A statement in terms of the inherited weight w_{2d+1} is false: for $\theta = \log_2 3$, $r_5 = 8/5$ carries inherited weight 1 but is a convergent, and $r_3 = 5/3$ carries inherited weight 2 but is a semi-convergent.

Theorem 11.8 (Semi-convergent characterisation of r_{2d})**.** *Assume θ is irrational. For $d \geq 1$ such that depth d is an approximation depth, r_{2d} is a semi-convergent of θ if and only if $w_{2d} > w_{2d+1}$, i.e. the inherited weight of r_{2d} exceeds that of its sibling r_{2d+1}.*

Proof. Let r_{d-1}^-, r_{d-1}^+ be the children formed at depth $d-1$, with absolute scaled errors $x = |e_{d-1}^- - \theta o_{d-1}^-|$ and $y = |e_{d-1}^+ - \theta o_{d-1}^+|$ and $o_{d-1}^- < o_{d-1}^+$. By convention, r_{2d} is the worse child (larger $|\varepsilon|$) and r_{2d+1} is the better child.

($\Leftarrow$) Suppose $w_{2d} > w_{2d+1}$, so this is a gender-changing generation and $r_{2d} = r_{d-1}^+$ is the older sibling (worse estimate). Then $x/o_{d-1}^- < y/o_{d-1}^+$, and since $o_{d-1}^- < o_{d-1}^+$ we get $x < y$.

Since (r_{d-1}^-, r_{d-1}^+) is a bounded pair (Corollary 8.2), their signed errors $e_{d-1}^- - \theta o_{d-1}^-$ and $e_{d-1}^+ - \theta o_{d-1}^+$ have opposite signs. Write them as $-x$ and $+y$ (i.e. r_{d-1}^- underestimates and r_{d-1}^+ overestimates; the argument is symmetric if the signs are reversed). The signed error of the unweighted mediant $r_{d-1}^- \oplus r_{d-1}^+$ is then

$$(e_{d-1}^- + e_{d-1}^+) - \theta(o_{d-1}^- + o_{d-1}^+) = (-x) + y = y - x > 0,$$

which has the same sign as the error of r_{d-1}^+. Hence θ still lies on the r_{d-1}^- side of $r_{d-1}^- \oplus r_{d-1}^+$, so the SB path continues beyond r_{d-1}^+: r_{d-1}^+ is not the last node of its run. Hence $r_{2d} = r_{d-1}^+$ is a semi-convergent.

($\Rightarrow$) We proceed by induction on $d \geq 1$.

Base case $d = 1$**.** r_2 is a child of the depth-0 pair, formed at weight $w_0^+ - 1$ or w_0^+. By the $d = 0$ case of Theorem 11.6, $w_0^+ \geq 2$ always. If $w_2 \leq w_3$, then this is a gender-preserving generation and $r_2 = r_0^-$ is the older sibling at weight $w = w_0^+ - 1 \geq 1$. By the block structure of Theorem 11.3, r_2 is the last same-side node at stage 0, hence a convergent. ($w_0^- = 0$ would force $w_0^+ = 1$, contradicting $w_0^+ \geq 2$, so this sub-case cannot arise at $d = 1$.)

Inductive step. Suppose ($\Rightarrow$) holds at all depths $< d$. Suppose $w_{2d} \leq w_{2d+1}$, so this is a gender-preserving generation and $r_{2d} = r_{d-1}^-$ is the older sibling, i.e. $r_{d-1}^- = r_{2d-2} + (w_{d-1}^+ - 1)\, r_{2d-1}$.

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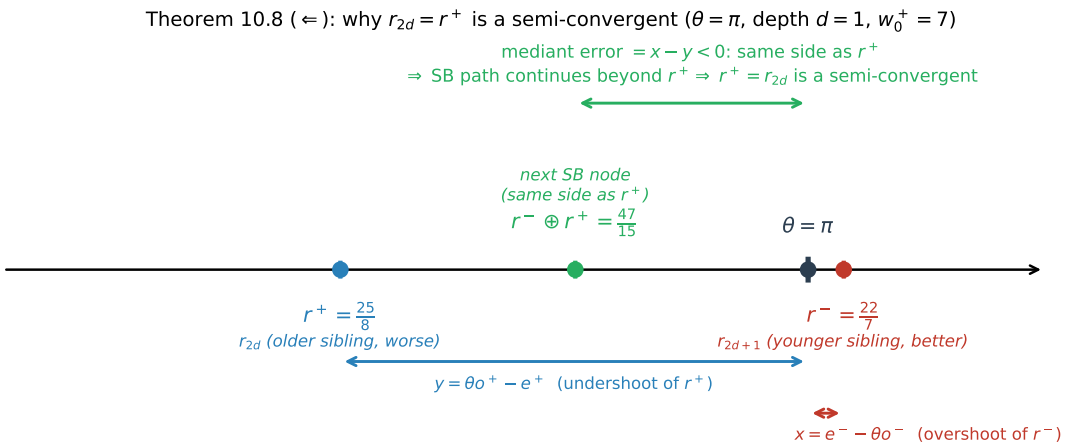


Figure 2: Geometry of the (⇐) direction of Theorem 11.8, illustrated for $\theta = \pi$ at depth $d = 1$ (weight $w_0^+ = 7$). The depth-0 generation is gender-changing: $r_0^- = 22/7$ is the younger sibling and $r_0^+ = 25/8$ the older; r_0^+ lies on the opposite side of θ from r_{2d-2}. The worse child $r_0^+ = 25/8 = r_{2d}$ underestimates θ, and the unweighted mediant $r_0^- \oplus r_0^+ = 47/15$ also underestimates θ (mediant error $= x - y < 0$, same side as r_0^+). Since the next SB-path node has not yet crossed θ, the run continues beyond r_0^+, so $r_0^+ = r_{2d}$ is a semi-convergent rather than a convergent. By contrast, $r_0^- = 22/7 = r_{2d+1}$ *is* a convergent of π: it is the last node before the SB-path crosses θ, which is exactly the definition of a convergent. This is consistent with Theorem 11.6: since $w_1^+ = 15 \geq 2$, $r_{2d+1} = r_0^- = 22/7$ is a convergent, not a semi-convergent.

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- If $w_{d-1}^- \geq 1$: by the block structure established in the proof of Theorem 11.3, the SB path visits the nodes $r_{2d-2} + w\, r_{2d-1}$ for $w = 1, \ldots, w_{d-1}^+$ in order within stage $d-1$. All nodes with $w < w_{d-1}^+$ lie on the same side of θ as r_{2d-2}; the node at $w = w_{d-1}^+$ (namely r_{d-1}^+) lies on the other side. Since $r_{d-1}^- = r_{2d-2} + (w_{d-1}^+ - 1)\, r_{2d-1}$ is the node at $w = w_{d-1}^+ - 1 \geq 1$, it is the last same-side node before the crossing. Hence $r_{2d} = r_{d-1}^-$ is a convergent.

- If $w_{d-1}^- = 0$: then $w_{d-1}^+ = 1$ and $r_{2d} = r_{2d-2}$ (the coarser bound is carried forward unchanged). Since $d \geq 2$ (the case $d = 1$ is handled above), the gender-changing case of Theorem 11.6 shows that $r_{2d-1} = r_{d-2}^-$ would imply $|A_{d-1}| > |B_{d-1}|$ and hence $w_{d-1}^+ \geq 2$; by contraposition, $w_{d-1}^+ = 1$ forces $r_{2d-1} = r_{d-2}^+$, so the generation at depth $d-2$ is gender-preserving with $r_{2d-2} = r_{d-2}^-$ as older sibling. Therefore $w_{2d-2} \leq w_{2d-1}$, and by the induction hypothesis (⇒) at depth $d-1$, r_{2d-2} is a convergent. Hence $r_{2d} = r_{2d-2}$ is a convergent.

In both sub-cases r_{2d} is a convergent, completing the induction. □

All steps in both proofs are argued by hand.

Denominator growth and the depth bound

The two lemmas below are the analytical engine behind the $O(\log N)$ depth bound (Theorem 11.15): the first gives a lower bound on weight values, the second converts that into exponential denominator growth.

Lemma 11.9 (No two consecutive unit weights)**.** *$w_0^+ \geq 2$ always (the depth-0 weight is at least 2), and for no d does $w_d^+ = w_{d+1}^+ = 1$ hold simultaneously.*

Proof. Write A_d, B_d for the signed errors at depth d, with r_{2d+1} finer (smaller relative error $|B_d|/o_{2d+1} < |A_d|/o_{2d}$). The weight is $w_d^+ = \lfloor |A_d|/|B_d| \rfloor + 1$, so $w_d^+ = 1$ iff $|A_d| < |B_d|$.

Base case. At $d = 0$, the initialisation sets $o_0 = o_1 = 1$ and r_1 is chosen as the finer estimate, so $|B_0| < |A_0|$. Hence $|A_0|/|B_0| > 1$, giving $w_0^+ \geq 2$, and $\mathcal{I}(0)$ holds vacuously.

Invariant. Define $\mathcal{I}(d)$: if $|A_d| < |B_d|$ then $|B_d| < 2|A_d|$. We show $\mathcal{I}(d) \Rightarrow \mathcal{I}(d+1)$ for all $d \geq 0$.

The two child candidates produced from depth d are

$$r^- = \text{Mediant}(r_{2d},\, r_{2d+1},\, w_d^+ - 1), \quad \text{error } A_d^- = A_d + (w_d^+ - 1)B_d, \quad o_d^- = o_{2d} + (w_d^+ - 1)\, o_{2d+1},$$
$$r^+ = \text{Mediant}(r_{2d},\, r_{2d+1},\, w_d^+), \qquad \text{error } B_d^+ = A_d + w_d^+ B_d, \quad o_d^+ = o_{2d} + w_d^+\, o_{2d+1}.$$

The depth-$(d+1)$ pair assigns $r_{2(d+1)+1}$ to whichever child has the smaller relative error (finer), and $r_{2(d+1)}$ to the other (coarser).

Case A ($w_d^+ = 1$). Then $w_d^- = 0$, so $r^- = r_{2d}$ is carried forward unchanged: $A_d^- = A_d$ and $o_d^- = o_{2d}$. For r^+: $B_d^+ = A_d + B_d$ and $|B_d^+| = |B_d| - |A_d|$ (since A_d and B_d have opposite signs and $|A_d| < |B_d|$). The relative errors satisfy

$$\frac{|B_d^+|}{o_d^+} = \frac{|B_d| - |A_d|}{o_{2d} + o_{2d+1}} < \frac{|B_d|}{o_{2d+1}} < \frac{|A_d|}{o_{2d}} = \frac{|A_d^-|}{o_d^-},$$

so r^+ is strictly finer. The assignment is therefore unambiguous: $A_{d+1} = A_d^- = A_d$ and $B_{d+1} = B_d^+ = A_d + B_d$. Thus $|A_{d+1}| = |A_d|$ and $|B_{d+1}| = |B_d| - |A_d|$. By $\mathcal{I}(d)$, since $|A_d| < |B_d|$, we have $|B_d| < 2|A_d|$, so $|B_{d+1}| = |B_d| - |A_d| < |A_d| = |A_{d+1}|$. Hence $w_{d+1}^+ \geq 2$ and $\mathcal{I}(d+1)$ holds vacuously.

Case B ($w_d^+ = k \geq 2$, i.e. $|A_d| > |B_d|$). Let $\delta = \text{frac}(|A_d|/|B_d|) \in [0, 1)$ be the fractional part of $|A_d|/|B_d|$. If $\delta = 0$ (i.e. $-A_d/B_d \in \mathbb{Z}$) the step is exact and depth $d+1$ is terminal, so there is nothing to prove. If $\delta = 1/2$ then $|A_d^-| = |B_d^+|$, and r^+ is finer since $o_d^+ > o_d^-$; hence $|A_{d+1}| = |B_{d+1}|$ and $\mathcal{I}(d+1)$ holds vacuously (depth $d+1$ is then an exact step with $w^- = 1$). Otherwise $\delta \in (0, 1) \setminus \{1/2\}$. Thus $|A_d^-| = \delta|B_d|$ and $|B_d^+| = (1-\delta)|B_d|$, with $|A_d^-| + |B_d^+| = |B_d|$ and $|A_d^-| \neq |B_d^+|$. If the ordering at depth $d+1$ gives $|A_{d+1}| \geq |B_{d+1}|$, then $w_{d+1}^+ \geq 2$ and $\mathcal{I}(d+1)$ holds vacuously; done.

Assume therefore $|A_{d+1}| < |B_{d+1}|$, so $|A_{d+1}| = \min(|A_d^-|, |B_d^+|)$ and $|B_{d+1}| = \max(|A_d^-|, |B_d^+|)$. We must show $|B_{d+1}| < 2|A_{d+1}|$.

Subcase $\delta < 1/2$: then $|A_d^-| < |B_d^+|$, so $|A_{d+1}| = \delta|B_d|$. For this assignment, r^+ must be finer than r^-, i.e. $|B_d^+|/o_d^+ < |A_d^-|/o_d^-$. Since $k \geq 2$ and $o_{2d} \geq 1$, we have $o_d^- = o_{2d} + (k-1)o_{2d+1} \geq o_{2d} + o_{2d+1} > o_{2d+1}$, so this gives $(1-\delta)o_d^- < \delta\, o_d^- + \delta\, o_{2d+1}$, hence $(1-2\delta)o_d^- < \delta\, o_{2d+1} < \delta\, o_d^-$, so $1 - 2\delta < \delta$, i.e. $\delta > 1/3$. Then $|B_{d+1}| = (1-\delta)|B_d| < \frac{2}{3}|B_d| < 2\delta|B_d| = 2|A_{d+1}|$.

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Subcase $\delta > 1/2$: then $|B_d^+| < |A_d^-|$. If $\delta \geq 2/3$, then $|A_d^-| = \delta|B_d| \geq \frac{2}{3}|B_d|$ and $|B_d^+| = (1-\delta)|B_d| \leq \frac{1}{3}|B_d|$, so $|A_d^-|/o_d^- > |B_d^+|/o_d^+$ (the denominator $o_d^+ > o_d^-$ only reinforces this), forcing r^+ to be finer. The assignment then gives $A_{d+1} = A_d^-$ and $B_{d+1} = B_d^+$, so $|A_{d+1}| = \delta|B_d| > |B_{d+1}| = (1-\delta)|B_d|$, contradicting $|A_{d+1}| < |B_{d+1}|$. Hence $\delta < 2/3$, and then $|B_{d+1}| = \delta|B_d| < \frac{2}{3}|B_d| < 2(1-\delta)|B_d| = 2|A_{d+1}|$.

In all sub-cases $\mathcal{I}(d+1)$ holds, completing the induction. Since Case A gives $w_{d+1}^+ \geq 2$ whenever $w_d^+ = 1$, two consecutive unit weights are impossible. □

Lemma 11.10 (Younger-sibling weight)**.** *For every depth $d \geq 1$, the younger sibling r_{2d+1} inherits weight $w_{2d+1} \geq 1$.*

Proof. The inherited weights of the depth-d pair are $\{w_{2d}, w_{2d+1}\} = \{w_{d-1}^+ - 1,\, w_{d-1}^+\}$. The younger sibling r_{2d+1} is the better (smaller relative error) child.

In a *gender-preserving* generation r^+ is the better child, so $w_{2d+1} = w_{d-1}^+ \geq 1$.

In a *gender-changing* generation r^- is the better child, so $w_{2d+1} = w_{d-1}^+ - 1$. By Lemma 11.9 (Case A), $w_{d-1}^+ = 1$ implies gender-preserving, so a gender-changing generation must have $w_{d-1}^+ \geq 2$, giving $w_{2d+1} \geq 1$. □

Corollary 11.11 (Older-sibling weight)**.** *For every depth $d \geq 1$, the older sibling r_{2d} inherits weight $w_{2d} \geq 0$.*

Proof. $w_{2d} = w_{d-1}^+ - 1 \geq 0$ since $w_{d-1}^+ \geq 1$ by definition. □

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Lemma 11.12 (Pair-sum invariant)**.** *For every depth $d \geq 0$,*

$$o_{2d} \;<\; 2\, o_{2d+1}.$$

Proof. At $d = 0$: $o_0 = o_1 = 1$, so $1 < 2$.

For $d \geq 1$ the flat-sequence recurrences give

$$o_{2d} = o_{2d-2} + w_{2d}\, o_{2d-1}, \qquad o_{2d+1} = o_{2d-2} + w_{2d+1}\, o_{2d-1},$$

where $\{w_{2d},\, w_{2d+1}\} = \{w_{d-1}^+ - 1,\, w_{d-1}^+\}$, so the two weights differ by exactly 1. By Lemma 11.10, $w_{2d+1} \geq 1$. Hence

$$2\, o_{2d+1} - o_{2d} \;=\; o_{2d-2} + (2w_{2d+1} - w_{2d})\, o_{2d-1}.$$

Case A ($w_{2d} = w_{2d+1} + 1$): $2\, o_{2d+1} - o_{2d} = o_{2d-2} + (w_{2d+1} - 1)\, o_{2d-1} \geq o_{2d-2} \geq 1 > 0$.

Case B ($w_{2d} = w_{2d+1} - 1$): $2\, o_{2d+1} - o_{2d} = o_{2d-2} + (w_{2d+1} + 1)\, o_{2d-1} \geq o_{2d-2} + 2\, o_{2d-1} \geq 3 > 0$.

In both cases $2\, o_{2d+1} > o_{2d}$. □

Lemma 11.13 (Mirror invariant)**.** *If $w_d^+ = 1$, then $o_{2d+1} < 2\, o_{2d}$ (equivalently, $3\, o_{2d} > S_d$).*

Proof. At $d = 0$: $o_0 = o_1 = 1$, so $o_1 = 1 \le 2 = 2\,o_0$.

For $d \ge 1$: since $w_d^+ = 1$, Lemma 11.9 (Case A) forces $w_{d-1}^+ \ge 2$. The flat-sequence recurrences give

$$o_{2d} = o_{2d-2} + w_{2d}\,o_{2d-1}, \qquad o_{2d+1} = o_{2d-2} + w_{2d+1}\,o_{2d-1},$$

where $\{w_{2d},\, w_{2d+1}\} = \{w_{d-1}^+ - 1,\, w_{d-1}^+\}$. Subtracting twice the first equation from the second:

$$o_{2d+1} - 2\,o_{2d} = (o_{2d-2} + w_{2d+1}\,o_{2d-1}) - 2(o_{2d-2} + w_{2d}\,o_{2d-1}) = -o_{2d-2} + (w_{2d+1} - 2w_{2d})\,o_{2d-1}.$$

Since $d \ge 1$ the depth-$(d-1)$ pair exists and $o_{2d-2} \ge 1$.

Case A ($w_{2d} = w_{d-1}^+,\ w_{2d+1} = w_{d-1}^+ - 1$): $o_{2d+1} - 2\,o_{2d} = -o_{2d-2} - (w_{d-1}^+ + 1)\,o_{2d-1} < 0$.

Case B ($w_{2d} = w_{d-1}^+ - 1,\ w_{2d+1} = w_{d-1}^+$): $o_{2d+1} - 2\,o_{2d} = -o_{2d-2} + (2 - w_{d-1}^+)\,o_{2d-1}$. Since $w_{d-1}^+ \ge 2$, the second term is ≤ 0, so the whole expression $\le -o_{2d-2} \le -1 < 0$.

In both cases $o_{2d+1} - 2\,o_{2d} < 0$. □

Lemma 11.14 (Denominator growth)**.** *For every depth $d \ge 0$, both denominators of the depth-d pair satisfy*

$$o_{2d},\, o_{2d+1} \;\ge\; \tfrac{2}{3} \cdot 3^{\lfloor d/2 \rfloor} \;\ge\; 2^{\lfloor d/2 \rfloor}.$$

Proof. Let $S_d = o_{2d} + o_{2d+1}$.

Base cases. At $d = 0$: $o_0 = o_1 = 1$, so $S_0 = 2$ and both $\ge 2^0 = 1$. At $d = 1$: since $w_0^+ \ge 2$, the child denominators are $o_2 = o_0 + (w_0^+ - 1)o_1 \ge 1 + 1 = 2$ and $o_3 = o_0 + w_0^+ o_1 \ge 3$, so both $\ge 2^1 = 2$.

Recurrence $S_{d+2} \ge 3S_d$. We split on which of w_d^+, w_{d+1}^+ is ≥ 2 (Lemma 11.9 guarantees at least one).

Case 1 ($w_d^+ \ge 2$). The two depth-$(d+1)$ denominators are $o_d^- \ge o_{2d} + o_{2d+1} = S_d$ and $o_d^+ \ge o_{2d} + 2o_{2d+1} > S_d$. At depth $d+2$, each member's denominator is at least the sum of its two depth-$(d+1)$ parents' denominators (for any weight ≥ 1), so both are $\ge o_d^- + o_d^+ \cdot 1 \ge S_d$. More precisely, one member at $d+2$ has denominator $\ge o_d^- + w_{d+1}^+ o_d^+ \ge S_d + S_d = 2S_d$ and the other $\ge o_d^- \ge S_d$, giving $S_{d+2} \ge 3S_d$.

Case 2 ($w_d^+ = 1$, forcing $w_{d+1}^+ \ge 2$). At depth $d+1$: $o_d^- = o_{2d}$ (the coarser parent is carried) and $o_d^+ = S_d$. With $w_{d+1}^+ \ge 2$, the depth-$(d+2)$ denominators satisfy $o_{d+1}^- \ge o_{2d} + S_d$ and $o_{d+1}^+ \ge o_{2d} + 2S_d$, so $S_{d+2} \ge (o_{2d} + S_d) + (o_{2d} + 2S_d) = 2o_{2d} + 3S_d > 3S_d$.

In both cases $S_{d+2} \ge 3S_d$, so by induction $S_d \ge 2 \cdot 3^{\lfloor d/2 \rfloor}$.

Individual bounds. For $d \le 1$: at $d = 0$ both $o_0 = o_1 = 1 \ge \frac{2}{3} \cdot 1$; at $d = 1$ we showed $o_2 \ge 2$ and $o_3 \ge 3$, both $\ge \frac{2}{3} \cdot 3^0 = \frac{2}{3}$. For $d \ge 2$, every member at depth d is a child produced at a step where the weight was ≥ 2 (by Lemma 11.9 applied at depth $d-1$ or $d-2$), so its denominator is at least the sum S_{d-2} of both depth-$(d-2)$ denominators. Setting $k = \lfloor d/2 \rfloor \ge 1$:

$$\min(o_{2d},\, o_{2d+1}) \;\ge\; S_{d-2} \;\ge\; 2 \cdot 3^{k-1} \;=\; \tfrac{2}{3} \cdot 3^k \;=\; \tfrac{2}{3} \cdot 3^{\lfloor d/2 \rfloor}.$$

Since $(3/2)^{k-1} \ge 1$ for $k \ge 1$, we also have $\frac{2}{3} \cdot 3^k = 2^k \cdot (3/2)^{k-1} \ge 2^k = 2^{\lfloor d/2 \rfloor}$. Hence both $o_{2d}, o_{2d+1} \ge \frac{2}{3} \cdot 3^{\lfloor d/2 \rfloor} \ge 2^{\lfloor d/2 \rfloor}$. □

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Theorem 11.15 (Linear depth bound)**.** *Let $\theta > 1$ be irrational. For every depth d and both members $e/o \in \{r_{2d}, r_{2d+1}\}$ of the depth-d pair,*

$$d \;\le\; \frac{2}{\log_2 3}\,\log_2(e+o) + 2, \qquad \frac{2}{\log_2 3} \approx 1.262.$$

The depth grows at most logarithmically in $e+o$, so $O(\log N)$ iterations of the mediant-tree algorithm suffice to produce an approximant of size $N = e+o$.

Proof. We combine Lemmas 11.9 and 11.14. Let e/o be either member of the depth-d pair. From Lemma 11.14, $o \ge \frac{2}{3}\cdot 3^{\lfloor d/2\rfloor}$, so $3^{\lfloor d/2\rfloor} \le \frac{3}{2}\,o \le \frac{3}{2}(e+o)$. Taking logarithms base 3:

$$\lfloor d/2\rfloor \;\le\; \log_3\!\big(\tfrac{3}{2}(e+o)\big) \;=\; \frac{\log_2(e+o) + \log_2(3/2)}{\log_2 3}.$$

Hence $d \le 2\lfloor d/2\rfloor + 1 \le \frac{2}{\log_2 3}\log_2(e+o) + \frac{2\log_2(3/2)}{\log_2 3} + 1$. Since $\frac{2\log_2(3/2)}{\log_2 3} + 1 < 2$, this gives $d \le \frac{2}{\log_2 3}\log_2(e+o) + 2$. □

The optimal per-target slope m^*_θ depends on θ through the average size of partial quotients: large partial quotients drive faster denominator growth and hence a smaller m^*_θ. For $\theta = \phi$ (all partial quotients equal to 1), the asymptotic slope is $1/(2\log_2\phi) \approx 0.72$; for $\theta = \log_2 3$ the empirical slope is approximately 0.45.

Remark 11.16. Defining m^*_θ as the infimum of slopes m for which there exists b with $d \le m\log_2(e+o) + b$ for all $d \ge 0$ and both members e/o of the depth-d pair, Theorem 11.15 gives $d \le \frac{2}{\log_2 3}\log_2(e+o) + 2$ for every irrational $\theta > 1$, and hence $m^* = \sup_\theta m^*_\theta \le 2/\log_2 3 < 2$.

Conjecture 11.17 (Two-step pair-sum growth)**.** *For every irrational $\theta > 1$ and every depth $d \ge 0$,*

$$S_{d+2} \;\ge\; \tfrac{10}{3}\,S_d,$$

where $S_d = o_{2d} + o_{2d+1}$ is the pair-sum of denominators at depth d. Consequently, for every depth $d \ge 0$ and both members $e/o \in \{r_{2d}, r_{2d+1}\}$,

$$d \;\le\; \frac{2}{\log_2(10/3)}\,\log_2(e+o) + 3, \qquad \frac{2}{\log_2(10/3)} \approx 1.151,$$

and hence $m^* \le 2/\log_2(10/3) < 2/\log_2 3$.

Remark 11.18. A proof of Conjecture 11.17 has been provided by the reviewer (Aristotle, A-43), including a Lean formalisation (`pair_sum_growth_ten_thirds`). Inclusion of the proof is deferred to a later revision so that the current review round can focus on clearing the remaining open items in the manuscript.

Figure 3 shows that slope varies with partial-quotient size. Irrationals whose partial quotients are predominantly small (such as ϕ and $\sqrt{2}$) tend to produce steeper curves (more depth per bit of $\log_2(e+o)$), while those with predominantly large partial quotients (such as π and $\log_3 8$) tend to produce shallower curves. This is a qualitative tendency, not a monotone relationship: the slope m^*_θ is not a monotone function of partial-quotient size. The proved

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bound $d \le (2/\log_2 3)\log_2(e+o)+2$ of Theorem 11.15 and the conjectured tighter bound of Conjecture 11.17 both apply to every curve. Figure 4 illustrates the per-target (m_θ, b_θ) pairs for $\log_2 3$ and ϕ, alongside both bounds.

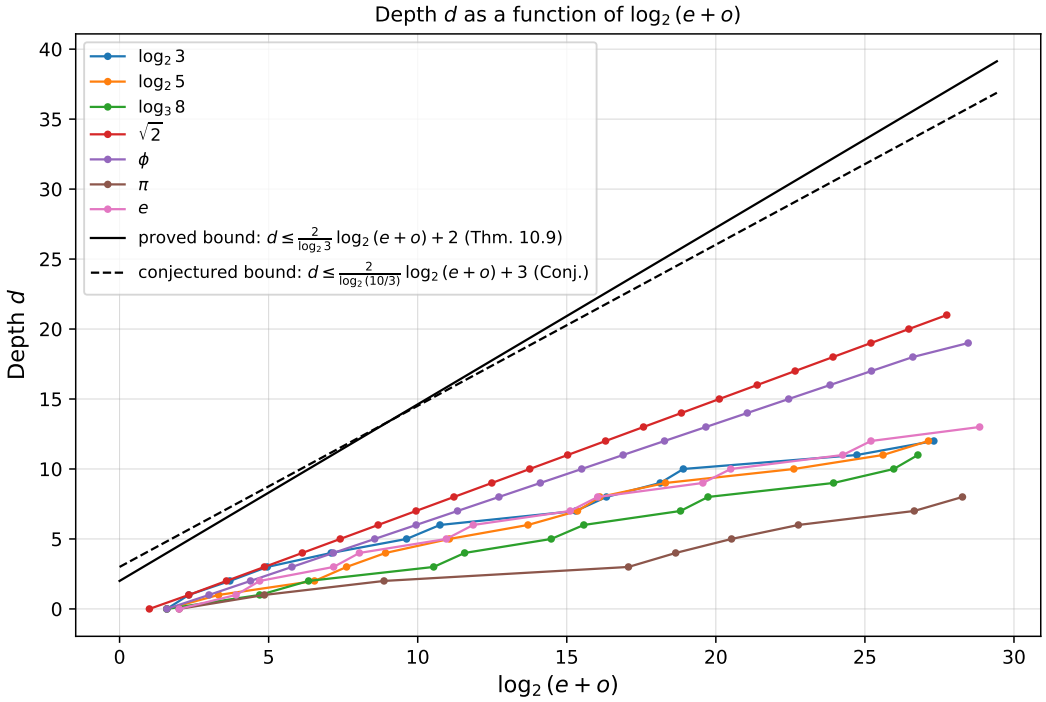


Figure 3: Depth d vs. $\log_2(e+o)$ for the better mediant-tree estimate at each depth, across seven irrationals. Each curve is approximately linear, consistent with Theorem 11.15. The solid black line is the proved bound $d \le \frac{2}{\log_2 3}\log_2(e+o)+2$ (Theorem 11.15, slope ≈ 1.262); the dashed black line is the conjectured tighter bound $d \le \frac{2}{\log_2(10/3)}\log_2(e+o)+3$ (Conjecture 11.17, slope ≈ 1.151). The slope of each curve reflects partial-quotient size: irrationals with small partial quotients (e.g. ϕ, $\sqrt{2}$) converge slowly and have steeper curves; those with large partial quotients (e.g. π, $\log_3 8$) converge quickly and have shallower curves.

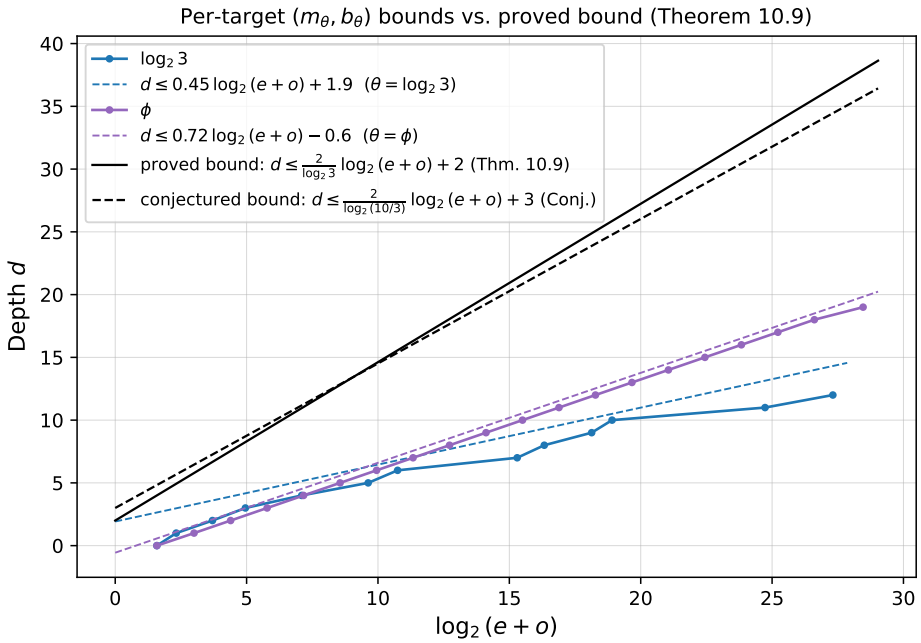


Figure 4: Per-target (m_θ, b_θ) bounds for $\theta = \log_2 3$ and $\theta = \phi$, together with the proved upper bound $d \le \frac{2}{\log_2 3} \log_2(e+o) + 2$ of Theorem 11.15 (solid black line, slope ≈ 1.262) and the conjectured tighter bound $d \le \frac{2}{\log_2(10/3)} \log_2(e+o) + 3$ of Conjecture 11.17 (dashed black line, slope ≈ 1.151). Each coloured dashed line is an empirical upper bound fitted to its own curve only; the slopes $m_{\log_2 3} \approx 0.45$ and $m_\phi \approx 0.72$ are both well below the proved ceiling $m = 2/\log_2 3$.

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12 Relationship to Sturmian Words

Standing notation for this section. Throughout, $\theta > 1$ is real and

$$\alpha = \frac{1}{1+\theta}$$

is the Sturmian slope; s denotes the Sturmian word with slope α. For any approximant $r = (e, o)$ produced by the mediant tree, set $N = e + o$ (the length of the mediant word $\pi(r)$). For a non-terminal bounded pair (r_{2d}, r_{2d+1}) at depth d (i.e. $\Theta \neq 0$, so neither estimate equals θ exactly), set $n_i = e_i + o_i$ for $i \in \{2d, 2d+1\}$, and define

$$\hat{\alpha}_i \;=\; \frac{1}{1+r_i} \;=\; \frac{o_i}{e_i+o_i} \;=\; \frac{o_i}{n_i}.$$

Since $x \mapsto 1/(1+x)$ is order-reversing and r_{2d}, r_{2d+1} strictly bracket θ (both conditions hold at non-terminal depths), the values $\hat{\alpha}_{2d}$ and $\hat{\alpha}_{2d+1}$ strictly bracket α:

$$\min(\hat{\alpha}_{2d},\, \hat{\alpha}_{2d+1}) \;<\; \alpha \;<\; \max(\hat{\alpha}_{2d},\, \hat{\alpha}_{2d+1}).$$

Since r_{2d} and r_{2d+1} are Farey-adjacent (Definition 6.4), Lemma 9.2 gives that $\hat{\alpha}_{2d}$ and $\hat{\alpha}_{2d+1}$ are also Farey adjacent.

12.1 Structural properties of mediant words

Before stating the main Sturmian identity we record four structural results about finite mediant words.

Theorem 12.1 (Every mediant word ends in O)**.** *For every estimate r produced by the mediant tree, the mediant word $\pi(r)$ ends in O.*

Proof. By structural induction on the mediant tree.

Base case. $\pi(e/1) = \mathsf{E}^e\mathsf{O}$, which ends in O.

Inductive step. Let (r_{2d}, r_{2d+1}) be the parent bounded pair and assume $\pi(r_{2d})$ and $\pi(r_{2d+1})$ both end in O. For each $w^\dagger \in \{w^-, w^+\}$, set $r^\dagger = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^\dagger)$. Write $u = \pi(r_{2d})$ and $v = \pi(r_{2d+1})$. By Definition 10.2: if $\Delta > 0$ then $\pi(r^\dagger) = v^{w^\dagger} \cdot u$, which ends in u; if $\Delta < 0$ and $w^\dagger \geq 1$ then $\pi(r^\dagger) = u \cdot v^{w^\dagger}$, which ends in v; and if $\Delta < 0$ and $w^\dagger = 0$ (which occurs when $w^+ = 1$) then $\pi(r^\dagger) = u$. In every case $\pi(r^\dagger)$ ends in u or v, both of which end in O by the inductive hypothesis. □

Theorem 12.2 (Overestimate criterion)**.** *Let $r = (e, o)$ be an approximant produced by the mediant tree.*

(1) *If $e \neq \theta o$ (i.e. $r \neq \theta$ exactly), the Sturmian symbol s_{N-1} satisfies*

$$s_{N-1} = \begin{cases} \mathsf{O} & \text{if } e > \theta o \text{ (} r \text{ an overestimate),} \\ \mathsf{E} & \text{if } e < \theta o \text{ (} r \text{ an underestimate).} \end{cases}$$

(2) *If $e \neq \theta o$, since $\pi(r)[-1] = \mathsf{O}$ (Theorem 12.1), $\pi(r)[-1] = s_{N-1}$ if and only if r is an overestimate.*

(3) *If $e = \theta o$ (so θ is rational and $r = \theta$), then $s_{N-1} = \mathsf{O} = \pi(r)[-1]$.*

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Proof. **Parts (1) and (2).** The Sturmian word satisfies $s_j = \mathsf{O}$ iff the fractional part $f_j := \{(j+1)\alpha\}$ lies in $[0, \alpha)$. So $s_{N-1} = \mathsf{O}$ iff $\{(e+o)\alpha\} \in [0, \alpha)$.

Set $\Theta = o(e - \theta o)$, so $e = \theta o + \Theta/o$. Since $\alpha = 1/(1+\theta)$:

$$(e+o)\alpha = o + \frac{\Theta\alpha}{o}.$$

Since o is a positive integer, $\{(e+o)\alpha\} = \{\Theta\alpha/o\}$. Since r is a good estimate (Lemma 8.3), $|\Theta|/o < 1$; with $\alpha < 1/2$ we have $|\Theta\alpha/o| < \alpha < 1/2$, so:

- $\Theta > 0$: $\{\Theta\alpha/o\} = \Theta\alpha/o \in (0, \alpha)$, so $s_{N-1} = \mathsf{O}$.
- $\Theta < 0$: $\{\Theta\alpha/o\} = 1 + \Theta\alpha/o \in (1-\alpha, 1) \subset (\alpha, 1)$, so $s_{N-1} = \mathsf{E}$.

For a tree estimate, $\pi(r)[-1] = \mathsf{O}$ by Theorem 12.1. Combined with Part (1), $\pi(r)[-1] = s_{N-1}$ iff both equal O, i.e. iff r is an overestimate.

Part (3). If $e = \theta o$ then $(e+o)\alpha = (\theta o + o)/(1+\theta) = o \in \mathbb{Z}$, so $\{(e+o)\alpha\} = 0 \in [0, \alpha)$, giving $s_{N-1} = \mathsf{O}$. Since $r = \theta$ is produced as $r^- = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^-)$ at the exact step, it is a mediant tree approximant, so $\pi(r)[-1] = \mathsf{O}$ by Theorem 12.1. □

Remark 12.3. When $r = \text{Mediant}(r_{2d}, r_{2d+1}, w^+)$ is the child at the bracketing weight, the overestimate condition $e > \theta o$ is equivalent to $\Delta(r_{2d}, r_{2d+1}) > 0$: since w^+ is the bracketing weight, $\text{sgn}(e - \theta o) = \text{sgn}(f(w^+)) = \text{sgn}(B) = \text{sgn}(\Delta)$ (where $f(w) = A + wB$ as in Section 8, and $\Delta = \Delta(r_{2d}, r_{2d+1})$ is the determinant of the *parent pair*). For the companion child $r = \text{Mediant}(r_{2d}, r_{2d+1}, w^-)$ (when $w^+ = 1$, the companion is r_{2d} itself), the relationship is reversed, provided $r \neq \theta$ (i.e. $-A/B \notin \mathbb{Z}$). The sign of Δ of the parent pair is therefore not a uniform criterion for a general good estimate; the sign of $\Theta = o(e - \theta o)$ is.

Theorem 12.4 (Universal EE prefix)**.** *For every non-leaf estimate r produced by the mediant tree (i.e. $|\pi(r)| > \lceil\theta\rceil + 1$), the mediant word begins $\pi(r)[0{:}2] = \mathsf{EE}$.*

Proof. By Lemma 10.3, $\pi(r) = \text{Ch}^-(e, o)$. Every non-leaf estimate $r = e/o$ lies strictly between the two leaf estimates $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, so in particular $e/o > \lfloor\theta\rfloor \geq 1$, giving $e > o$. By Definition 5.1, $\text{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor(i+1)o/(e+o)\rfloor > \lfloor i \cdot o/(e+o)\rfloor$. For $i = 0$: $1 \cdot o/(e+o) < 1$ (since $o < e + o$), so $\lfloor o/(e+o)\rfloor = 0 = \lfloor 0\rfloor$, giving $\pi(r)[0] = \mathsf{E}$. For $i = 1$: $e > o$ implies $e + o > 2o$, so $2o/(e+o) < 1$, giving $\lfloor 2o/(e+o)\rfloor = 0$, and $\pi(r)[1] = \mathsf{E}$. Hence $\pi(r)[0{:}2] = \mathsf{EE}$. □

Theorem 12.5 (Palindrome invariant)**.** *For every estimate r produced by the mediant tree with $N \geq 2$, the interior slice $\pi(r)[1{:}N{-}1]$ is a palindrome.*

Proof. By Lemma 10.3, $\pi(r) = \text{Ch}^-(e, o)$. By Definition 5.1, $\text{Ch}^-(e, o) = \mathsf{E} \cdot U \cdot \mathsf{O}$ where U is the palindromic interior (Berstel [5], Chapter 1). Therefore $\pi(r)[1{:}N{-}1] = U$ is a palindrome. □

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Theorem 12.6 (Sturmian identity)**.** *Assume θ is irrational. For every estimate r produced by the mediant tree,*

$$\pi(r)[i] = s_i \qquad \text{for all } 0 \leq i \leq N - 2.$$

The terminal position $i = N - 1$ is excluded: by Theorem 12.1 the mediant always emits O *there, while by Theorem 12.2 the Sturmian emits* O *if r is an overestimate and* E *if r is an underestimate.*

Proof. By Lemma 10.3, $\pi(r) = \text{Ch}^-(e, o)$. By Definition 5.1, $\text{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor(i+1)o/N\rfloor > \lfloor i \cdot o/N\rfloor$, and the Sturmian word satisfies $s_i = \mathsf{O}$ iff $\lfloor(i+1)\alpha\rfloor > \lfloor i\alpha\rfloor$. It suffices to show $\lfloor j \cdot o/N\rfloor = \lfloor j\alpha\rfloor$ for all $1 \leq j \leq N - 1$.

The estimate $r = e/o$ belongs to a bounded pair (r_{2d}, r_{2d+1}) at some depth d, so $r = r_i$ for some $i \in \{2d, 2d+1\}$, giving $o/N = \hat{\alpha}_i$ and $N = n_i$.

Key lemma (Farey gap). No fraction m/j with $0 < j < N$ lies strictly between α and o/N. *Proof:* Suppose m/j with $0 < j < N$ lies strictly between α and o/N. Since $o/N = \hat{\alpha}_i$ is one endpoint of the interval $[\min(\hat{\alpha}_{2d}, \hat{\alpha}_{2d+1}), \max(\hat{\alpha}_{2d}, \hat{\alpha}_{2d+1})]$ and α lies strictly in its interior, m/j lies strictly between $\hat{\alpha}_{2d}$ and $\hat{\alpha}_{2d+1}$. By the Farey mediant property (Definition 3.1, Section 3), any fraction strictly between two Farey-adjacent fractions has denominator at least $n_{2d} + n_{2d+1} > N$, contradicting $j < N$.

Agreement. Fix $1 \leq j \leq N - 1$ and suppose for contradiction that $\lfloor j\alpha\rfloor \neq \lfloor j \cdot o/N\rfloor$. Then there exists an integer m with $j\alpha < m \leq j \cdot o/N$ or $j \cdot o/N < m \leq j\alpha$ (the two cases are

symmetric). In either case m/j has denominator $j < N$. Note that $m/j = o/N$ is impossible since $\gcd(o, N) = 1$ and $0 < j < N$, and $m/j = \alpha$ is impossible since α is irrational. Hence m/j lies *strictly* between α and o/N, contradicting the Key lemma. Hence $\lfloor j\alpha \rfloor = \lfloor j \cdot o/N \rfloor$ for all $1 \leq j \leq N-1$, and therefore $\mathrm{Ch}^-(e, o)[i] = s_i$ for $0 \leq i \leq N-2$.

Terminal position. The exclusion of $i = N-1$ follows from Theorems 12.1 and 12.2: $\pi(r)$ ends in O, while $s_{N-1} = \mathsf{O}$ iff r is an overestimate ($e/o > \theta$). □

Corollary 12.7 (Mismatch characterisation)**.** *Assume θ is irrational. The mediant word $\pi(r)$ and the Sturmian prefix $s[0{:}N]$ are identical if and only if r is an overestimate ($e/o > \theta$). When r is an underestimate ($e/o < \theta$), they agree at every position except $i = N-1$, where the mediant emits* O *and the Sturmian emits* E*.*

The single terminal discrepancy is an artefact of truncation, not a genuine disagreement: the mediant sequence closes its last primitive word with O at position $N-1$, whereas the Sturmian word — which continues indefinitely — may still be in the interior of a longer word at that depth. As the depth increases and $N \to \infty$, both sequences converge to the same infinite word.

Remark 12.8. The identification of finite approximants of a Sturmian word with lower Christoffel words is classical (Lothaire [3], Berstel et al. [5]). The novel content of Theorem 12.6 is the proof via the mediant tree, which gives an explicit construction of these approximants together with a precise characterisation of the terminal discrepancy (the last-letter criterion, Theorems 12.1 and 12.2).

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This has been verified computationally for several irrational values of θ, including $\sqrt{2}$, $\log_2 3$, $\log_2 5$, ϕ, and π. For $\theta = \log_2 3$ the verification extends to sequences of length up to 27,869,189 symbols: at every depth the two sequences agree everywhere except possibly the last symbol, where the mediant emits O and the Sturmian emits E.

Remark 12.9 (The mediant sequence is the better finite approximation). At depths where the two sequences differ, the mediant's finite mediant word is a closer approximation to $\alpha = 1/(\theta+1)$ in terms of running O-density. The terminal O emitted by the mediant sequence gives a density slightly *above* α, whereas the Sturmian's terminal E gives a density substantially *below* α.

This reflects a structural asymmetry in the two methods. The Sturmian algorithm at position $N-1$ is mid-word in a longer primitive word, so it emits E — inherently favouring the above-θ estimate and leaving the running density below α. The mediant tree, by contrast, explicitly selects at each step the estimate with smaller $|\varepsilon|$, i.e. r_{2d+1}, whichever side of θ it falls on. The terminal O is a consequence of the Christoffel word structure: every lower Christoffel word (and hence every mediant word; Theorem 12.1) ends in O, regardless of which side of θ the estimate falls on.

For $\theta = \log_2 3$, the improvement at the discrepancy depths is substantial: at $N = 1719$ the mediant density error is 1.4×10^{-8} against the Sturmian's 5.8×10^{-4} — roughly four orders of magnitude better.

The O-density of $\pi(r)$ for an estimate e/o is

$$\frac{o}{e+o} = \frac{1}{e/o+1} \xrightarrow{e/o\to\theta} \frac{1}{\theta+1} = \alpha,$$

consistent with the Sturmian slope.

Efficiency and lazy generation. Although the mediant words are Sturmian, the mediant tree is a far more efficient way to generate them. To produce a prefix of N symbols, the naive Sturmian algorithm must evaluate $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ at each of the N positions, requiring $O(N)$ real-arithmetic operations.

The mediant algorithm instead builds a compressed representation of the same prefix. Each iteration requires a constant number of real-arithmetic operations to compute the next bounded pair and weight; after d iterations the tree encodes a Sturmian prefix of length $N = e_{2d+1} + o_{2d+1}$. N grows exponentially with d (Theorem 11.15), so $O(\log N)$ iterations suffice to encode a prefix of length N, each requiring $O(1)$ oracle evaluations.

Once the tree is built, enumerating the N symbols requires *no real arithmetic at all*: each node specifies an integer weight w and two child words, and reading off the sequence is a recursive depth-first traversal governed entirely by integer counts. No auxiliary memory beyond the call stack is required.

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OO-avoidance. Since $\theta > 1$ implies $\alpha = 1/(\theta+1) < 1/2$, the Sturmian word avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. The mediant tree gives an independent structural proof: every mediant word ends in O and begins with E (since $e \geq 1$; Lemma 10.3 and Definition 5.1), so every concatenation boundary within a branch word $v^w \cdot u$ or $u \cdot v^w$ reads $\cdots$OE$\cdots$; OO never arises at any boundary, and by induction never arises anywhere.

Remark 12.10. The propagated weights w_d^+ are related to the partial quotients of the continued fraction expansion of θ, though they are not in general identical to them. A large weight at depth d corresponds to a large partial quotient, producing a long run of one symbol type in the mediant word — the same phenomenon that gives rise to the irregular spacing of O's and E's in an irrational Sturmian word. Conversely, small weights correspond to small partial quotients, and the mediant word advances by only a short block at each such step.

Connection to continued fractions

The connection between the mediant algorithm and continued fractions is a direct consequence of the Sturmian identity (Theorem 12.6). In combinatorics on words, the standard sequence for a Sturmian word with slope $\alpha = 1/(1+\theta)$ is generated by the recurrence $s_n = s_{n-1}^{b_n} \cdot s_{n-2}$, where the weights b_n are precisely the partial quotients of the continued-fraction expansion of α (Lothaire [3], Ch. 2; Widmer [4], Proposition 2.4.6); here $b_1 = a_0$ and $b_n = a_{n-1}$ for $n \geq 2$, with $\alpha = 1/(1+\theta) = [0;\, a_0+1,\, a_1,\, a_2,\, \ldots]$. The mediant word sequence agrees with this standard sequence at all interior positions (Theorem 12.6). This does *not* imply that the algorithm's weights w^+ equal the partial quotients b_k: the Sturmian identity constrains only interior positions of the words, not which weights the algorithm selects.

What does follow is that the two children at weights w^- and w^+ always have denominators $o_{2d} + w^- \cdot o_{2d+1}$ and $o_{2d} + w^+ \cdot o_{2d+1}$ (one even-indexed, one odd-indexed depending on which is the better estimate), mirroring the structure of the classical recurrence $q_{k+1} = b_k q_k + q_{k-1}$. Exponential denominator growth is established in Section 11 (Theorem 11.15).

The MT fractions are not in general the same as the CF convergents. The standard CF recurrence jumps by the full partial quotient a_k at each step, landing on a convergent p_k/q_k. The MT algorithm picks the optimal bracketing weight w^+, which need not equal any partial quotient.

13 Extension to $\theta \leq 1$

The algorithm of Sections 9–10 requires $\theta > 1$. We now show how to handle all other real θ by algebraic reduction.

If one were content to accept OO in the mediant word, no special treatment would be needed: the algorithm and the Sturmian definition extend formally to any $\theta > 0$, and for $\theta < 1$ the slope $\alpha = 1/(1+\theta) > 1/2$ makes OO unavoidable. The reductions below are motivated by applications (described in [7]) that require OO-free mediant words.

Notation. Let $r(\theta)$ denote the rational approximant produced by the algorithm for $\theta > 1$, i.e. the sequence of bounded pairs (r_{2d}, r_{2d+1}) and their limit. Each subsection below defines $r(\theta)$ for the remaining cases by reducing to a $\theta' > 1$ and applying a simple transformation.

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13.1 Case $0 < \theta < \frac{1}{2}$

For $\theta \in (0, \frac{1}{2})$, define $\theta' = (1-\theta)/\theta > 1$ and set

$$r(\theta) = \frac{o}{e+o},$$

where $e/o = r(\theta')$. That is, if $r(\theta') = e/o$ then $r(\theta)$ is obtained by swapping the role of e and o: numerator o (O-count), denominator $e+o$ (total length).

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+(1-\theta)/\theta} = \theta,$$

which is less than $\frac{1}{2}$ since $\theta < \frac{1}{2}$. Consequently the OO-avoidance result of Section 12 applies: the mediant word for θ' is OO-free, and its O-density converges to θ.

Sturmian identity. The Sturmian word with slope $\alpha' = \theta$ is the infinite mediant word produced by the mediant tree for θ' (Theorem 12.6). This Sturmian word encodes θ directly: the O-density of any long prefix approximates θ with the same precision as the underlying mediant approximation of θ'.

13.2 Case $\frac{1}{2} < \theta < 1$

For $\theta \in (\frac{1}{2}, 1)$, define $\theta' = \theta/(1-\theta) > 1$ and set

$$r(\theta) = \frac{e}{e+o},$$

where $e/o = r(\theta')$.

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+\theta/(1-\theta)} = 1-\theta < \tfrac{1}{2},$$

so it is OO-free and its Sturmian slope is $1-\theta$. The sequence encodes θ'; the rational approximant $e/(e+o)$ to θ is read off from the E- and O-counts of that sequence.

13.3 Case $\theta \in \mathbb{Z}$

For integer $\theta = n$, the seed itself is the answer: $r_0 = r_1 = n/1$ and depth 0 is an exact step (as noted in Section 10.1). No algorithm iterations are needed; the mediant word is $\mathsf{E}^n\mathsf{O}$. (The reductions above may produce an integer θ'; this subsection covers those cases too.)

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13.4 Case $\theta < 0$

For $\theta < 0$, set

$$r(\theta) = -r(-\theta).$$

13.5 Degenerate cases $\theta \in \{0, \frac{1}{2}, 1\}$

These three values sit at the boundary between regimes and are defined directly:

$$r(0) = 0/1, \qquad r(\tfrac{1}{2}) = 1/2, \qquad r(1) = 1/1.$$

The corresponding mediant words are E^∞, $(\mathsf{EO})^\infty$, and $(\mathsf{EO})^\infty$ respectively (at $\theta = 1$ the O-density is $1/2$, consistent with the $\theta \to 1^+$ limit of $1/(1+\theta) \to 1/2$). Note the one-sided limit: as $\theta \to 1^-$ in the $(\frac{1}{2}, 1)$ regime, the Sturmian slope is $1-\theta \to 0$, so the words tend to E^∞, not $(\mathsf{EO})^\infty$.

Summary. Table 1 collects the reductions for all ranges of θ.

Figures 5 and 6 plot the reduction map and the resulting Sturmian slope across all regimes.

14 Conclusion and Open Problems

We have presented a theory of rational approximation for real numbers based on bounded pairs and the weighted mediant construction. The core algorithm applies to $\theta > 1$; all other real numbers are handled by algebraic reduction (Section 13).

Table 1: Reductions to the $\theta > 1$ algorithm for all ranges of θ

Range of θ	Reduction θ'	Approximant to θ	O-density
$\theta > 1$	$\theta' = \theta$	e/o	$1/(1+\theta) < \frac{1}{2}$
$0 < \theta < \frac{1}{2}$	$\theta' = (1-\theta)/\theta$	$o/(e+o)$	$\theta < \frac{1}{2}$
$\frac{1}{2} < \theta < 1$	$\theta' = \theta/(1-\theta)$	$e/(e+o)$	$1-\theta < \frac{1}{2}$
$\theta < 0$	$\theta' = -\theta$ (or above)	negate	—
$\theta \in \mathbb{Z}$	exact seed: $r_0 = r_1 = \theta/1$	$\theta/1$	$1/(1+\theta)$
$\theta = 0$	trivial: word E^∞	$0/1$	0
$\theta = \frac{1}{2}$	trivial: word $(\mathsf{EO})^\infty$	$1/2$	$\frac{1}{2}$
$\theta = 1$	trivial: word $(\mathsf{EO})^\infty$	$1/1$	$\frac{1}{2}$

What is new. The weighted mediant step is the classical Stern–Brocot step. The first novelty is Theorem 8.1 (Section 8): the unique optimal weight w^+ is identified via a sign-change bracket on the linear error function, giving a closed-form formula. That each iterate (except, for rational θ, the final companion r^+ of the exact step) lies on the Stern–Brocot path of θ follows from the weighted-mediant Farey structure (Theorem 11.3).

At the next logical level, the denominator growth lemmas (Lemma 11.14 and Lemma 11.9, Section 11) establish that denominators grow geometrically, bounding the depth by $O(\log N)$. Lemma 10.3 (every mediant word is a lower Christoffel word, Section 10) follows from the standard Christoffel concatenation theorem (Berstel et al. [5]); the contribution is the identification that the mediant algorithm *naturally* produces lower Christoffel words, without any explicit reference to the Christoffel tree.

Building on these, Lemma 8.4 (Section 8) gives the exact characterisation of the unique rational terminal depth in eight parts; it is the analytic pivot on which the terminal invariants rest.

The terminal invariants follow: the Sturmian identity (Theorem 12.6), the palindrome invariant (Theorem 12.5), the MT-never-skips property (Theorem 11.4), and the semi-convergent characterisations (Theorems 11.8 and 11.6). The Sturmian identity and palindrome are consequences of the Christoffel word theory (Lothaire [3], Berstel et al. [5]); the contribution is a direct proof via the mediant construction, together with the last-letter criterion (Theorems 12.1 and 12.2) which is new. The mediant tree taken as a whole is a *compressed transducer*: it encodes the construction history of a length-N Sturmian prefix in $O(\log N)$ iterations (Theorem 11.15), each requiring $O(1)$ oracle evaluations, after which the symbols are enumerated by a purely integer recursive traversal.

Open problems. The following items are genuinely open.

1. **Characterisation of the MT fractions.** Every MT fraction is a convergent or semi-convergent of θ (except, for rational θ, the final companion r^+ of the exact step, which lies on the same side of θ as r_{2d+1} and is not a convergent or semi-convergent); the MT visits the subsequence of those whose consecutive mediants straddle θ. Characterise this subsequence in terms of the continued-fraction expansion of θ, and determine when an MT fraction equals a CF convergent.

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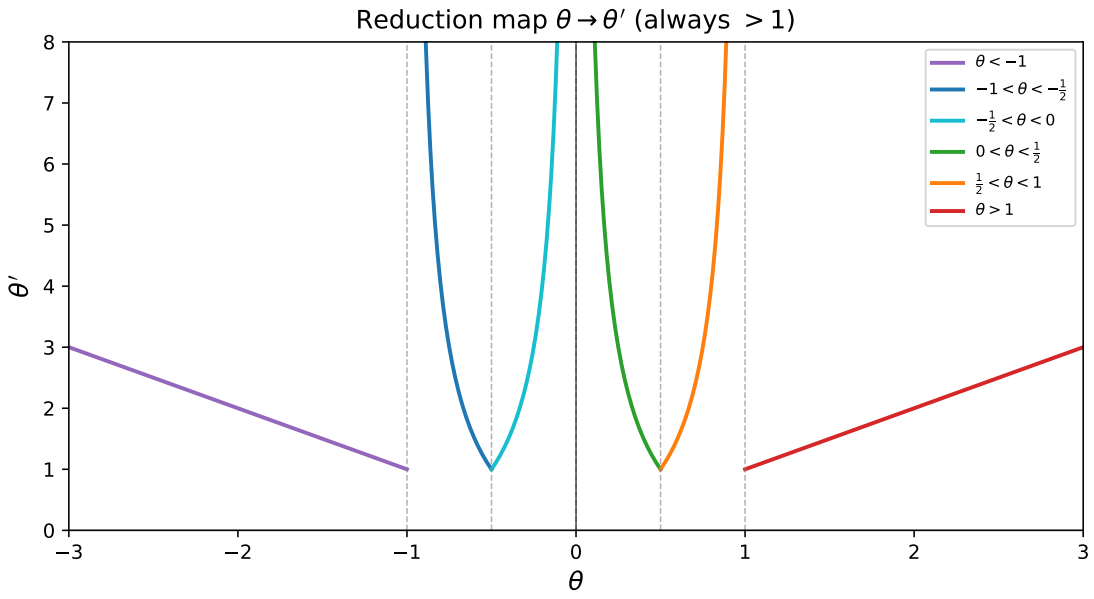


Figure 5: The reduction map $\theta \to \theta'$, which is always greater than 1. Colours identify the regimes; vertical dashed lines mark the boundary values $\theta \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$.

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2. **Cost comparison with CF.** The CF algorithm produces convergents in $O(\log N)$ arithmetic steps; the MT does likewise but visits different fractions. Give a precise comparison of the approximation quality per step between the two algorithms.

3. **Tighter depth-bound constants.** Theorem 11.15 proves $d \leq (2/\log_2 3)\log_2(e + o) + 2$ with universal constants. The exact value of $m^* = \sup_\theta m^*_\theta \leq 2/\log_2 3$ (Remark 11.16) is unknown; in particular ϕ has been shown not to attain the supremum.

The following item has a proof that was deferred from this paper.

4. **Two-step growth conjecture.** Conjecture 11.17 proposes the tighter bound $d \leq (2/\log_2(10/3))\log_2(e + o) + 3$ (slope ≈ 1.151). A proof exists but its inclusion would substantially increase the length of this paper; it is deferred to a future work.

References

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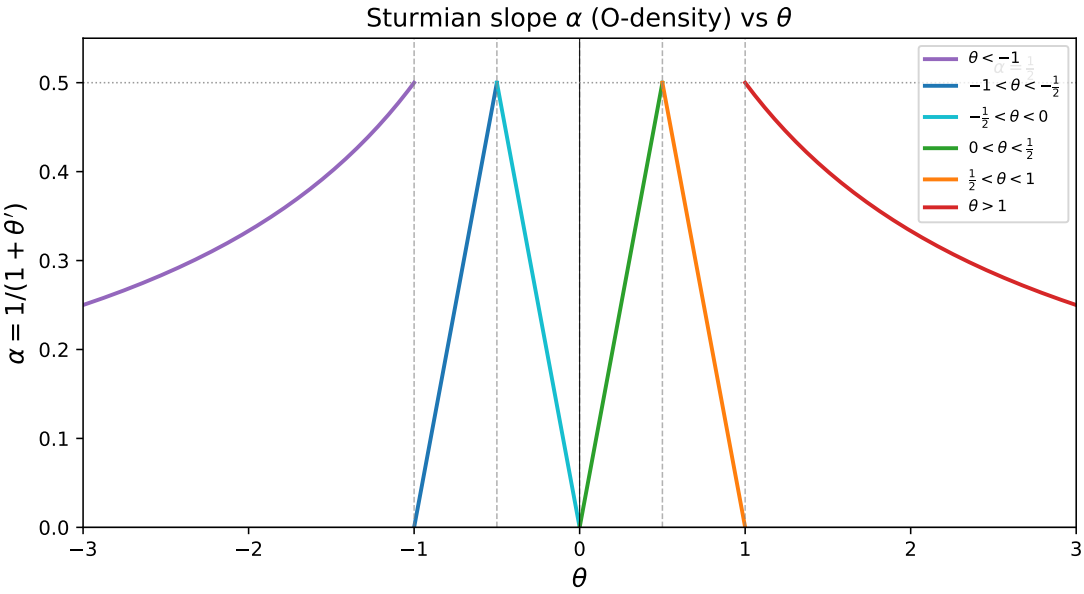


Figure 6: The Sturmian slope $\alpha = 1/(1+\theta')$ (the O-density of the mediant word), which is always at most $\frac{1}{2}$ (equalling $\frac{1}{2}$ at $\theta = 1$). Colours and boundary lines as in Figure 5.

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[4] S. Widmer, *Topics in Word Complexity*, PhD thesis, Université Claude Bernard Lyon 1, 2010. HAL: `tel-00812583`.

[5] J. Berstel, A. Lauve, C. Reutenauer, and F. Saliola, *Combinatorics on Words: Christoffel Words and Repetitions in Words*, CRM Monograph Series **27**, American Mathematical Society, Providence, RI, 2009.

[6] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, fifth edition, Oxford University Press, 1979. (See §11.10: Hurwitz's theorem and the role of ϕ as the hardest irrational to approximate by rationals.)

[7] J. Seymour, *Mediants and Collatz: Constructing Generalised Collatz Trajectories via Rational Approximation*, `https://github.com/wildducktheories/collatz`, Paper 77 (2026).

A Examples and Computational Results

A.1 Worked example: mediant words for $\log_2 3$

Example A.1 (Mediant words and trees for $\log_2 3$)**.** Starting from the initial bounded pair $(r_0, r_1) = (1/1,\ 2/1)$ with $\theta = \log_2 3 \approx 1.58496$, the algorithm produces the following sequence of bounded pairs. At each depth d, propagated weight w_d^+ is used to compute the two candidates $r^- = \text{Mediant}(r_{2d}, r_{2d+1}, w_d^+ - 1)$ and $r^+ = \text{Mediant}(r_{2d}, r_{2d+1}, w_d^+)$; the better estimate (smaller $|\varepsilon|$) becomes r_{2d+3}.

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w_d^+	r^-	r^+
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53

The leaf words are $\pi(r_0) = \texttt{EO}$ and $\pi(r_1) = \texttt{EEO}$. Reading the word decomposition bottom-up:

Level 1 ($d = 0$, $w_0^+ = 2$, $\Delta > 0$, $\pi(r_3) = \pi(r_1)^2 \cdot \pi(r_0)$, $\pi(r_2) = \pi(r_1)^1 \cdot \pi(r_0)$):

$$\underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_3)=\texttt{EEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_2)=\texttt{EEOEO}}$$

Level 2 ($d = 1$, $w_1^+ = 1$, $\Delta > 0$, $\pi(r_5) = \pi(r_3)^1 \cdot \pi(r_2)$, $\pi(r_4) = \pi(r_2)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEO}}_{\pi(r_3)}\,\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_5)=\texttt{EEOEEOEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_4)=\texttt{EEOEO}}$$

Level 3 ($d = 2$, $w_2^+ = 3$, $\Delta > 0$, $\pi(r_7) = \pi(r_5)^2 \cdot \pi(r_4)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\,\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\,\underbrace{\texttt{EEOEO}}_{\pi(r_4)}}_{\pi(r_7)}$$

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The same estimates as mediant trees (depths $d = 1$ and $d = 2$):

$$\begin{aligned}
r_2 &= 3/2 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 1),\\
r_3 &= 5/3 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 2),\\
r_4 &= 3/2 = \text{Mediant}(\text{Mediant}(1,\,2,\,1),\ \text{Mediant}(1,\,2,\,2),\ w = 0),\\
r_5 &= 8/5 = \text{Mediant}(\text{Mediant}(1,\,2,\,1),\ \text{Mediant}(1,\,2,\,2),\ w = 1).
\end{aligned}$$

A weight of $w = 0$ means the left subtree is used directly (contributing zero copies of the right subtree).

A.2 Approximating $\log_2 3$

Table 2 shows 22 steps of the algorithm for $\theta = \log_2 3$, starting from $r_0 = 1/1$, $r_1 = 2/1$. Note the large weight $w = 24$ at depth $d = 6$ and $w = 55$ at depth $d = 10$, corresponding to large partial quotients of the continued fraction of $\log_2 3$, which produce large jumps in precision.

Table 2: Successive bounded pairs approximating $\log_2 3 \approx 1.58496\ldots$

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w^+	r^-	r^+
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53
4	65/41	84/53	$+2.54\times10^{-4}$	-3.59×10^{-5}	6	485/306	569/359
5	485/306	569/359	$+3.04\times10^{-6}$	-2.70×10^{-6}	1	485/306	1054/665
6	485/306	1054/665	$+3.04\times10^{-6}$	-5.98×10^{-8}	24	24727/15601	25781/16266
7	25781/16266	24727/15601	-1.42×10^{-9}	$+1.06\times10^{-9}$	2	50508/31867	75235/47468
8	75235/47468	50508/31867	$+2.10\times10^{-10}$	-2.08×10^{-10}	2	125743/79335	176251/111202
9	125743/79335	176251/111202	$+4.20\times10^{-11}$	-2.95×10^{-11}	2	301994/190537	478245/301739
10	478245/301739	301994/190537	-1.07×10^{-11}	$+3.08\times10^{-13}$	55	16785921/10590737	17087915/10781274
11	16785921/10590737	17087915/10781274	-4.50×10^{-15}	$+1.03\times10^{-15}$	5	85137581/53715833	102225496/64497107
12	102225496/64497107	85137581/53715833	$+1.23\times10^{-16}$	-5.88×10^{-17}	3	272500658/171928773	357638239/225644606
13	272500658/171928773	357638239/225644606	$+9.47\times10^{-18}$	-6.79×10^{-18}	2	630138897/397573379	987777136/623217985
14	987777136/623217985	630138897/397573379	-2.30×10^{-18}	$+2.42\times10^{-19}$	15	9809721694/6189245291	10439860591/6586818670
15	9809721694/6189245291	10439860591/6586818670	-1.41×10^{-20}	$+1.40\times10^{-21}$	10	103768467013/65470613321	114208327604/72057431991
16	114208327604/72057431991	103768467013/65470613321	$+6.96\times10^{-23}$	-6.41×10^{-23}	2	217976794617/137528045312	321745261630/202998658633
17	321745261630/202998658633	217976794617/137528045312	-1.67×10^{-23}	$+5.95\times10^{-24}$	5	1193652440098/753110839881	1411629234715/890638885193
18	1411629234715/890638885193	1193652440098/753110839881	$+7.97\times10^{-25}$	-1.44×10^{-25}	7	8573543875303/5409303924479	9767196315401/6162414764360
19	8573543875303/5409303924479	9767196315401/6162414764360	$+1.11\times10^{-26}$	-7.83×10^{-27}	2	18340740190704/11571718688839	28107936506105/17734133453199
20	28107936506105/17734133453199	18340740190704/11571718688839	-2.06×10^{-27}	$+1.01\times10^{-27}$	4	83130157078217/52449289519716	101470897268921/64021008208555
21	101470897268921/64021008208555	83130157078217/52449289519716	$+1.63\times10^{-28}$	-2.51×10^{-29}	8	683381996816440/431166034846567	766512153894657/483615324366283

A.3 Approximating π

Table 3 shows 22 steps for $\theta = \pi$, starting from $r_0 = 4/1$, $r_1 = 3/1$. The classical approximant $355/113$ is produced from the depth-1 pair and appears as r_5, the better member of the depth-2 pair; it has $\varepsilon \approx +8.49\times10^{-8}$, and the weight $w = 293$ at depth $d = 2$ produces an immediate jump of roughly three orders of magnitude (from $|\varepsilon| \approx 8.49\times10^{-8}$ to $|\varepsilon| \approx 1.06\times10^{-10}$).

Table 3: Successive bounded pairs approximating $\pi \approx 3.14159\ldots$

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w^+	r^-	r^+
0	4/1	3/1	$+0.273$	-0.045	7	22/7	25/8
1	25/8	22/7	-5.28×10^{-3}	$+4.02\times10^{-4}$	15	333/106	355/113
2	333/106	355/113	-2.65×10^{-5}	$+8.49\times10^{-8}$	293	103993/33102	104348/33215
3	103993/33102	104348/33215	-1.84×10^{-10}	$+1.06\times10^{-10}$	2	208341/66317	312689/99532
4	208341/66317	312689/99532	-3.89×10^{-11}	$+9.28\times10^{-12}$	3	833719/265381	1146408/364913
5	833719/265381	1146408/364913	-2.77×10^{-12}	$+5.13\times10^{-13}$	4	4272943/1360120	5419351/1725033
6	4272943/1360120	5419351/1725033	-1.29×10^{-13}	$+7.05\times10^{-15}$	15	80143857/25510582	85563208/27235615
7	85563208/27235615	80143857/25510582	$+2.74\times10^{-16}$	-1.84×10^{-16}	2	165707065/52746197	245850922/78256779
8	165707065/52746197	245850922/78256779	$+5.22\times10^{-17}$	-2.49×10^{-17}	2	411557987/131002976	657408909/209259755
9	411557987/131002976	657408909/209259755	$+6.16\times10^{-18}$	-5.45×10^{-18}	1	411557987/131002976	1068966896/340262731
10	411557987/131002976	1068966896/340262731	$+6.16\times10^{-18}$	-9.77×10^{-19}	3	2549491779/811528438	3618458675/1151791169
11	2549491779/811528438	3618458675/1151791169	$+1.76\times10^{-19}$	-1.65×10^{-19}	1	2549491779/811528438	6167950454/1963319607
12	2549491779/811528438	6167950454/1963319607	$+1.76\times10^{-19}$	-2.43×10^{-20}	3	14885392687/4738167652	21053343141/6701487259
13	14885392687/4738167652	21053343141/6701487259	$+9.94\times10^{-21}$	-8.33×10^{-23}	85	1783366216531/567663097408	1804419559672/574364584667
14	1804419559672/574364584667	1783366216531/567663097408	-5.85×10^{-25}	$+3.91\times10^{-25}$	2	3587785776203/1142027682075	5371151992734/1709690779483
15	3587785776203/1142027682075	5371151992734/1709690779483	-1.00×10^{-25}	$+6.28\times10^{-26}$	2	8958937768937/2851718461558	14330089761671/4561409241041
16	14330089761671/4561409241041	8958937768937/2851718461558	$+2.20\times10^{-26}$	-2.46×10^{-27}	15	139755218526789/44485467702853	148714156295726/47337186164411
17	148714156295726/47337186164411	139755218526789/44485467702853	-9.99×10^{-29}	$+5.13\times10^{-29}$	3	428224593349304/136308121570117	567979811876093/180793589272970
18	567979811876093/180793589272970	428224593349304/136308121570117	$+1.17\times10^{-29}$	-1.21×10^{-30}	13	5706674932067741/1816491048114374	6134899525417045/1952799169684491
19	5706674932067741/1816491048114374	6134899525417045/1952799169684491	$+7.43\times10^{-32}$	-1.55×10^{-32}	5	30246273033735921/9627687726852338	36381172559152966/11580486896536829
20	30246273033735921/9627687726852338	36381172559152966/11580486896536829	$+1.45\times10^{-33}$	-1.40×10^{-33}	1	30246273033735921/9627687726852338	66627445592888887/21208174623389167
21	30246273033735921/9627687726852338	66627445592888887/21208174623389167	$+1.45\times10^{-33}$	-1.07×10^{-34}	7	430010946591069243/136876735467187340	496638392183958130/158084910090576507

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Disclosures

Use of Artificial Intelligence

Large language models played a substantial role in the development of this paper, and the author considers it important to describe that role honestly.

Review and gap identification. Claude (Anthropic) and Gemini (Google) were used as critical reviewers across multiple structured review rounds. Each model was presented with the manuscript and asked to identify mathematical gaps, proof obligations, notational

inconsistencies, and expository weaknesses. The resulting issue lists directly drove successive revisions and are documented in the paper's revision history.

Theorem formulation. The boundary between the author identifying a result and an AI model identifying it is not always sharp. In several instances the need for a theorem or lemma was recognised during AI-assisted review rather than by the author independently. The author regards this as a genuine intellectual contribution by the models involved, not merely automated checking.

Proof development. On several occasions, Aristotle (the code name used for the AI reviewer in this project's review workflow) did not merely flag a conjecture as unproven but supplied the proof that elevated it to a theorem. Theorem 11.6 and Theorem 11.8 are examples where the final rigorous proofs emerged from this collaborative process. Claude also drafted proofs and theorem statements at various points in the development; in some cases these drafts were adopted with revision, in others they were rejected or replaced.

Exposition. Claude was used to assist with drafting, restructuring, and checking revised passages of mathematical exposition. No AI-generated text was incorporated without review and revision by the author.

Responsibility. The overall research programme, the framing of results, all decisions about what to accept, defer, or reject, and final responsibility for the correctness of the paper rest with the author. The author does not regard AI assistance as diminishing the paper's validity, but believes readers are entitled to a complete account of how it was produced.

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Lean 4 Verification Status

During peer review, a reviewer produced an independent Lean 4 formalisation of selected results from this paper, including the exponential denominator growth theorem and the weighted-mediant identity. The author is grateful for this contribution, which helped identify and correct several proof gaps during the review process.

The author has chosen not to publish the Lean 4 development at this stage. Accordingly, **this paper makes no claim of Lean 4 certification or machine-checked verification**. Every theorem, lemma, and proof in this paper stands solely on its hand-argued proof as written in the text. Readers should not infer formal verification status from any part of this manuscript. A complete Lean 4 formalisation of the results in this paper is left for future work.

Reviewer's Comments

The following items identify open proof obligations, gaps, and vulnerabilities in this manuscript. Each item records its source, the manuscript version in which it was raised, and its current status. **Raised** items are newly identified and unaddressed. **Pending** items are scheduled for the next revision. **Deferred** items have been noted but action intentionally postponed. Reviewers should not expect Deferred items to be actioned in the next revision; raising new comments solely about the content of Deferred items is unproductive. **Resolved** items are addressed and awaiting reviewer confirmation. **Stale** items were resolved in an earlier version whose fix was subsequently superseded; the body records the history but the item is

subsumed by a later entry and need not be reviewed independently. **Rejected** items have had their proposed resolution rejected and must return to **Pending** or **Raised** before the next review. **Dismissed** items are those for which the author has determined no further action is warranted. **Closed** items are fully accepted by all parties. Only **Dismissed** and **Closed** items are candidates for removal from future versions.

Versioning convention. The *since* field records the exact distributed version in which an issue was first present. The *version* field on a Fix or Rejected entry records the exact distributed version in which that fix or rejection was first included. Version strings match exactly the tag used in the manuscript's date line at the time of distribution: `v0.18` is correct when no patch suffix was assigned; `v0.18` would be incorrect if the distributed version was `v0.18.1`.

Guidance for reviewers. Reviewers are invited to comment on any aspect of the manuscript's mathematical content and exposition: new issues, concerns about existing fixes, or verdicts on **Resolved** items. Reviewers should not attempt to micromanage the correctness of the issue management process itself (versioning labels, status transitions, preamble wording, etc.); such concerns are the author's responsibility and comments about them are unproductive.

ID	Status	Title
J-9	Deferred	Review implications of using a finite-precision oracle with the mediant algorithm
J-8	Deferred	Section 10 conjectures and lemma: irrationality hypotheses may be too strong
J-7	Deferred	Section 9 prose: weaken "irrational" to "non-integer" at l. 984
J-6	Deferred	Deploy age/height/fit/alpha primitives in theorem and lemma statements
J-4	Deferred	Reconsider placement and title of Section 12 (Sturmian sequences)
J-2	Resolved	Prevent Section 13 diagrams from floating beyond Section 13
J-11	Resolved	Move Thm 8.1 (Optimal weight formula) from Appendix A back into Section 8
J-1	Resolved	Remove Chapter 11 examples; relocate forward reference to appendix
A-127	Resolved	Thm 10.1 parenthetical cites Lem. 8.4(5) for $\lvert\varepsilon^-\rvert = 0$; should cite (3)
A-126	Resolved	Seed tie-break sentence reuses $r^{\pm}$ for lower/upper leaf, clashing with weight-indexed usage
A-125	Resolved	Thm 10.4 identity: ticket calls it both `conj:semi-r0` and "MT never skips a convergent"
A-124	Resolved	$r(\theta') = \theta'/1$ for integer θ' is a definitional extension, not an algorithm output
A-123	Resolved	Table 13 boundary rows: $k \geq 1$ includes $k = 1$ giving $\theta = 1/2$, $\theta' = 1 \notin \mathbb{Z}_{\geq 2}$
A-122	Resolved	Thm 10.6 exclusion note and §6 sentence: only r_{2d+3} makes exact depths convention-dependent — false
A-121	Resolved	Lem 10.9 Case B uses δ before defining it (expository)
A-120	Closed	Residual terminology and scoping errors (minor): five sites

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ID	Status	Title
A-119	Closed	§14 Open Problem 1 omits the exact-step exception (fifth A-108 site)
A-118	Resolved	§13 reductions can produce integer θ'; excluded by the new $\theta \notin \mathbb{Z}$ assumption
A-117	Closed	"brackets θ" in Thm 7.3 / Cor 7.4 can be read non-strictly, making Cor 7.4 first conclusion false at exact step with $B < 0$
A-116	Closed	Thm 10.4 depends on the CF convention for rational θ; not listed in §6 exception list
A-115	Closed	Tie case $\{\theta\} = 1/2$: "either ordering" makes algorithm output ill-defined; strict-ordering claims fail at depth 0
A-114	Resolved	Thm 10.6 exclusion note and §6 sentence wrong: "$r_{2d+1} = \theta$" false at exact depth; convention direction reversed
A-112	Resolved	Thm 10.6 is CF-convention-dependent when $-A/B = 1$ at the exact step
A-110	Resolved	Seed is degenerate for $\theta \in \mathbb{Z}$ (leaves coincide) and $\{\theta\} = 1/2$ (equal errors)
A-103	Closed	Terminology: "non-terminal step" ($\Theta \neq 0$) includes the terminal-producing Case 2 step
A-43	Deferred	the tightness conjecture (v0.17) is false: $m^*_\theta \leq 2/\log_2(10/3)$ holds
RC-22	Deferred	Conj. 10.10 (bound via e) refuted; replaced by Remark 11.16
RC-16	Deferred	Inter-paper circular coupling: Paper 76 errors propagate to Paper 77
RC-15	Deferred	High notation density may impede referee comprehension
RC-14	Deferred	Base-specific structural dependencies limit generality
RC-13	Deferred	Weight propagation bound slackness uncharacterised
N17	Deferred	Ticket uses "Thm 10.8" and "Figure 10.1" but PDF has Thm 10.7 / Fig. 2

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J-9 (v0.19). Review implications of using a finite-precision oracle with the mediant algorithm. *Source: jon — Since: v0.19 — Status: Deferred*

The mediant algorithm is presented for an exact real $\theta > 1$, but any practical implementation has access only to a finite-precision approximation of θ — a rational oracle. This raises several questions:

- **Correctness under rounding.** If the oracle returns $\tilde{\theta} = \theta + \delta$ for some bounded error $|\delta| < \epsilon$, under what conditions does the algorithm still produce the correct bounded pair and bracketing weight? When can rounding cause the algorithm to select the wrong child or the wrong weight?
- **Termination.** For a rational oracle the algorithm may terminate early (the oracle's θ is exactly representable as a mediant before the true θ is). How does this interact with the convergence guarantees of Section 10?

- **Error propagation.** The invariants Δ and Θ are computed from θ. How do finite-precision errors in θ propagate into these invariants, and do the sign conditions $\operatorname{sgn}(\Theta) = \operatorname{sgn}(\Delta)$ remain reliable?

- **Relationship to irrationality assumptions.** Several theorems assume θ irrational (Section 6, "Rational vs. irrational θ"). A finite-precision oracle is always rational; the practical algorithm therefore operates in exactly the regime where these assumptions fail. The paper should acknowledge this and characterise how the algorithm degrades gracefully.

To be addressed in a future revision, potentially as a dedicated subsection in Section 13 (Extension) or Section 14 (Conclusion and Open Problems).

J-8 (v0.19). Section 10 conjectures and lemma: irrationality hypotheses may be too strong. *Source: jon — Since: v0.19 — Status: Deferred*

Three sites in Section 10 (Convergence Analysis) use irrationality where a weaker condition may suffice:

(1) Proof of Lemma 11.9, Case B: "Since θ is irrational, $|A_d|/|B_d| \notin \mathbb{Z}$". The conclusion requires only that A_d/B_d is non-integer, which follows from irrationality of θ but may hold more broadly. The weakest sufficient condition should be identified and stated.

(2) Conjecture 11.15 (linear depth bound): hypothesis "Let $\theta > 1$ be irrational". The conjecture may hold for all real $\theta > 1$; the irrationality restriction should be reviewed once a proof is available.

(3) Conjecture 11.17 (two-step growth): hypothesis "For every irrational $\theta > 1$". Same as (2).

To be revisited when proofs of the conjectures are attempted.

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J-7 (v0.19). Section 9 prose: weaken "irrational" to "non-integer" at l. 984. *Source: jon — Since: v0.19 — Status: Deferred*

Section 9 (Mediant Words and the Mediant Tree) contains the running-prose claim "Since θ is irrational, $\lceil\theta\rceil = \lfloor\theta\rfloor + 1$". The conclusion $\lceil\theta\rceil = \lfloor\theta\rfloor + 1$ requires only that $\theta \notin \mathbb{Z}$ (i.e. θ is non-integer), not full irrationality. The hypothesis should be weakened to "Since θ is non-integer…", consistent with the principle that irrationality should only be assumed where strictly necessary. To be fixed in a future revision.

J-6 (v0.19). Deploy age/height/fit/alpha primitives in theorem and lemma statements. *Source: jon — Since: v0.19 — Status: Deferred*

The ordering primitives age, height, fit, and alpha (Definitions 6.6–6.9) have been defined in Section 6 but not yet deployed. Many theorems and lemmas currently state ordering hypotheses verbosely or implicitly; these could be tightened using the destructuring idiom $(u, v) = f(r, r')$.

Key candidate sites identified during review:

- `fit`: `thm:mediant-convergence` ($|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$ hypothesis), `cor:children-bounded-pair`, `conj:semi-r1`, `thm:last-sturmian`.

- `age`: `thm:optimal-w`, `thm:convergence`, `lem:no-consecutive-unit-weights`.
- `height`: `lem:pair-sum-invariant`, `thm:last-sturmian`.
- `alpha`: any argument that transfers a result from θ to Sturmian slope $\alpha = 1/(1+\theta)$.

To be assessed in a future revision, potentially after reviewer feedback.

J-4 (v0.19). Reconsider placement and title of Section 12 (Sturmian sequences). *Source: jon — Since: v0.19 — Status: Deferred*

Section 12 (Sturmian sequences) may be better placed earlier in the paper, or its title may need revision to more accurately reflect its content and its relationship to the preceding algorithm sections. The section's position and heading should be reconsidered as part of a structural review. To be addressed in a future minor version.

J-2 (v0.19). Prevent Section 13 diagrams from floating beyond Section 13. *Source: jon — Since: v0.19 — Status: Resolved*

Figures in Section 13 are currently permitted to float beyond the end of Section 13, which disrupts the reading experience. Add appropriate float barriers (e.g. `\FloatBarrier` or section-level placement constraints) to keep all Section 13 diagrams within that section. To be addressed in a future minor version.

Fix (v0.20.1): Added `\usepackage{placeins}` to the preamble and inserted `\FloatBarrier` immediately before the Extension section heading, preventing all Section 13 figures from floating beyond that section boundary.

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J-11 (v0.19.5). Move Thm 8.1 (Optimal weight formula) from Appendix A back into Section 8. *Source: jon — Since: v0.19.5 — Status: Resolved*

Theorem 8.1 (the optimal weight formula) was moved to Appendix A early in the paper's development, at a time when the appendix housed several long proofs. That justification no longer applies: the appendix now contains only the proof of Thm 8.1 itself, and every other major result (Thm 7.3, Cor 8.2, Lem 8.3, Lem 8.4, Thm 10.3, Thm 10.6, etc.) has its proof in the main body. Thm 8.1 is no longer special in length or difficulty relative to its neighbours. Moving the proof back into §8 would remove the forward reference at the theorem statement, eliminate the appendix entirely, and make the paper's structure more uniform. The move is mechanical (cut the appendix proof, paste after the theorem statement in §8, renumber any cross-references to Appendix A), but it will change page numbering throughout and should be done in a dedicated revision.

Fix (v0.20.1): Moved the proof of Thm 8.1 from Appendix A into Section 8 immediately after the theorem statement. Removed Appendix A entirely. Updated the roadmap paragraph in the Introduction to remove the sentence pointing to Appendix A. The `\app:optimal-w` label and its edges were removed from graph.yml.

J-1 (v0.19). Remove Chapter 11 examples; relocate forward reference to appendix. *Source: jon — Since: v0.19 — Status: Resolved*

The examples in Chapter 11 are lengthy and interrupt the flow from the algorithm sections to the Sturmian sections. They should be removed from Chapter 11 and the forward reference they anchor should be relocated to an appropriate place in the appendix. To be addressed in a future minor version.

Fix (v0.20.1): Removed the stub `\section{Examples}` (Section 11) entirely. The section contained only a single sentence forwarding the reader to Appendix B; that forward reference already exists in Section 10 (Convergence Analysis), so no content was lost. Section numbering from the former Section 12 onward shifts by one.

A-127 (v0.20). Thm 10.1 parenthetical cites Lem. 8.4(5) for $|\varepsilon^-| = 0$; should cite (3). *Source: aristotle — Since: v0.20 — Status: Resolved*

Minor. The Thm 10.1 proof parenthetical cites Lemma 8.4(5) for $|\varepsilon^-| = 0$. But part (5) gives $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$; $|\varepsilon^-| = 0$ follows from part (3) ($r^- = \theta$ exactly). Fix: cite Lemma 8.4(3) (or "(3),(5)").

Fix (v0.20.1): Changed citation from Lemma 8.4(5) to (3) in the Thm 10.1 proof parenthetical.

A-126 (v0.20). Seed tie-break sentence reuses $r^\pm$ for lower/upper leaf, clashing with weight-indexed usage. *Source: aristotle — Since: v0.20 — Status: Resolved*

Minor (expository). The seed paragraph tie-break sentence reads "$r_1 = \lfloor\theta\rfloor/1$ (the lower leaf as r^-, the upper as r^+)". Elsewhere r^- and r^+ are the children at weights w^- and w^+. Using them here for the initial leaves conflates two uses. Fix: write "($r_0 = \lceil\theta\rceil/1$, $r_1 = \lfloor\theta\rfloor/1$)" instead.

Fix (v0.20.1): Replaced "(the lower leaf as r^-, the upper as r^+)" with "(so $r_0 = \lceil\theta\rceil/1$)", removing the $r^\pm$ notation from the seed paragraph entirely.

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A-125 (v0.20). Thm 10.4 identity: ticket calls it both `conj:semi-r0` and "MT never skips a convergent". *Source: aristotle — Since: v0.20 — Status: Resolved*

The A-116 fix description in the review ticket calls Thm 10.4 `conj:semi-r0` ("Semi-convergent characterisation of r_{2d}"), while the Q5 answer refers to "MT never skips a convergent" (which is Thm 10.3, label `thm:sb-path` or similar). These are different theorems. Verify that the irrationality hypothesis "Assume θ is irrational" was added to the intended theorem, and that the §6 exception list names the right labels. Needs PDF to resolve.

Fix (v0.20.1): Confirmed: the irrationality hypothesis was on conj:semi-r0 (semi-convergent characterisation of r_{2d}), not on conj:never-skips ("MT never skips a convergent"). Both theorems are convention-dependent for rational θ for the same reason (SB-path reasoning, A-112/A-122). Added "Assume θ is irrational" to conj:never-skips. Updated the §6 exception list first bullet to name both conj:never-skips and conj:semi-r0 explicitly, and the terminal-depth bullet to note that both are restricted to irrational θ. Updated graph.yml accordingly.

A-124 (v0.20). $r(\theta') = \theta'/1$ for integer θ' is a definitional extension, not an algorithm output. *Source: aristotle — Since: v0.20 — Status: Resolved*

The new §13 boundary subsection says "$r(\theta') = \theta'/1$ directly (the algorithm is not needed)". But for integer θ' the seed has $B = \varepsilon_{r_1} = 0$, so Step 4's formula $w^+ = \lfloor -A/B \rfloor + 1$ is undefined (division by zero). The algorithm cannot produce $\theta'/1$; it must be stated as a definitional extension of r (as the A-110 history suggests), not as a consequence of the algorithm not being needed. Lean-verified: `init_natCast` gives $B = 0$ for every integer seed. Fix: restate as "we extend the definition of r to integer inputs by setting $r(\theta') = \theta'/1$ directly".

Fix (v0.20.1): The boundary subsection and its "algorithm is not needed" claim were removed entirely (joint fix with A-110, A-118, A-123). Integer θ is now handled by a subsection that appeals directly to the seed construction (Section 10.1): $r_0 = r_1 = n/1$, depth 0 exact. No claim is made about algorithm output for $B = 0$.

A-123 (v0.20). Table 13 boundary rows: $k \geq 1$ includes $k = 1$ giving $\theta = 1/2$, $\theta' = 1 \notin \mathbb{Z}_{\geq 2}$. *Source: aristotle — Since: v0.20 — Status: Resolved*

The two new rows added for the integer-boundary case say $k \geq 1$. But $k = 1$ gives $\theta = 1/(1+1) = 1/2$ (resp. $\theta = 1/(1+1) = 1/2$), which lies in neither open interval $(0, 1/2)$ nor $(1/2, 1)$, and the reduction gives $\theta' = 1 \notin \mathbb{Z}_{\geq 2}$. Lean-verified: `half_reduces_to_one`. Fix: change $k \geq 1$ to $k \geq 2$ in both rows, or handle $\theta = 1/2$ (and $\theta = 1$) separately.

Fix (v0.20.1): Removed both boundary rows from Table 13 entirely (joint fix with A-110, A-118, A-124). Integer θ is now handled by the exact-seed subsection; $\theta = 1/2$ and $\theta = 1$ remain in the degenerate-cases subsection.

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A-122 (v0.20). Thm 10.6 exclusion note and §6 sentence: only r_{2d+3} makes exact depths convention-dependent — false. *Source: aristotle — Since: v0.20 — Status: Resolved*

The rewritten Thm 10.6 exclusion note and the §6 exception sentence claim that convention-dependence at an exact depth comes "entirely" from $r_{2d+3} = \theta$, and that r_{2d+1} is "a generic output". This is false. When the exact step has $w^- = 1$, r_{2d+1} is a convergent in one CF expansion and only a semi-convergent in the other. Lean-verified counterexample: $\theta = 16/7 = [2; 3, 2] = [2; 3, 1, 1]$, depth 1 exact with $c = w^- = 1$. Here $r_3 = 9/4$ is a convergent of $[2; 3, 1, 1]$ but only a semi-convergent of $[2; 3, 2]$ (theorem `ex167_cf`). This affects both the Thm 10.6 exclusion note and the §6 exception sentence (A-116 exception wording may also need updating). Fix: state that both r_{2d+1} (when $w^- = 1$) and $r_{2d+3} = \theta$ depend on the CF convention at an exact depth.

Fix (v0.20.1): Rewrote the Thm 10.5 exclusion note and the §6 exception sentence using SB-path reasoning: r_{2d+1} is one of the two SB parents of θ at an exact depth, and which parent is the "last same-side node" (convergent) depends on the CF expansion. This covers all convention-dependent fractions without naming them individually. Joint fix with A-112 and A-114.

A-121 (v0.19.5). Lem 10.9 Case B uses δ before defining it (expository). *Source: aristotle — Since: v0.19.5 — Status: Resolved*

Nit. In the proof of Lem. 10.9, Case B, the two new branches ($\delta = 0$ and $\delta = 1/2$) are written before the definition "$\delta = \mathrm{frac}(|A_d|/|B_d|)$". The reader encounters the symbol δ

without knowing what it denotes. Fix: define δ at the start of Case B, before the three-way split.

Fix (v0.19.6): Moved the definition "Let $\delta = \mathrm{frac}(|A_d|/|B_d|) \in [0, 1)$ be the fractional part of $|A_d|/|B_d|$" to the start of Case B, before the three-way split on $\delta = 0$, $\delta = 1/2$, and otherwise.

Note (v0.20): Status stays Resolved. Aristotle confirms the range $[0, 1)$ is correct and $\delta = 0$ is the exact-step case. Full close pending PDF verification that δ is defined before every use.

A-120 (v0.19.5). Residual terminology and scoping errors (minor): five sites. *Source: aristotle — Since: v0.19.5 — Status: Closed*

Five minor residues from the v0.19.5 fixes.

1. **Thm. 10.1 proof parenthetical**: says "At the terminal step for rational θ, $r^- = \theta \ldots$". Under the new Step-types terminology this must be "exact step". Also add a citation to Rem. 7.5 or Lem. 8.4(5). (This is the remaining blocker for A-103.)

2. **Thm. 10.3 proof, last sentence**: "Both r^- and r^+ are produced this way, so both lie on the path." Add "at a generic step": at the exact step r^+ is not on the SB path.

3. **§8 "The continuous optimum" subsection**: at the exact step there is no integer in $(c, c+1)$ and $w^* = c + \frac{1}{2}$ is equidistant from w^- and w^+. Qualify the subsection to generic steps.

4. **Def. 9.2 propagated weights**: "the w^-, w^+ of Theorem 8.1 applied to (r_{2d}, r_{2d+1})" — Thm. 8.1 does not apply at an exact depth. Replace with "computed by Step 4 (Theorem 8.1 at generic depths, Lemma 8.4(1) at the exact depth)".

5. **§6 "At the terminal depth" bullet**: it uses the exact-depth indices r_{2d+2}, r_{2d+3} while describing the terminal depth; and it lists Thm. 12.2 as an exception, which it is not (Thm. 12.2 applies at the terminal depth via Part (3)). Either move the Thm. 12.2 remark out of the exception list or label it "(applies)".

Fix (v0.19.6): Applied all five sub-fixes: (1) Thm 10.1 proof parenthetical: "terminal step" changed to "exact step" with Lemma 8.4(5) citation (merged into A-103 fix). (2) Thm 10.3 proof last sentence: added "at a generic step" qualifier and a parenthetical for the exact step. (3) §8 continuous-optimum subsection: added a Remark qualifying the analysis to generic steps and explaining the exact-step case. (4) Def 9.2 propagated weights: replaced "Theorem 8.1 applied to" with "computed by Step 4 (Theorem 8.1 at generic depths; Lemma 8.4(1) at the exact depth)". (5) §6 terminal-depth bullet: verified indices r_{2d+2}, r_{2d+3} are correct for the terminal depth; Thm 12.2 correctly listed as applying (not an exception) in current text.

A-119 (v0.19.5). §14 Open Problem 1 omits the exact-step exception (fifth A-108 site). *Source: aristotle — Since: v0.19.5 — Status: Closed*

Nit. Section 14, Open Problem 1 states: "Every MT fraction is a convergent or semiconvergent of θ." This is a fifth site of the A-108 type: the statement is false for rational

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θ because the final companion r^+ of the exact step is neither a convergent nor a semi-convergent. The four sites fixed in v0.19.5 were §1 Contribution 4, §14 "What is new", Thm. 10.3, and Thm. 10.8. Open Problem 1 was missed. Fix: add "(except, for rational θ, the final companion r^+ of the exact step)" to the Open Problem 1 statement.

Fix (v0.19.6): Added the exact-step exception to §14 Open Problem 1: "(except, for rational θ, the final companion r^+ of the exact step, which lies on the same side of θ as r_{2d+1} and is not a convergent or semi-convergent)".

A-118 (v0.19.5). §13 reductions can produce integer θ'; excluded by the new $\theta \notin \mathbb{Z}$ assumption. *Source: aristotle — Since: v0.19.5 — Status: Resolved*

The seed paragraph (A-110 fix) now assumes $\theta \notin \mathbb{Z}$. Section 13 maps $\theta \in (0, \frac{1}{2})$ to $\theta' = (1-\theta)/\theta$, which is an integer for $\theta = 1/(k+1)$ (e.g. $1/3 \mapsto 2$). It maps $\theta \in (\frac{1}{2}, 1)$ to $\theta' = \theta/(1-\theta)$, which is an integer for $\theta = k/(k+1)$ (e.g. $2/3 \mapsto 2$). Integer inputs $\theta \geq 2$ themselves are also not covered by any row of Table 13. This contradicts the claims "all other real numbers are handled" (§13 and §14) and "universal encoder" (§1). Lean-verified: $1/3 \mapsto 2$ under the §13 map. Fix: add a row to Table 13 for $\theta' \in \mathbb{Z}_{\geq 2}$, with $r(\theta') = \theta'/1$ and its leaf word; then verify that the encoder handles all rational inputs.

Fix (v0.19.6): Added a new subsection "Boundary: $\theta' \in \mathbb{Z}_{\geq 2}$" in §13 handling $\theta = 1/(k+1)$ and $\theta = k/(k+1)$, and added two rows to Table 13 for these integer-boundary cases.

Rejected (v0.20): Two defects in the new Table 13 rows. (1) Both rows say $k \geq 1$, but $k = 1$ gives $\theta = 1/2$ (resp. $\theta = 1/2$) which reduces to $\theta' = 1 \notin \mathbb{Z}_{\geq 2}$, outside the stated domain. Fix: use $k \geq 2$. See A-123. (2) The subsection says $r(\theta') = \theta'/1$ "directly (the algorithm is not needed)", but for integer θ' the seed has $B = 0$ so Step 4's formula $\lfloor -A/B \rfloor$ is undefined. This must be stated as a definitional extension of r, not an algorithm output. See A-124.

Fix (v0.20.1): Replaced the boundary subsection and its two Table 13 rows with a clean integer-θ subsection: for $\theta = n \in \mathbb{Z}$ the seed is exact ($r_0 = r_1 = n/1$, depth 0 exact), no algorithm iterations are needed, and the mediant word is $\mathsf{E}^n\mathsf{O}$. Joint fix with A-110, A-123, A-124.

A-117 (v0.19.5). "brackets θ" in Thm 7.3 / Cor 7.4 can be read non-strictly, making Cor 7.4 first conclusion false at exact step with $B < 0$. *Source: aristotle — Since: v0.19.5 — Status: Closed*

Theorem 7.3's hypothesis says the input pair "brackets θ". The paper uses "bracket" non-strictly elsewhere: Def. 6.8 uses $u/\theta < 1$, $v/\theta \geq 1$, and Def. 6.4 uses $e_l/o_l < \theta \leq e_u/o_u$. Under that reading, at an exact step with $B < 0$ the pair $(r^-, r^+) = (\theta, r^+)$ *does* bracket θ, and Cor. 7.4's first conclusion $\varepsilon^-\varepsilon^+ < 0$ would be asserted as true — but it is false ($\varepsilon^- = 0$). Rem. 7.5 and the proof assume the strict condition $f(w-1)f(w) < 0$, which is consistent with the intent. Fix: state Thm. 7.3's hypothesis as "strictly brackets θ, i.e. $(\lambda^- - 1)(\lambda^+ - 1) < 0$", matching the §8 preamble's wording.

Fix (v0.19.6): Changed the hypothesis of Thm 7.3 (thm:mediant-convergence) from "brackets θ" to "strictly brackets θ (i.e. $\varepsilon^-\varepsilon^+ < 0$, where $\varepsilon^\pm = e^\pm - \theta\, o^\pm$ are the signed errors of the two candidates)".

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A-116 (v0.19.5). Thm 10.4 depends on the CF convention for rational θ; not listed in §6 exception list. *Source: aristotle — Since: v0.19.5 — Status: Closed*

Theorem 10.4 ("MT never skips a convergent") depends on the continued-fraction convention for rational θ. Def. 2.1 does not fix an expansion, and the Thm. 10.6 note explicitly says the paper does not. Under the expansion whose last partial quotient is 1 the theorem is false. Lean-verified counterexample: $\theta = 5/3 = [1; 1, 1, 1]$ has convergent $3/2$. The run is $(1/1, 2/1) \to (7/4, 5/3)$ (depth 0 exact, $w^- = 2$). The fraction $3/2$ appears in no MT pair. In general, under the $[\ldots, 1]$ expansion, the extra convergent $r_{2d} + (w^- - 1)r_{2d+1}$ is missed whenever the exact step has $w^- \geq 2$. (The general statement is informal; only the $5/3$ instance is machine-checked.) The proof does not treat the exact stage at all. By the proof's own characterisation of "last node in a run before the path crosses", $3/2$ is a convergent but the run never visits it. Fix (two options): (A) Adopt the $a_n \geq 2$ convention in Def. 2.1. Thm. 10.4 then holds; the proof should add the exact stage: p_{n-1}/q_{n-1} is r_{2d+1} if $w^- \geq 2$, or a member of the exact-depth pair if $w^- = 1$, and θ is in the terminal pair. Under this convention the Thm. 10.6 exclusion is genuinely needed (A-114). (B) Restrict Thm. 10.4 to irrational θ and add it to the §6 exception list.

Fix (v0.19.6): Applied option (B): added "Assume θ is irrational" to the hypothesis of Thm 10.4 (conj:semi-r0). Added it to the §6 exception list alongside Thm 10.3.

A-115 (v0.19.5). Tie case $\{\theta\} = 1/2$: "either ordering" makes algorithm output ill-defined; strict-ordering claims fail at depth 0. *Source: aristotle — Since: v0.19.5 — Status: Closed*

When $\theta = n + \frac{1}{2}$, the two leaves have equal $|\varepsilon|$. The seed paragraph says "either ordering may be used", but the two orderings give different runs and different terminal pairs. Lean-verified: for $\theta = 3/2$, seed $(1/1, 2/1)$ gives terminal pair $(5/3, 3/2)$ (not bounded, $B > 0$); seed $(2/1, 1/1)$ gives $(4/3, 3/2)$ (bounded, $B < 0$). The parity words $\pi(r_2)$ differ too. The algorithm's output is therefore not well-defined by the seed paragraph. Several strict-ordering statements also fail at depth 0 in the tie case:

1. The §1 depth-index bullet ("older sibling always the worse estimate").
2. The Index convention ("always orient so $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$").
3. The Def. 6.6 parenthetical.
4. The conclusion $|\varepsilon_{2d+1}| < |\varepsilon_{2d}|$ of Thm. 7.3 (via Rem. 7.5).
5. The hypothesis of Lem. 8.4 ($|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$ strict); Lem. 8.4(5)'s last inequality becomes equality.
6. The base case of Lem. 10.9 ("$|B_0| < |A_0|$"; conclusion $w_0^+ = 2$ still holds).
7. Rem. 10.5: "the better leaf r_1 is always a convergent" fails when $r_1 = (n+1)/1$, which is a semi-convergent of $[n; 2]$.

Fix: adopt the tie-break $r_1 = \lfloor\theta\rfloor/1$ (the lower leaf). This gives $B < 0$ and a bounded terminal pair. Either weaken Lem. 8.4's strict fit-ordering hypothesis to $\geq$ (with part (5)

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reading $|\varepsilon^+| < |\varepsilon_{2d+1}| \le |\varepsilon_{2d}|$), or add a note that depth 0 with $\{\theta\} = 1/2$ satisfies the hypothesis with equality and handle it as a sub-case.

Fix (v0.19.6): Replaced "either ordering may be used" in the seed paragraph with an explicit tie-break: $r_1 = \lfloor\theta\rfloor/1$ (lower leaf as r^-, upper as r^+) when $\{\theta\} = 1/2$.

A-114 (v0.19.5). Thm 10.6 exclusion note and §6 sentence wrong: "$r_{2d+1} = \theta$" false at exact depth; convention direction reversed. *Source: aristotle — Since: v0.19.5 — Status: Resolved*

Two errors in the Thm 10.6 proof note "Exclusion of exact depths" and the matching sentence in the §6 exception list. (1) Both say "$r_{2d+1} = \theta$" at an exact depth. This is false: at an exact depth d, $\theta = r_{2d+3}$ (the younger child), not r_{2d+1}. (2) Both claim that under the "last partial quotient ≥ 2" convention the biconditional holds at exact depths. The opposite is true: under that convention the biconditional *fails* at exact depths with $w^- = 1$. What is convention-dependent is the status of r_{2d+1} (the better estimate at that depth) when $w_d^- = 1$. Then $\theta = r_{2d} \oplus r_{2d+1}$, so r_{2d} and r_{2d+1} are θ's two SB parents. r_{2d+1} is the parent with the larger denominator, i.e. $((a_n - 1)p_{n-1} + p_{n-2})/((a_n - 1)q_{n-1} + q_{n-2})$ for $\theta = [a_0; \ldots, a_n]$ with $a_n \ge 2$; that is a *semi-convergent* under the $a_n \ge 2$ convention. When $w_d^- \ge 2$, $r_{2d+1} = p_{n-1}/q_{n-1}$ is a convergent under both conventions and the biconditional holds. Lean-verified counterexample: $\theta = 7/5 = [1; 2, 2]$, depth 1 exact, $w^- = 1$, $w^+ = 2$. Here $r_3 = 4/3$ is a semi-convergent (not a convergent) of $[1; 2, 2]$, so $w_1^+ = 2$ correctly says "not a semi-convergent" — but the biconditional predicts otherwise. (The general argument above is informal; only the 7/5 instance is machine-checked.) Fix: rewrite the Thm 10.6 proof note: "At an exact depth with $w^- = 1$ and $d \ge 1$, r_{2d+1} is the larger-denominator SB parent of θ, which is a semi-convergent under the $a_n \ge 2$ convention and a convergent under the other; the biconditional is therefore convention-dependent and the exact depth is excluded. When $w^- \ge 2$ no exclusion is needed, but the restriction is retained for uniformity." Make the corresponding correction in §6.

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Fix (v0.19.6): Rewrote the Thm 10.6 exclusion note: corrected "$r_{2d+1} = \theta$" to "$r_{2d+3} = \theta$" (Lemma 8.4(3)) and clarified that the convention-dependence applies to r_{2d+3}, not r_{2d+1}. Corrected matching sentence in §6 exception list.

Rejected (v0.20): Same false justification as A-112: the rewritten note claims convention-dependence comes only from $r_{2d+3} = \theta$, but r_{2d+1} is also convention-dependent when $w^- = 1$. Lean-verified counterexample: $\theta = 16/7$, $r_3 = 9/4$ (theorem `ex167_cf`). See A-122.

Fix (v0.20.1): Joint fix with A-112 and A-122: rewrote the Thm 10.5 exclusion note and the §6 exception sentence to use SB-path reasoning that covers both r_{2d+1} (as a SB parent of θ) and $r_{2d+3} = \theta$ without assigning the convention-dependence to a single fraction.

A-112 (v0.19.4). Thm 10.6 is CF-convention-dependent when $-A/B = 1$ at the exact step. *Source: aristotle — Since: v0.19.4 — Status: Resolved*

Minor. Theorem 10.6 asserts that r_{2d+1} is a semi-convergent of θ if and only if $w_d^+ = 1$. At the exact step with $-A/B = w^- = 1$ (so $w^+ = 2$), the theorem predicts r_{2d+1} is *not* a semi-convergent. Lean-checked: for $\theta = 8/5$ at depth $d = 1$ one has $w^+ = 2$ but $r_3 = 5/3$ is a semi-convergent of $[1; 1, 1, 2]$. The result is convention-dependent: under the other standard expansion $[1; 1, 1, 1, 1]$ (last partial quotient ≥ 2) $5/3$ is not a semi-convergent. Fix (B3-15):

Either fix Def 2.1 to require the last partial quotient ≥ 2 and restrict Thm 10.6 to non-exact depths, or restrict Thm 10.6 to irrational θ.

Fix (v0.19.5): Option A (no Def 2.1 change needed): restricted Thm 10.6 to non-exact depths. Added proof note: at exact depths $w^+ \geq 2$ (Lem. 8.4(1)) and semi-convergent status of r_{2d+1} is CF-convention-dependent; the algorithm naturally produces $w^+ \geq 2$, consistent with last-partial-quotient ≥ 2 convention, but since the paper does not fix a CF convention for rational θ the exact depth is excluded. Added exception to §6 exact-step bullet.

Rejected (v0.19.5): The restriction is correct and necessary, but the justification is wrong in two ways (A-114). (1) The proof note says "$r_{2d+1} = \theta$" at an exact depth; in fact $r_{2d+1} \neq \theta$ ($\theta = r_{2d+3}$). (2) The convention claim is backwards: under the "last partial quotient ≥ 2" convention the biconditional *fails* at exact depths with $w^- = 1$. Counterexample (Lean-verified): $\theta = 7/5 = [1; 2, 2]$, depth 1 exact with $w^- = 1$; $r_3 = 4/3$ is a semi-convergent, but $w_1^+ = 2$ predicts otherwise. Fix: rewrite the proof note to say that at an exact depth with $w^- = 1$ and $d \geq 1$, r_{2d+1} is the larger-denominator SB parent of θ, a semi-convergent under the $a_n \geq 2$ convention, so the biconditional fails and the exact depth is excluded.

Fix (v0.19.6): Rewrote the Thm 10.6 exclusion note (via A-114 fix): corrected the wrong index ($r_{2d+3} = \theta$, not r_{2d+1}) and removed the incorrect convention-direction claim. The new note states that $r_{2d+3} = \theta$ is the last convergent and its semi-convergent status is CF-convention-dependent, justifying the exclusion without making a false claim about which convention validates the biconditional.

Rejected (v0.20): The index correction ($r_{2d+3} = \theta$) is right, but the new justification is false. At an exact depth with $w^- = 1$, r_{2d+1} is also convention-dependent (e.g. $\theta = 16/7$, $r_3 = 9/4$ is a convergent under $[2; 3, 1, 1]$ but only a semi-convergent under $[2; 3, 2]$). The exclusion is correct but the stated reason ("only r_{2d+3} makes the depth convention-dependent") is false. See A-122.

Fix (v0.20.1): Rewrote the Thm 10.5 exclusion note using SB-path reasoning: at an exact depth $\theta = r_{2d} \oplus r_{2d+1}$ (Lem. 8.4(3)), so r_{2d+1} is one of the two Stern–Brocot parents of θ, and which parent is the "last same-side node" (i.e. convergent) depends on the CF expansion. Since the paper does not fix a CF convention for rational θ, the convergent/semi-convergent label of r_{2d+1} is undefined and the exact depth is excluded. Updated the §6 exception sentence to match (joint fix with A-114, A-122).

A-110 (v0.19.4). Seed is degenerate for $\theta \in \mathbb{Z}$ (leaves coincide) and $\{\theta\} = 1/2$ (equal errors). *Source: aristotle — Since: v0.19.4 — Status: Resolved*

Minor. The §9.1 paragraph asserts "since θ is irrational, $\lceil\theta\rceil = \lfloor\theta\rfloor + 1$" inside a theory claimed for all real $\theta > 1$. Two degenerate cases arise. (1) For $\theta \in \mathbb{Z}$ the two leaves coincide. The seed is not a bounded pair ($\Delta = 0$) and the algorithm halts at depth 0 without any exact step. Lean-checked: $\theta = 2$ (theorem `seed_integer_2`). (2) For $\{\theta\} = 1/2$ the leaves have equal $|\varepsilon|$, so "r_1 is the better leaf" is ambiguous. Remark 10.5's dichotomy ($w_0^+ = 1$ or $w_0^+ \geq 2$), the base case of Lemma 10.9 ($|B_0| < |A_0|$), and the $d = 0$ case of Theorem 10.6 (max/min > 1) all fail. Depth 0 is then already an exact step with $-A/B = 1$. Lean-checked: $\theta = 3/2$ (theorem `seed_tie_3_2`). Fix (B3-11): Add "We assume $\theta \notin \mathbb{Z}$" to the seed paragraph; note that $\{\theta\} = 1/2$ makes depth 0 an exact step with $w^- = 1$; adjust Rem. 10.5, Lem. 10.9 base case, and Thm. 10.6 case $d = 0$ accordingly.

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Fix (v0.19.5): Applied B3 prose fixes; paper rebuilt cleanly at 72 pages.

Rejected (v0.19.5): Two defects remain. (1) Tie case ($\{\theta\} = 1/2$, A-115): "either ordering may be used" makes the algorithm's output ill-defined; the two orderings give different terminal pairs ($\theta = 3/2$ verified in Lean). Fix: adopt tie-break $r_1 = \lfloor\theta\rfloor/1$. (2) Integer reductions (A-118): the new $\theta \notin \mathbb{Z}$ assumption leaves the §13 reductions uncovered whenever a reduction produces $\theta' \in \mathbb{Z}$ (e.g. $1/3 \mapsto 2$).

Fix (v0.19.6): Fixed both remaining defects (via A-115 and A-118 fixes): (1) Tie case: replaced "either ordering may be used" with an explicit tie-break $r_1 = \lfloor\theta\rfloor/1$ in the seed paragraph. (2) Integer reductions: added a new subsection in §13 handling $\theta' \in \mathbb{Z}_{\geq 2}$ (the boundary cases $\theta = 1/(k+1)$ and $\theta = k/(k+1)$) and two corresponding rows to Table 13.

Rejected (v0.20): Part 1 (tie-break) is closed via A-115. Part 2 (integer reductions) is rejected: inherits the A-118 defects ($k = 1$ gap and undefined $B = 0$ case). See A-123, A-124.

Fix (v0.20.1): Part 2 resolved via a different approach (joint fix with A-118, A-123, A-124): removed the boundary subsection and its Table 13 rows entirely. Replaced with a one-paragraph §13 subsection noting that for integer $\theta = n$ the seed itself is the answer ($r_0 = r_1 = n/1$, depth 0 exact, mediant word $\mathsf{E}^n\mathsf{O}$), referencing Section 10.1. Added a corresponding row to Table 13.

A-103 (v0.19.3). Terminology: "non-terminal step" ($\Theta \neq 0$) includes the terminal-producing Case 2 step. *Source: aristotle — Since: v0.19.3 — Status: Closed*

Minor. The fixes for A-96 use "non-terminal" to mean $\Theta \neq 0$ (i.e. the next pair exists), but the Case 2 terminal-producing step itself has $\Theta \neq 0$ at the time it is executed. Under the current definition, the Case 2 step *is* a non-terminal step. This creates two contradictions. (1) §1 w^+ definition says "at non-terminal steps, w^+ is the unique integer such that the mediants straddle θ". But at the Case 2 step $\Theta \neq 0$ and no straddling integer exists. (2) The §8 preamble note says Thm. 8.1 "is only invoked at non-terminal steps where $\Theta \neq 0$". But Thm. 8.1 does not apply at the Case 2 step, even though $\Theta \neq 0$ there. Fix: introduce distinct terms. *Terminal-producing step*: the step at which $-A/B \in \mathbb{Z}$ (i.e. Case 2; $\Theta \neq 0$ at entry). *Terminal depth*: the depth at which $\Theta = 0$ (no further children). Use $-A/B \notin \mathbb{Z}$ (or "non-terminal-producing") in §1 and §8 where the intent is to exclude the Case 2 step, and reserve "non-terminal depth" for contexts where the intent is to exclude $\Theta = 0$.

Fix (v0.19.5): Applied B3 prose fixes; paper rebuilt cleanly at 72 pages.

Rejected (v0.19.5): One downstream misuse remains: the proof of Thm. 10.1 still says "At the terminal step" where it means "At the exact step". Fix required: change "terminal step" to "exact step" in the Thm. 10.1 parenthetical and add a citation to Rem. 7.5 / Lem. 8.4(5).

Fix (v0.19.6): Changed "At the terminal step" to "At the exact step" in the Thm. 10.1 proof parenthetical and added citation to Lemma 8.4(5).

Note (v0.20): Closed. Non-blocking observation: the citation added in v0.19.6 cites Lem. 8.4(5); the correct part for $|\varepsilon^-| = 0$ is (3) (filed as A-127).

A-43 (v0.17). the tightness conjecture (v0.17) is false: $m_\theta^* \leq 2/\log_2(10/3)$ holds. *Source: aristotle — Since: v0.17 — Status: Deferred*

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Two steps of the mediant tree multiply the pair sum S_d by at least $10/3$, not just 3. Setting $a = o_{2d}$ (older) and $b = o_{2d+1}$ (younger), the auxiliary invariant "$3b \geq S_d$, and if $w_d^+ = 1$ then also $3a \geq S_d$" holds at every depth. A case split gives $S_{d+2} \geq (10/3)S_d$ in all cases:

- $w_d^+ \geq 2$, $w_{d+1}^+ \geq 2$: $S_{d+2} \geq 2S_{d+1} \geq 4S_d$.
- $w_d^+ \geq 2$, $w_{d+1}^+ = 1$: $S_{d+2} \geq 3a + 4b \geq 3S_d + b \geq (10/3)S_d$.
- $w_d^+ = 1$: $S_{d+2} \geq 2a + 3(a + b) = 3S_d + 2a \geq (11/3)S_d$.

Hence $d \leq \frac{2}{\log_2(10/3)} \log_2(e_d + o_d) + 3$ for every θ, giving $m_\theta^* \leq 2/\log_2(10/3) \approx 1.151 < 2/\log_2 3 \approx 1.262$ for *every* irrational $\theta > 1$. This directly refutes the v0.17 conjecture that $\sup_\theta m_\theta^* = 2/\log_2 3$. Theorem 11.15 and Remark 11.16 must also be updated to the tighter slope $2/\log_2(10/3)$.

Note (v0.17.4): The reviewer's proof has been acknowledged and the tighter bound stated as Conjecture 11.17, with Remark 11.18 noting that inclusion of the proof is deferred to a later revision. The figures and open problem 3 have been updated to reference the conjectured bound. Promotion of the conjecture to a theorem, and the consequent update of Theorem 11.15 and Remark 11.16 to the tighter slope, are deferred pending inclusion of the proof.

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RC-22 (v0.14). Conj. 10.10 (bound via e) refuted; replaced by Remark 11.16. *Source: gemini — Since: v0.14 — Status: Deferred*

The original conjecture that the constants (m_e, b_e) for $\theta = e$ serve as universal bounds is false and ill-posed. Aristotle (`golden_violates_small_slope`) proves that for $\theta = \phi$ no line $m \cdot \log_2(e_d + o_d) + b$ with $m \leq 0.7$ bounds the depth for all d, regardless of b. The crossover is not at extreme depth: at $d = 5$ the pair for ϕ is $233/144$ and $0.43 \cdot \log_2 377 + 1.19 < 5$ already (`golden_depth5_crossover`). Furthermore e is asymptotically *near the bottom* of the family, not the top, because its weights grow without bound. The original conjecture has been replaced by Remark 11.16, which gives the proved inequality $m^* \leq 2/\log_2 3 < 2$. The question of whether any tighter bound or exact value for m^* can be established remains open (see A-5, A-43).

RC-16 (v0.14). Inter-paper circular coupling: Paper 76 errors propagate to Paper 77. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): this comment rests on the hallucinated content of RC-13–RC-15. The cross-paper dependency between Papers 76 and 77 is real and acknowledged, but no "circular coupling" exists: Paper 77 uses definitions from Paper 76 unidirectionally, and the errors identified in the review process were independent in both papers.

RC-15 (v0.14). High notation density may impede referee comprehension. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the phrase "stream-tagging frameworks" does not appear in the manuscript. The density concern may be legitimate in principle but the specific evidence cited is not drawn from this paper.

RC-14 (v0.14). Base-specific structural dependencies limit generality. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the paper does not employ 2-adic or 3-adic valuations, nor does it treat generalised Collatz variants. These concepts are not present in the manuscript; the comment appears to be directed at a different paper.

RC-13 (v0.14). Weight propagation bound slackness uncharacterised. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the phrase "trajectory spikes" does not correspond to any concept in the manuscript. The paper makes no asymptotic claims that depend on weight-propagation slackness; the relevant complexity bound is now Theorem 11.15.

N17 (v0.14). Ticket uses "Thm 10.8" and "Figure 10.1" but PDF has Thm 10.7 / Fig. 2. *Source: aristotle — Since: v0.14 — Status: Deferred*

The Round 14 review ticket refers to "Thm 10.8" (the PDF has Thm 10.7; 10.8 is a Remark), "Figure 10.1" (the PDF has Figure 2), and quotes the bounding-line parameters as 0.454/1.92 while the PDF prints 0.45/1.9. This is an organisational issue with how future review tickets are generated, not a defect in the paper itself. Deferred to the review-process workflow.

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